LILONGWE UNIVERSITY OF AGRICULTURE AND NATURAL RESOURCES

BASIC SCIENCES DEPARTMENT

MODULE 2: CHEMISTRY II - CHE 121

Peter P. Mumba

MacDonald Jose

Felix Daire Kumwenda

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UNIT 1: HYDROCARBONS

1.0.0 Introduction

This unit initially gives you an introduction to Organic Chemistry. I will briefly discuss how carbon forms its bonds. Having grasped this we will then discuss the hydrocarbons followed by an overview of the nomenclature (naming) of organic compounds which will help you grasp the naming of compounds we will introduce in this and in subsequent units. After this, I will walk you through the chemical and properties of alkanes and alkenes.

1.1.0 Unit Objectives

By the end of this unit, you should be able to

- Define Organic Chemistry
- Define hybridization and explain sp^3, sp^2 and sp type of hybridization of carbon atom
- Define functional group
- Give the chemical and physical properties of alkanes
- Use the set rules to name Alkanes.
- Give the chemical and physical properties of alkanes

1.1.1 Unit Key Terms

- Organic Chemistry
- Hybridization
- IUPAC Rules
- Functional group
- Alkane

1.2.0 Organic Chemistry

The name ‘’Organic Chemistry’’ started from the fact that in the early days, compounds were divided into two categories, **Inorganic** and **Organic**, depending on their source. Inorganic compounds were those that were derived from mineral sources and organic compounds were those derived from vegetable and animal sources, i.e., from living systems. Indeed at that time many scientists believed that organic compounds could only come from living systems and could not be made in the laboratory or from inorganic sources. **The compounds from living systems had one thing in common: they all contained the element Carbon**. Organic chemistry deals primarily with the chemistry of the carbon atom and its compounds. However, later on it was discovered that some so called ‘’organic compounds’’ could be made from inorganic compounds and even from non-living systems. Despite this, it was felt convenient to retain the word organic and this has stayed till today. You should note that all organic compounds contain the element carbon but not all carbon containing compounds are organic compounds. For example, carbon dioxide (CO_2) is a compound that contains carbon but it is not an organic compound.

1.3.0 THE CARBON ATOM

In the module on atomic theory and the structure of the atom we mentioned that there are eight main groups in the periodic table. In this init, and the rest to follow, we are concerned with group four elements and in particular, the **carbon atom**. As we said earlier, Organic Chemistry deals with the chemistry of carbon and its compounds. Interest in the chemistry of carbon lies in the fact that this element combines with more other elements in the periodic table than any other atom. A carbon atom can also form strong bonds with another carbon atom; either single, double or triple bonds can be formed so that several compounds are possible for this atom. Probably, a more important reason for studying the chemistry of carbon is that carbon compounds are involved in life processes; the proteins, carbohydrates, lipids (or fats) and many others are all important carbon compounds found in living systems. Also, many of the drugs, e.g. aspirin,

quinine, etc, that we take contain carbon. How about the clothes we wear, the wood on the roofs of our houses, all these also contain carbon.

In the second row of the periodic table, in which carbon appears, the outermost shell can have a maximum of eight electrons. A carbon has got only four electrons in its outermost shell and so it (i.e., the shell) is not full. *All elements, which have their outermost shells incomplete, will strive to fill them completely so as to acquire an inert gas configuration* (i.e., configuration of group 8a elements; two electrons in the K-shell, eight in the L-shell, eighteen in the M-shell, and so on). This is why they react (or combine) with other elements to form compounds (see the module on bonding you covered in the first semester). For carbon, there are three ways in which it can get the inert gas configuration:

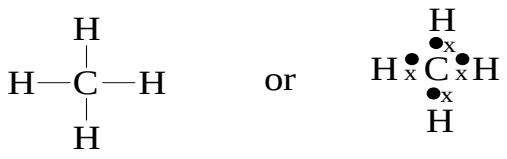
(a) It can gain four other electrons to form the C^{4-} ion so that it has the neon configuration.

(b) It can lose its four outer-shell electrons and form the C^{4+} ion so that it has the helium configuration.

(c) It can share its four outer-shell electrons with other elements (which also have incompletely filled orbitals) so that it has the neon configuration.

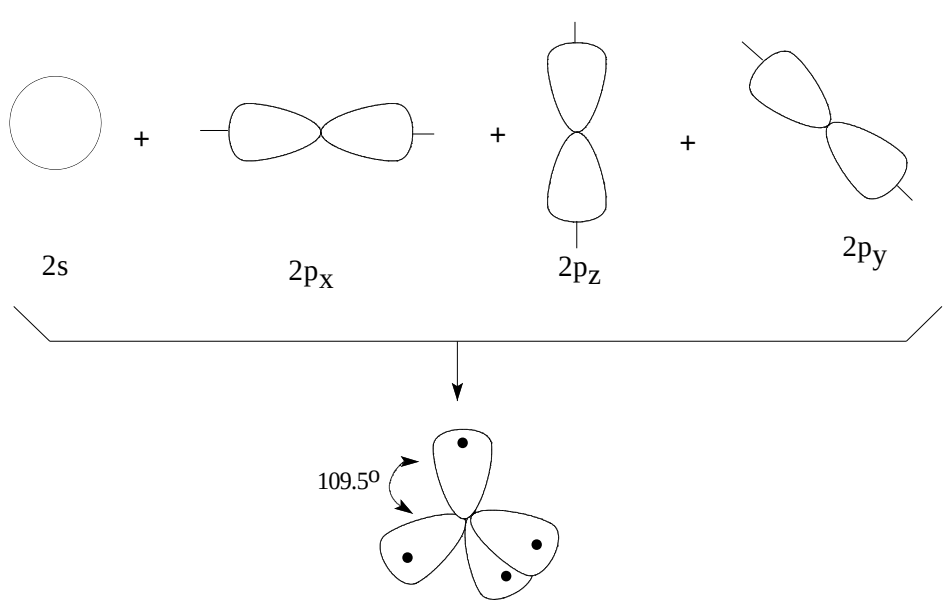
Case (a) is not possible for carbon because of the small nature of the atom the +4 charge on it would be much. If the C^{4-} ion existed, it would be very reactive in an attempt to reduce the charge. As such the ion, C^{4-}, does not exist. Case (b) is the same as case (a), the C^{4-} if it existed, would be very reactive in an attempt to reduce that charge. Case (c) is the only possible one; carbon shares its valence electrons with other atoms that also, have some of their shells not full. Carbon compounds are therefore formed through covalent bonding.

A bond is formed from two electrons, and so carbon, with its four electrons can form a maximum of four bonds (i.e., each of the four valence electrons in carbons shares with another electron from another atom to form a bond). Remember that a covalent bond is formed from overlap of two orbitals, one from each of the participating atoms. Multiple bonds are also possible in which two or three electrons in carbon, share with an equal number of electrons from another atom to form double or triple bonds respectively. The bonds can be represented by solid lines or by dots and crosses (a *Lewis structure*,) as shown for single bonds in methane below:



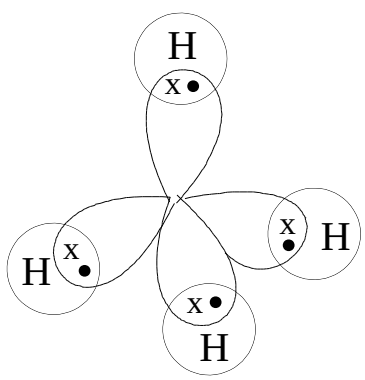
1.3.1 HYBRIDIZATION IN CARBON

With the Pauli Exclusion Principle mentioned in the unit on atomic theory and the structure of the atom , we would write the electronic structure of carbon as $1s^2 2s^2 2p^2$. Recall that there are three of the p-orbitals which are degenerate (of the same energy), and situated at right angles in three dimensions along the x, y, and z-axes. They are therefore represented as, np_x, np_y and np_z. For carbon we, have $2p_x$, $2p_y$ and $2p_z$. Also remember that occupation of such degenerate orbitals by electrons follows Hund's rule. The ground state electronic structure of carbon is therefore more correctly written as $1s^2 2s^2 2p_x^1 2p_y^1$ with the $2p_z$ orbital, empty. Since there are now only two incompletely filled orbitals in the ground state of carbon, one might be tempted to think that carbon would form only two bonds in its compounds. *Well*, compounds in which carbon is divalent, have been detected but they are not stable. As we have already mentioned, carbon forms four bonds in all its stable compounds. For this to happen, carbon must have four electrons in its outer-shell. Carbon achieves this by promoting one electron from its 2s orbital to the empty $2p_z$ orbital. This gives rise to a carbon atom in the excited state with the configuration $1s^2 2s^1 2p_x^1 2p_y^1 2p_z^1$. With four half-filled orbitals, carbon can now form four bonds rather than two. However, if the three half-filled 2p-orbitals and the one half-filled 2s orbital were used to form bonds as they are, there would be three equivalent bonds at right angles to each other (from the three 2p orbitals) and one different bond (from the 2s orbital). Experimental evidence shows however, that all the four single bonds in carbon compounds, like methane, are identical and that all the bond angles are 109.5^o to each other. *This is achieved by mixing together the three 2p orbitals and the one 2s orbital to yield four new identical orbitals called* **sp^3 hybrid atomic orbitals**. The mixing process is purely mathematical and it is termed **hybridization**. Visualize this mixing process as the addition of the four half-filled orbitals to form four new orbitals as in the diagram below:



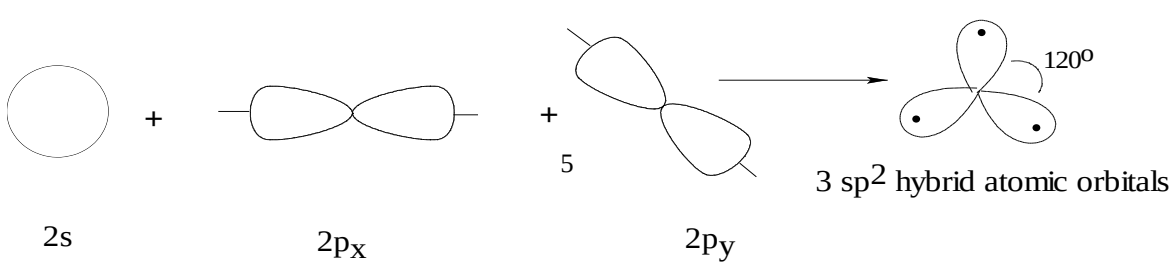
4 sp^3 hybrid atomic orbitals

Each of the four sp^3 hybrid orbitals gets one electron and shares it with an orbital electron from another atom (e.g., hydrogen) to form a bond. In a compound such as methane (CH_4) for example, we imagine an s-orbital of a hydrogen atom overlapping end on with one of the sp^3 hybrid orbitals of carbon and the two orbitals share their electrons. Since there are four hybrid orbitals in carbon, we need four hydrogen atoms, each sharing its electron with one of those in carbon, thus:



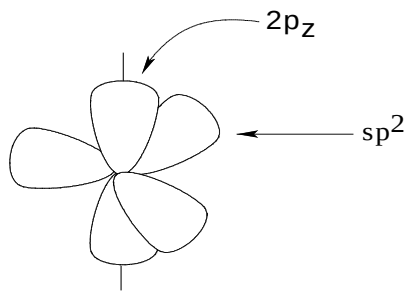
C-H or C-C single bonds (and the other single bonds of carbon compounds) are formed in that way, with the overlap of the orbitals being along the internuclear axes (i.e., the bond cuts across the nuclei of the two atoms joined). Each overlap results in a *molecular orbital* called a **sigma** (σ) **orbital** and *the bond formed is called a sigma bond* (i.e., σ-bond). For example, in methane, there are four σ-bonds.

This is not the only way a carbon can form its bonds. It is also possible for carbon to mix its 2s



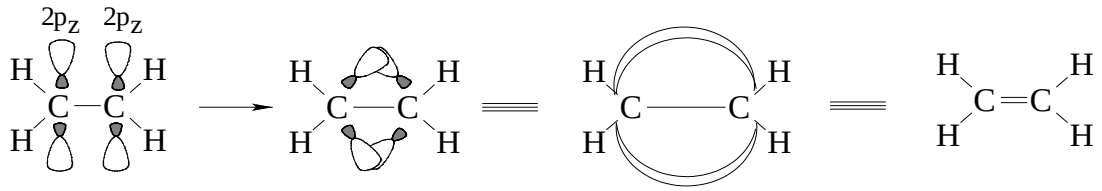
(a)

orbital with only two of its 2p (e.g., $2p_x^1$ and $2p_y^1$) orbitals. This gives *three identical hybrid atomic orbitals* called **sp^2 hybrid orbitals**. These orbitals are in one plane (i.e., are planar or flat) and disposed at an angle of 120° to each other (see (a) below). The other 2p orbital (unhybridized $2p_z$ in this case), remains at right angles to the plane of the three sp^2 orbitals (as shown in (b) :



(b) 3 sp^2 hybrid atomic orbitals at right angles with a $2p_z$ orbital

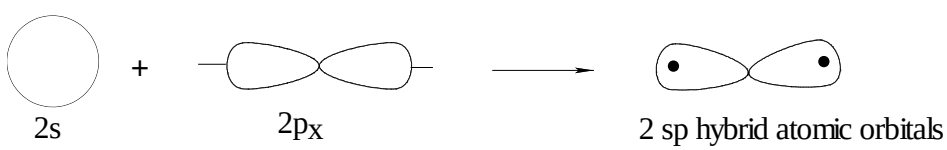
In molecules such as ethene, the 1s orbitals of two hydrogen atoms overlap, each with an sp^2 orbital while the remaining sp^2 orbital from each carbon overlaps end on with a similar one from a different carbon. All the three overlaps result in three sigma bonds. The remaining 2p orbitals (one on each carbon) overlap sideways with each other above and below the plane of the sp^2 orbitals. This latter overlap leads to a molecular orbital called a **pi-orbital** (π) *and the bond formed is called a* **π-bond**. The sideways overlap of the p orbitals is not as effective as the end on overlap between two sp^2 hybrid orbital overlap. Consequently, the pi bond is weaker than the sigma bond. The pi bond does not pass through the centers of the two atoms that are joined together. As such it is not symmetric like the sigma bond is. The formation of a pi bond means that there are now two bonds between the carbon atoms, i.e. there is a carbon to carbon double bond, which in ethene can be depicted as follows:



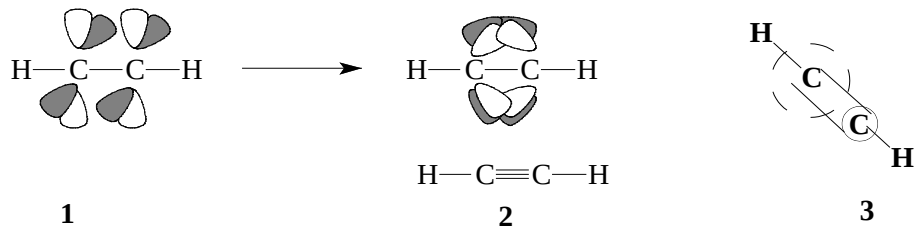
$2p_z$ orbitals overlap above and below the plane of sp^2 hybrid orbitals

The electrons occupying the π-orbitals are called **pi electrons**. You should note here that because the sp^2 hybrid orbitals are planar, the atoms or groups attached to the central carbon atom together with the carbon atom itself, will be coplanar and this makes it possible for the unhybridised p-orbitals (at right angles to the plane) to overlap easily. As can be seen in the structure of ethane above, the pi bond is exposed to the outside. There is thus a region of negative charge that can be attracted to anything that is positively charged. Such positively charged species are called **electrophiles** (Electro = electron and phile from Greek: *philen* = to love, so ''electron lover''). Let me *point out at this stage that the number of hybrid orbitals produced equals the number of atomic orbitals mixed*.

There is a third type of hybrid orbital that carbon forms. When the 2s and only one of the three 2p orbitals mix, two orbitals are formed and these are called **sp hybrid orbitals**. These are linear orbitals (disposed at an angle of 180° to each other):



The remaining two 2p orbitals ($2p_y$ and $2p_z$) are left at right angles to each other and to the sp hybrid orbitals. The pattern of bonding in ethyne is similar to that in ethene; a carbon to carbon σ-bond is formed from overlap of a hybrid orbital from one carbon with that of another carbon. In ethyne, this is one sp hybrid orbital on one carbon overlapping with another one from a different carbon. The remaining sp hybrid orbital on each carbon, overlaps with the 1s orbital of a hydrogen atom to form a sigma C-H bond. Unhybridised 2p orbitals on each carbon, overlap with the corresponding 2p orbitals on a second carbon atom (i.e., $2p_y$ on one carbon overlaps with a $2p_y$ on another carbon, etc.), and this results in the formation of two pi-bonds. Ethyne has therefore got three C-C bonds (two pi and one sigma bond), as shown in **1** and **2** below. The effect of these two overlaps is that the ethyne molecule appears cylindrical (**3**).



The linear nature of the sp orbitals makes the compounds having triple bonds (e.g., ethylene) linear as well.

1.4.0 HYDROCARBONS

The term hydrocarbon refers to compounds that contain carbon and hydrogen atoms only. Hydrocarbons can be classified into two large categories:

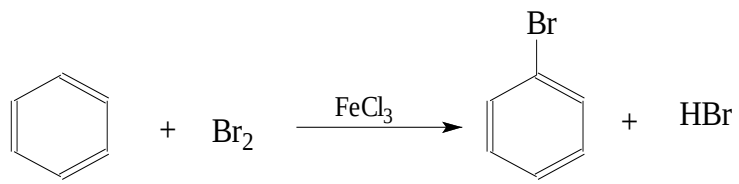
(a) ***Aliphatic*****:** These are hydrocarbons in which the carbon atoms are joined to form chains. The chains can also form rings. *Open chain ones are called* **acyclic**, while the cyclic ones are called **alicyclic**.

Aliphatic hydrocarbons are further divided into two groups: The **saturated** ones contain carbon to carbon single bonds and have the maximum number of hydrogen atoms. The alkanes for example, are saturated. **Unsaturated** aliphatic hydrocarbons contain either carbon to carbon double bonds (these are called alkenes) or carbon to carbon triple bonds (these are called alkynes). *Thus, the unsaturated hydrocarbons have multiple bonds*.

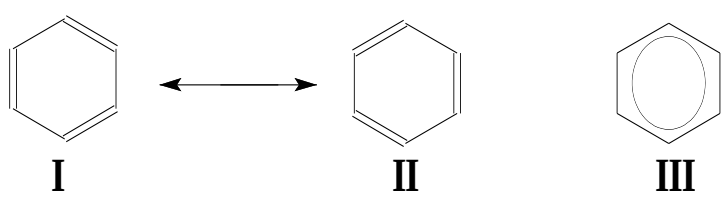
(b) ***Aromatic:*** In simple terms, these are cyclic rings containing carbon and hydrogen atoms only in which carbon to carbon double bonds alternate with carbon to carbon single bonds. Since the rings have double bonds, aromatic compounds may be said to be unsaturated. In contrast to the alkenes however, aromatic compounds exhibit special stability, a property that is absent in the alkenes. The simplest of the aromatic compounds is benzene. Benzene and other aromatic compounds have chemical and physical properties that are different from those of the alkenes. For example, the alkenes undergo addition reaction with bromine as shown with ethene below:

$$CH_2{=}CH_2 \;+\; Br_2 \;\rightarrow\; CH_2Br\text{-}CH_2Br$$

Under the same conditions, benzene undergoes a substitution reaction in the presence of a catalyst such as ferric chloride, i.e.,



The double bonds in a benzene ring are not affected in the reaction. There are several explanations for the stability of these aromatic rings and we will mention only one here. Benzene can be represented by two structures **1** and **II**; the actual structure being intermediate between these two, i.e.,



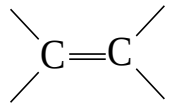
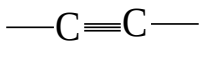
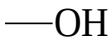
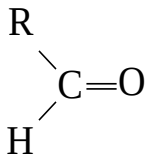
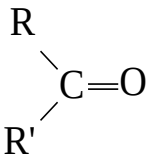
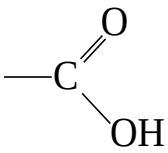
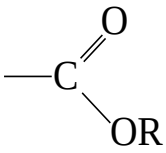
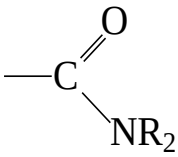
X-ray analysis has shown that the structure of benzene is planar and that all the six bond lengths are equal. This means that the single bonds in structure **I**, are not necessarily single and the double bonds are not double bonds. It is for this reason that structure **II** comes in. Note that the single bonds in structure **I** are written as double bonds in structure **II**. This implies that the bond lengths in benzene are intermediate between those of single and double. Because neither structure **I** nor **II** is the real structure of benzene, the two are connected by a double headed arrow. The two structures above are known as **canonical structures**, and the actual structure is called a resonance hybrid of the two. The π-electrons in benzene are said to be delocalized (or spread) over the entire ring. For this reason, benzene of often written with a ring inscribed inside

the hexagon as shown in structure **III**. The hydrogen atoms are sometimes left out on the structure but you must remember that each carbon bears one hydrogen atom.

1.5.0 FUNCTIONAL GROUPS

The ability of carbon atoms to bond to each other and to other different atoms or groups means that several carbon compounds can be made. You would then have to study the chemistry of all the known carbon compounds in order to understand them - a job too difficult to undertake!. Fortunately, each organic compound contains an atom or a group of atoms that makes the compound behave in a way characteristic of that atom or group of atoms. Such an atom or group of atoms that determines the properties (either physical or chemical) of a compound is called a **functional group**. This means that organic compounds can be classified according to the type of functional group that they contain. All compounds having the same functional group will therefore have similar properties. For example, compounds that contain carbon to carbon double bonds are placed in the same class (the alkenes) and their important reactions involve the double bond. If there were say, a million organic compounds to be studied you would find that these can be divided into ten families depending on their functional group. The important point is that, once you study the reactions of one member in each family then you know the reactions of the rest of the compounds in that family. Some commonly encountered functional groups and the corresponding classes of compounds are given in Table 1.1.

Table 1.1 Some classes of organic compounds and their functional groups

Class of Compounds	Functional group
Alkene	
Alkyne	
Alcohol	
Alkanal (Aldehyde)	 (R = C or H)
Alkanone (Ketone)	 (R,R' = C)
Carboxylic acid	
Anine	R_3N (R = C or H)
Ester	 (R = C)
Amide	 (R = C or H)

Take note in the table above that some compounds have two different functional groups attached to the same carbon. In compounds such as the carboxylic acids or amides, the presence of the two functional groups gives rise to properties that are neither for one nor the other functional

group. Such compounds are therefore placed in separate classes and they are said to be polyfunctional compounds.

1.6.0 AN OVERVIEW OF THE NOMENCLATURE OF ORGANIC COMPOUNDS

Before we discuss the different classes of organic compounds, it is perhaps appropriate to look at the general rules of naming them. This will help you to have a firm grasp and understanding of the principles, when we come to the naming of each specific class in the later units of this module. In the early days, many organic compounds were prepared and given names that reflected the source or the method of preparation. The names gave little, if any, information about the structures of the compounds and worse, the names had Latin or Greek roots. If this system of naming continued, people using other languages would face problems in understanding them. *Such common names (i.e., names that do not describe structures) are called* **trivial names**. The answer to this problem was provided by the **I**nternational **U**nion of **P**ure and **A**pplied **C**hemistry (IUPAC), which formulated rules for naming these organic compounds. *The rules give rise to names that describe structures* and are therefore understood by all chemists. *The names of organic compounds derived using* IUPAC *rules are called* **IUPAC *names*** *or* **systematic names.** A systematic name has three parts: the **root**, which is taken as the parent compound (or base compound), shows the number of carbon atoms in the parent compound; the **ending**, shows the nature of the functional group and the degree of unsaturation; the **prefix**, shows the groups (sometimes called the substituents) attached to the root. Table 1.2 shows the endings for some classes of compounds

When a hydrogen atom is removed from any of the top three groups (i.e., alkane, alkene or alkyne), the resulting radicals have the ending -yl. For example, alkyl in alkanes (e.g., methyl, CH_3), alkenyl in alkenes (e.g., propenyl, $CH_3CH{=}CH$-) and alkynyl in alkynes (e.g., ethynly CH=C-). When there are many double bonds (or triple bonds) in a molecule, they are indicated by the suffixes: di- (for two), tri- (for three), tetra- (for four), penta- (for five), etc.

Table 1.2: The endings for some classes of compounds

-ane	designates C-C single bond, as in alkanes
-ene	designates C=C double bond, as in alkenes
-yne	designates Carbon to Carbon triple bond, as in alkynes
-ol	indicates alcohol
-one	indicates alkanone (or ketone)
-al	indicates alkanal (or aldehyde)
-oic acid	indicates carboxylic acid
-oate	indicates ester

The general IUPAC rules listed below apply to all classes of compounds although only a few groups are given as examples. *It should be emphasized that these rules are always used in combination with each other*.

1. Choose the longest continuous chain in the compound; this is taken as the parent compound (or base compound) and given its name as appropriate (i.e., whether alkane or alkene, etc.).
2. Number the parent chain starting from the end nearest to the functional group[15] so that the position of the group has the lowest possible number. In substituted alkanes, numbering starts from the end near the substituent.
3. The functional group is located by the number of the carbon atom to which it is attached and this number is placed either before or inserted in the parent name before the ending (see text).
4. The name of the substituent is written before the parent name, preceded by the number of

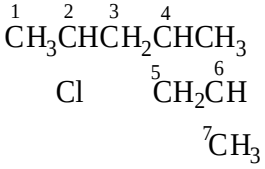
[15]When several different functional groups are present in a molecule, there is a set of rules governing the priorities in numbering but this is beyond the scope of this book. Interested readers should refer to further reading texts given in this chapter.

the carbon atom to which it is attached. Names of substituents are written in alphabetical order. Every substituent group must have its own number. Numbers are separated from each other by commas and are separated from words by hyphens.

5. If two or more substituents are similar, they are indicated by prefixes, **di** (for two), **tri** (for three), **tetra** (for four), **penta** (for five), etc. Each substituent must have a number locating its position on the carbon atom to which it attached.
6. In the alkenes or alkynes, since the double or triple bond is between two carbon atoms, the lower number is used to locate the position of the bond.
7. In the cyclic hydrocarbons, the prefix **cyclo-** is added to the parent name and the rules above then apply. Numbering starts from one carbon atom and proceeds round the ring in such a way that the numbers locating the substituents are as small as possible.

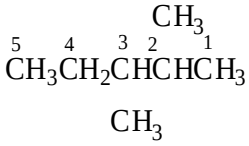
The examples below illustrate application of these rules:

ALKANES



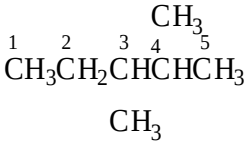
2-Chloro-4-methylheptane
(note: the longest chain is numbered- rule 1 and smaller numbers are taken- rule 2)

NOT 6-Chloro-4-methylheptane

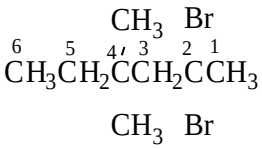


2,3-Dimethylpentane
(smaller numbers taken- rule 2, and two similar groups present hence dimethyl - rule 5)

NOT

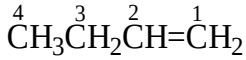


3.4- Dimethylpentane



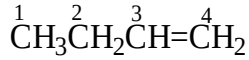
2.2-Dibromo-4.4-dimethylhexane
(two similar groups on one carbon; two numbers are required for each group- rule 5)

ALKENES

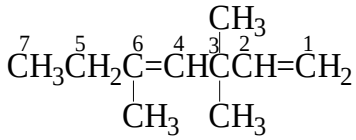


But-1-ene (rule 6)
(or 1-butene)

NOT

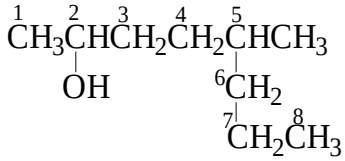


But-3-ene



3,3,5-Trimethylhepta-1,4-diene
(or 3,3,5-Trimethyl-1,4-heptadiene)

ALCOHOLS

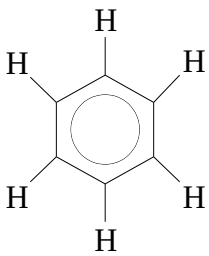


5-Methyloctan-2-ol
(or 5-Methyl-2-octanol)

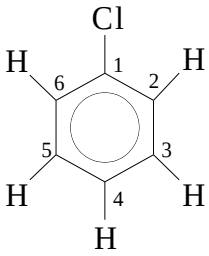
Note: Numbering starts from end close to functional group- rule 2

If the suffix in the name begins with a consonant such as in "diene", an "**a**" is placed before the ending. In the example of 3, 3, 5-trimethyl-1, 4-hept**a**diene, note the bolded letter "**a**"in the name. This is just for convenience as the name sounds better that way than without the "**a**" placed.

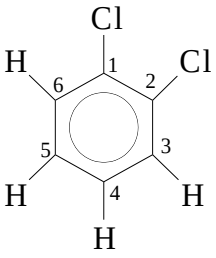
The aromatic compounds have their own system of names. The simplest aromatic compound is benzene (A):



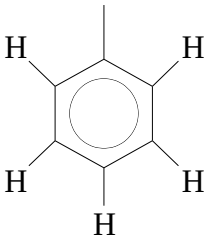
A



B



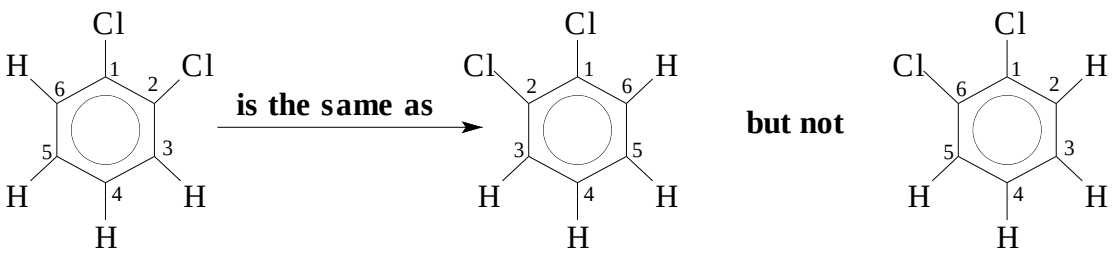
C



D $[C_6H_5]$

A substituted benzene molecule such as (B) in which one hydrogen atom is replaced by chlorine is called chlorobenzene (if the substituent was methyl (CH_3-), then it would be methylbenzene, etc.). In structure (C), two hydrogens on benzene have been replaced by chlorines. The ring is

numbered starting from a carbon atom to which a chlorine atom is attached as 1, and proceeds round the ring so that the other chlorine atom is given a small number. Since the chlorines are on positions 1 and 2, the molecule (C) is named 1, 2-dichlorobenzene (note, the rules 3 and 4 above, being applied here too). There can be 1, 3- and 1, 4-dichlorobenzene as well (try to write them). However, position 2 in structure (C) is the same as position 6 so that 1, 2-dichlorobenzene is the same as 1, 6-dichlorobenzene. From rule 2 above, we have to take the smaller numbers, so that the correct name is 1, 2-dichlorobenzene.



1 **2** **3**

You should note that structures **1** and **2** above are just the same and the numbering is in accord with the IUPAC rules. In structure **3** however, the numbering is wrong. We should number it the other way. It can similarly be argued that positions 3 and 5 in benzene are the same so that 1, 3- and 1, 5-dichlorobenzene are the same compound. Analogously, the correct name would be 1, 3-dichlorobenzene.

If a hydrogen atom is removed from a benzene ring, as in structure (D) on the previous page, the resulting radical or group is named phenyl. The general name of such aromatic radicals or groups is aryl (just like alkyl for alkane radicals.

1.7.0 THE ALKANES

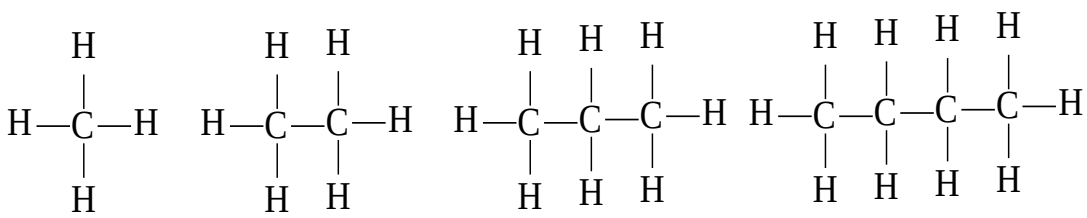
The alkanes are *aliphatic hydrocarbons that contain carbon to carbon* (C-C) *single bonds*. They are therefore saturated hydrocarbons and they have a general formula C_nH_{2n+2}. Some alkanes are folded into rings; these are called **cycloalkanes** and they have a general formula

C_nH_{2n}. We shall consider the chemistry of the cycloalkanes later but their naming will be considered here.

1.7.1 Nomenclature of the Alkanes

The saturated hydrocarbons occur as a continuous sequence of carbon atoms joined to each other. However, when there are more than three carbon atoms in a chain, branching can occur leading to a variety of isomeric structures (see unit 4). The hydrocarbons in which the carbon atoms are arranged in a continuous sequence are called **normal hydrocarbons** and for these the prefix, 'n', is added to their names.

Names of alkanes have the ending, **-ane** and the first four continuous chains have trivial names, methane (1 carbon), ethane (2 carbons), propane (3 carbons) and n-butane (4 carbons).



1 2 3 4

C_nH_{2n+2}: n = 1 n = 2 n = 3 n = 4

Above n = 4 (i.e., n □ 5 in the formula C_nH2_{n+2}), the names of the hydrocarbons have the prefixes pent-, hex-, hept-, oct-, etc., to show the number of carbon atoms. We have then; pentane (five carbon atoms), hexane (six carbon atoms), heptane (seven carbon atoms) decane (ten carbon atoms) and so on.

It should be noted in structures 1-4 above that each structure differs from the next higher one by a methylene group ($-CH_2-$). For example, structure 2 differs from structure 1 by CH_2 (i.e., adding a CH_2 group to structure 1, produces structure 2). We therefore have a series in which one molecule is bigger than the lower one by the same $-CH_2-$ group. Such a series is called a **homologous series**. If one hydrogen atom is removed from an alkane, then we have an *alkyl*

radical of general formula C_nH_{2n+1}. The names of the alky radicals are derived from the names of the corresponding alkanes, the ending **-ane** in the alkane being replaced by **-yl**. For the normal alkanes, the names are straightforward as shown below.

n	C_nH_{2n+2}	C_nH_{2n+1}
1	CH_4 methane	CH_3- methyl
2	CH_3CH_3 ethane	CH_3CH_2- ethyl
3	$CH_3CH_2CH_3$ propane	$CH_3CH_3CH_2$- propyl
4	$CH_3CH_2CH_2CH_3$ n-butane	$CH_3CH_2CH_2CH_2$- n-butyl
5	$CH_3CH_2CH_2CH_2CH_3$ n-pentane	$CH_3CH_2CH_2CH_2CH_2$- n-pentyl
6	$CH_3CH_2CH_2CH_2CH_2CH_3$ n-hexane	$CH_3CH_2CH_2CH_2CH_2CH_2$- n-hexyl

In the table above, the hydrogens are removed from the end of the chain. Radicals like the above, in which a hydrogen atom is removed from a carbon atom, which is attached to only one other carbon, are described as *primary radicals*. The other radicals are *secondary* (where a hydrogen atom is removed from a carbon atom attached to two other carbons) and *tertiary* (where a hydrogen atom is removed from a carbon attached to three other carbons). Examples are:

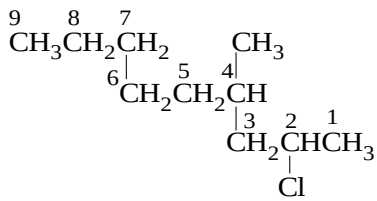
C_nH_{2n+2}		C_nH_{2n-1}	
$CH_3CH_2CH_2CH_3$	n-butane	$CH_3CH_2\underset{\vert}{C}HCH_3$	sec-butyl
$CH_3CH_2CH_3$	propane	$CH_3\underset{\vert}{C}HCH_3$	isopropyl
$CH_3-\overset{CH_3}{\underset{CH_3}{C}}-H$	tert-butane	$CH_3-\overset{CH_3}{\underset{CH_3}{C}}-$	tert-butyl

We have added the *isopropyl* radical to the butyl radicals above. *What is an **iso-** copmpound?* The prefix, "iso-", is used when two methyl groups are attached to a terminal carbon. For example, $(CH_3)_2CHCH_3$, is isobutane. When there are three methyl groups attached to a terminal carbon of

a long chain hydrocarbon, the prefix "neo-" is used. For example, $(CH_3)_3CCH_2CH_3$, is neohexane.

We can now apply the general rules given in the previous section to the naming of alkanes. Those rules are enough to describe the structures of alkanes. *We recommend that you continuously refer to them as you go over the examples below*. Here are some examples to

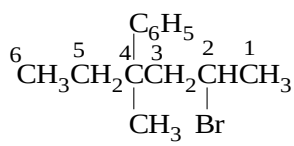
1



2-chloro-4-methylnonane

note: The longest chain is chosen and numbered (rule 1). This may not always be obvious. Look carefully!.

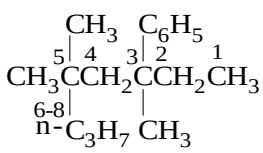
2



2-bromo-4-methyl-4-phenylhexane

note: The alphabetical order in naming substituents. Also note that C_6H_5, is phenyl ; a benzene ring, less one hydrogen and it is a substituent on the base chain

3

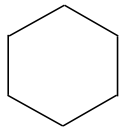


3,5,5-trimethyl-3-phenyloctane

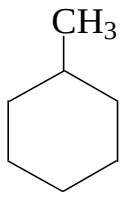
note: Look carefully!. The longest chain is the one numbered. n-C_3H_7 is normal propyl; -$CH_2CH_2CH_3$ (numbered 6-8), and must be a part of the long chain.

illustrate the application of the rules:

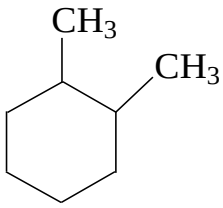
For the cycloalkanes, the prefix, **cyclo-** is attached to the name of the alkane with the same number of carbon atoms. When there are more than one substituents on the ring, numbering is done in such a way that gives all the substituents the lowest possible numbers (see rule 6). If only one substituent is on the ring, it is not necessary to locate its position (why?). The examples below illustrate this nomenclature system:



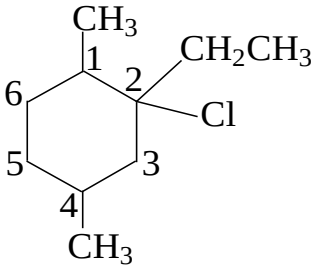
Cyclohexane



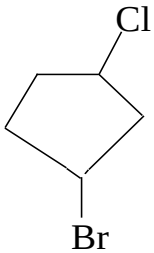
Methylcyclohexane
(**note**: no need to locate position of methyl group)



1,2-Dimethylcyclohexane



2-Chloro-2-ethyl-1,4-dimethylcyclohexane
(**note** the way the nubering goes (rule 6))

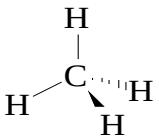


1-Bromo-3-chlorocyclopentane

Note that structures of cyclic hydrocarbons are most often written as polygons with the corners of the polygon representing the carbon atoms, and the hydrogens on each carbon are usually left out. This makes it less tedious to draw the structures.

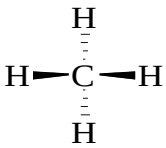
1.7.2 Conformations of acyclic alkanes

Drawing structures of alkanes can be easy if we realize that a carbon atom in these compounds is sp^3 hybridized. An sp^3 hybridized carbon has a tetrahedral shape. The bonds around this carbon are therefore, directed toward the corners of tetrahedron with the angles between any two of them at 109.5^o. In methane for example, the H-C-H angles are 109.5^o. In drawing a structure such as that of methane, we can either view it from the side (e.g., Structure **1**) or from the H-C-H angle (as in Structure **2**).



1

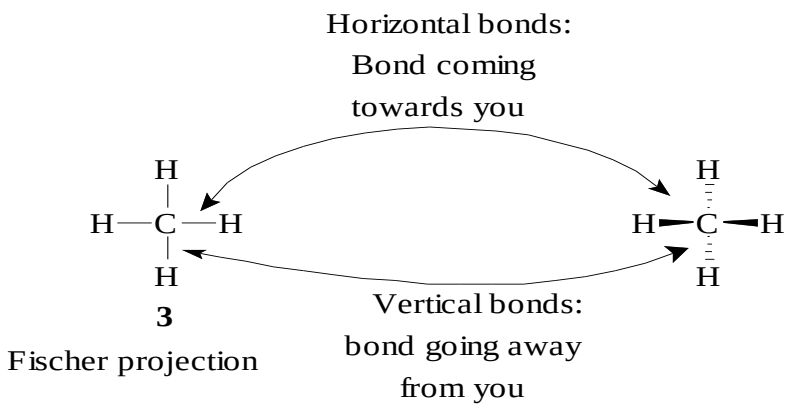
Side view



2

Viewed from the H-C-H angle

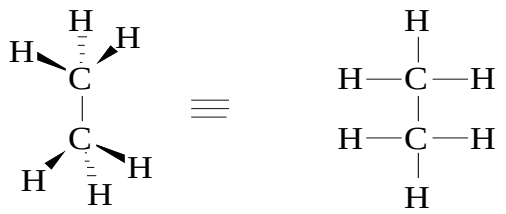
Molecular models can help to come up with the correct structure of a molecule. However, in their absence, you must imagine in your mind what shape a particular molecule will have. Both structure 1 and 2 are three-dimensional (3-D) presentations of methane. When viewing the structure from the side, you must orient the molecule in such a way that there are be bonds projecting towards you (the viewer), bonds projecting away and bonds lying in plane of the paper. In drawing such a molecule, the bonds which are in the plane of the paper are represented by solid lines, bonds coming out of paper towards you are written as thick wedges and the bonds going inside the paper and away from you are written as dotted lines as shown in structures **1** and **2** above for methane. Drawing structures as shown above can however, be cumbersome. Emil Fischer who worked on sugars, came up with a simpler way of representing compounds when viewed from an angle (e.g. structure **2**). The representations known as **Fischer projections** are written with a carbon atom at the center with vertical and horizontal solid lines attached to this carbon, representing the bonds as shown in structure **3**.



3

Fischer projection

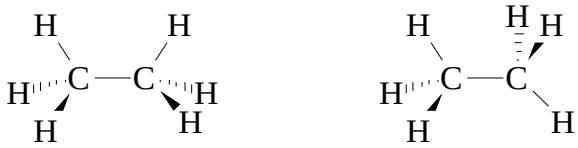
As you can see above, structure **3** is the same as structure **2** when we realize that the horizontal bonds are those that project out of paper (towards you, the viewer and represented by thick wedges in structure **2**) and vertical solid lines are those bonds that project away from you (represented by dotted lines in structure **2**). The 3-D representation and Fischer projection of ethane can therefore, be written as **4** and **5** respectively:



4 3-D structure of ethane

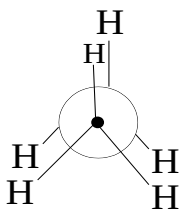
5 Fischer projection of ethane

In the case of ethane and higher alkanes, rotation of carbon-carbon bonds is possible. As a result, a particular molecule can have several different arrangements. These different arrangements representing the same molecule are known as **conformations.** Ethane for example, can assume two major conformations (there are others in between the two but we need not worry about them at the moment). One has each hydrogen on one carbon exactly in line (or aligned) with hydrogen on a second carbon. This arrangement is called the **eclipsed conformation** of ethane. The other conformation is where the hydrogen atoms on one carbon are not aligned with those on the other carbon. This arrangement is called the **staggered conformation**. These two conformations are written in the form:

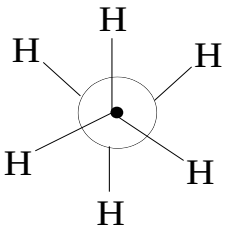


Eclipsed Staggered

The **Newman projection** is another way of representing conformations such as those of ethane and the other alkanes. Here, a molecule is viewed along the C-C bond axis with one carbon nearer you and the other carbon, away. The nearer carbon is drawn as a dot and the three C-H bonds extend from it. The far away carbon is indicated by a circle with the three C-H bonds joined to its circumference. Thus the eclipsed and staggered conformations of ethane can be represented in the Newman projections as follows:

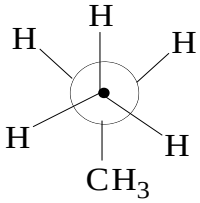


Eclipsed

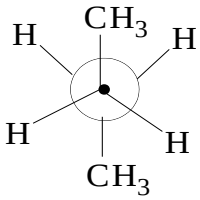


Staggered

The staggered conformations of propane and butane can therefore be written as:



Staggered form of propane



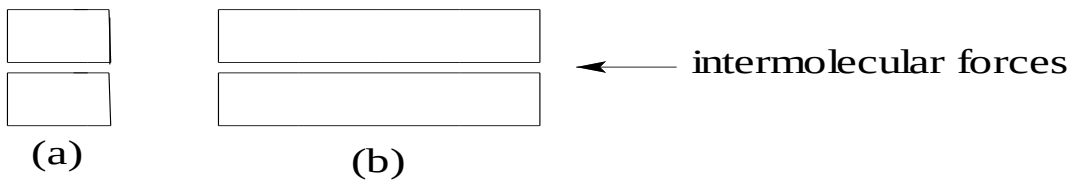
Staggered form of butane

Since, in the staggered forms, the atoms or groups of atoms are not aligned with each other on the two carbons, there is no interaction between them as is the case when these atoms or groups are aligned. Consequently, the staggered conformation is more stable than the eclipsed conformation.

1.7.3 Properties of acyclic alkanes

(**a**) ***Physical:*** For the normal alkanes, the low-molecular weight molecules (C_1-C_4) are gases under normal conditions. The medium ones (C_5-C_{17}) are liquids and those above C_{17} are solids at room temperature. This gradation of state is evident in the boiling and melting points of these compounds (Table 1.3). We can explain this difference in state by considering the sizes of the

molecules and the intermolecular forces operating. Let us consider for example, two different alkane molecules represented by the blocks below:



In alkanes, we have carbon to carbon bonds and carbon to hydrogen bonds only. The electronegativity values for carbon (2.5) and hydrogen (2.1) are nearly the same. As such, we can for all practical purposes, consider a carbon to hydrogen bond to be non polar. Consequently, the only forces that operate between any two alkane molecules are dispersion forces. In diagram (a) above, we see that the blocks are small. Consequently, the dispersion forces are operating on a smaller area than in diagram (b). This means that the forces in (a) are smaller than in (b). As a result the molecules represented by blocks in (a) would be in a different state from those represented by (b). Molecules of (a) would be gas while those of (b) would be liquid. If we imagine even bigger blocks than those of (b), we see that the dispersion forces would be even greater leading to a solid state.

Table 1.3: Boiling and Melting points of some normal alkanes

C_nH_{2n+2}	Name	Bp ^oC	Mp ^oC	state
n = 1	methane	-161.5	-183	gas
2	ethane	-88.6	-172	gas
3	propane	-42.1	-188	gas
4	butane	0.0	-138	gas
5	pentane	-36.1	-130	liquid
6	hexane	68.7	-95	liquid

7	heptane	98.4	-91	liquid
8	octane	125.7	-57	liquid
16	hexadecane	287	18	liquid
17	heptadecane	303	22	liquid
18	octadecane	306	28	solid
19	nonadecane	330	32	solid
20	eicosane	343	37	solid
30	triacontane	446.4	66	solid

From the table above we see that the normal alkanes show a smooth gradation of physical properties. For the alkane molecules with four or more carbon atoms however, the carbons can also join to form branches. For example, two butanes are known, called n-butane and isobutane. These are structural isomers (see unit 4). Branched alkanes, generally have lower melting and boiling points than their isomeric unbranched counterparts. For example, the boiling point of n-butane is 0^oC (see Table 1.3) while it is -12^oC for isobutane. The reason for this difference is obvious when we consider the shapes of the two compounds. n-Butane is $CH_3CH_2CH_2CH_3$ (linear) and isobutane is $CH_3CH(CH_3)CH_3$ (try to draw the full structure to see that it is T-shaped). The linear shape for the n-butane molecule means that these molecules can pack together more closely that the isobutane molecules which are T-shaped. Consequently, there are stronger intermolecular Van der Waals forces between n-butane molecules than between isobutane molecules. This results in higher boiling point for the former than for the latter. The same reasoning can be applied to other similar isomeric alkane molecules.

***(b) Chemical**:* Generally, alkanes are very inert; the C-C or C-H bonds are not readily attacked by strong acids or bases. Consequently, the alkanes are also called the **paraffins**, a term which means "of low affinity". However, at high temperatures (i.e., if we try to burn them), the alkane molecules react with oxygen. If oxygen is in excess, complete combustion occurs, producing carbon dioxide and water, and a blue flame. For example:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

If oxygen is not enough, incomplete combustion takes place, in which carbon dioxide and some fragments are produced, which give a yellow flame. It is the combustion of the alkanes that provides energy for vehicles.

In the presence of light and at room temperature, alkanes react with the first three halogens (fluorine, chlorine and bromine), according to:

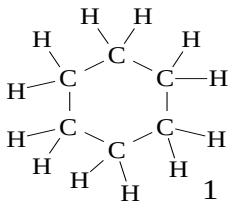
$$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$$

Further halogenation occurs, leading to dichloro-(e.g., CH_2Cl_2), trichloro-(e.g., $CHCl_3$) and tetrachloroalkanes (e.g., CCl_4). Note that in these reactions, a hydrogen on the alkane is replaced by a halogen. Reactions in which an atom (or a group of atoms) is replaced by another, are called **substitution reactions**. In the dark, at room temperature, such substitutions do not take place in the alkanes. Heavily halogenated alkanes do not react with oxygen; they are more inert than the alkanes themselves. Consequently, compounds such as carbon tetrachloride (CCl_4) are used as fire extinguishers.

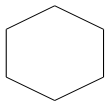
1.8.0 ***The Cycloalkanes***: We mentioned at the beginning of this unit that the hydrocarbon chains can also fold into rings. The cycloalkanes are hydrocarbons of this type; generally called *alicyclic hydrocarbons*. The cycloalkanes may be considered saturated cyclic alkanes although, they are not fully saturated. A normal alkane has a general formula, C_nH_{2n+2}. When a multiple bond is introduced in the structure, or, the structure itself is folded into a ring, the resulting structure becomes deficient in hydrogen atoms, and is said to have some degree of unsaturation. If the molecule is deficient in two hydrogens, the degree of unsaturation is one; when the molecule is deficient in four hydrogens, it has a degree of unsaturation of two, and so on. Therefore, cyclohexane has a degree of unsaturation of one.

Representations of cycloalkanes on paper can be one of two types; one in which the carbon and hydrogen atoms are included in the ring, as shown in structure**1**, below, for cyclohexane. This is however, tedious and we simplify it by drawing the rings as polygons in which the carbons and

hydrogens are omitted. Structure **2**, below is the same as structure**1**. The corners of the polygon represent the carbon atoms and the number of hydrogens on each carbon atom is understood.



1



2

cyclohexane

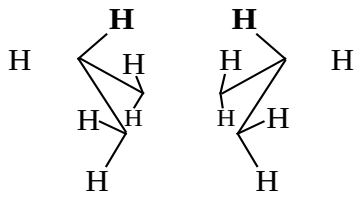
1.8.1 Properties of cycloalkanes.

The most abundant of the cycloalkanes are those with five and six carbons in the ring (cyclopentanes and cyclohexanes respectively). The physical properties of cycloalkanes are comparable with those of the corresponding open chain alkanes, although both the boiling and melting points tend to be higher in the former. The chemical reactions are also generally similar to those of the corresponding open chain alkanes. For example, cyclopentane undergoes a substitution reaction when reacted with chlorine, in the presence of light only.

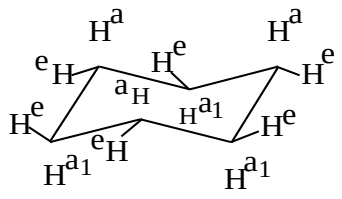
1.8.2 Strain theory.

If we consider the cyclohexane, say, structure 2 drawn above, we would say that the internal angles of this polygon should be 120°. Since the normal angles for a carbon atom are 109.5^o, we would be tempted to think that cyclohexane is a strained molecule. This is exactly what Adolf Baeyer (in 1885), had suggested for these ring compounds. Baeyer proposed that if the internal angles of a cyclic compound differed from those of a normal tetrahedral angle (109.5^o), for an sp^3 hybridized carbon, then the compound would be strained. This theory, called "Baeyers strain theory", assumed however, that the ring compounds are planar (i.e., flat). Cyclopropane is planar, so Baeyer's theory should apply to it. However, experiments show that cyclobutane and the higher cycloalkanes are quite stable, unlike what Baeyer predicted. Models of these

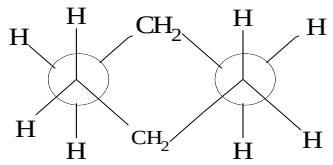
molecules have shown that their rings are not planer (or flat) but *puckered*. For example, two forms of cyclohexane have been identified, *the boat* and *the chair* forms:



boat



chair

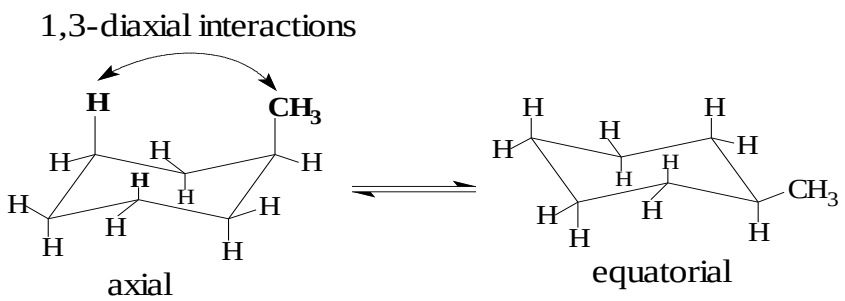


Newman projection of the chair form

Both forms are not planar structures. However, in the chair form, all the hydrogens are staggered (as can be seen in the Newman projection), and so, it is more stable than the boat form in which all the neighbouring hydrogen atoms are eclipsed (or, aligned). This causes nonbonded interactions (also called steric hinderance) to occur between the eclipsed atoms. In addition, the hydrogen atoms at the top of the boat, in **boldface** (sometimes called flagpole hydrogens), are so close that they interact and repel each other. The eclipsing of the hydrogens and the nonbonded interactions, make the boat form less stable. Let me point out here that although puckering will introduce strain in the ring (the so-called ring strain), this is overcome by the reduction in the nonbonded interactions between the neighbouring hydrogens.

The chair form has two different kinds of hydrogen atoms. These are labeled "a" and "e" in the structure above. As you can see in the structure, there are six hydrogens labeled "e", that form a belt round the plane of the ring and these are called **equatorial hydrogens**. These positions are also called equatorial positions. Any substituent occupying this position is called an equatorial substituent. The hydrogens labeled "a", are in two groups; three extend upwards (labeled, "a"), and three extend downwards (labeled, "a_1"). These are called **axial hydrogens** and their positions are said to be axial positions. Any substituent occupying this position, is called an axial substituent.

A mono substituted cyclohexane can therefore assume two different chair conformations; either with the substituent in the equatorial or axial position. For example, consider methylcyclohexane below.



The two forms are usually in equilibrium with each other. Experiments have shown that the form with the methyl group equatorial, is more stable than the one with the methyl group axial, i.e., equilibrium lies to the right hand side in the equation above. The reason for this is that when the methyl group is in the axial position, there is steric hindrance (or sometimes called nonbonded interactions) arising from interactions between the methyl group and the axial hydrogens (**in boldface**) on the same side (the so-called 1,3-diaxial interactions, see structure above). This makes the structure less stable. However, when the methyl group is in the equatorial position, there is no steric hindrance, since all the hydrogen atoms and methyl group in these positions are staggered. This results in a stable structure. The effect is more pronounced with even larger groups than the methyl. Consequently, larger substituents are found exclusively in the equatorial positions and not in the axial positions.

1.9.0 Unit summary

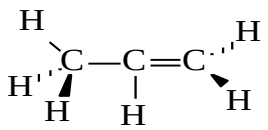
We started this Unit with the definition of Organic Chemistry. We stated that Organic Chemistry is the chemistry of carbon and its compounds. We then went on to briefly look at the carbon atom itself. It was stated that a carbon atom forms four bonds through hybridization of its orbitals. After this we covered hydrocarbons and said that these fall into two groups: aliphatic and aromatic hydrocarbons. It was then stated that apart from carbon and hydrogen, an organic compound may also contain an atom or group of atoms that define the its chemical and physical properties. We called this atom or group of atoms the functional group. We looked at an

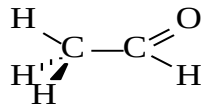
overview of the nomenclature of organic compounds. Finally we have discussed the chemical and physical properties of alkanes and alkenes.

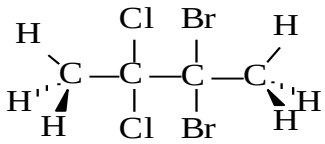
1.10.0 Unit test

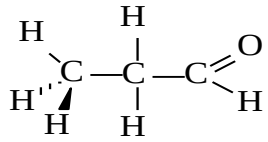
1. Define/explain the following:
    a. Hydrocarbons
    b. Functional group
    c. Monomer

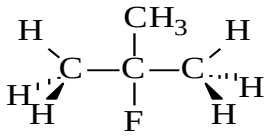
2. Write the Newman projections of the following structures.

(a) 

(c) 

(b) 

(d) 

(e) 

5. Explain why the boat form of cyclohexane is less stable than the chair form of cyclohexane

UNIT 2: Unsaturated Hydrocarbons (Alkenes and Alkynes)

2.0.0 Introduction

In this unit we will discuss the alkenes, compounds containing one or more carbon to carbon double bonds (i.e., -C=C-), which determine their properties. Therefore *the double bond is the functional group for this class of compounds.* The alkenes are therefore unsaturated compounds. Open chain alkenes have a general formula C_nH_{2n}. Alkenes are sometimes called olefins which means "oil formers", and the C=C bond is sometimes also called the olefin linkage. When double bonds alternate with single bonds (i.e., -C-C=C-C=C-, etc.), the alkene is said to be conjugated. Later on we will briefly look at the compounds containing a carbon to carbon triple bond, alkynes.

2.1.0 Unit Objectives

By the end of this unit you should be able to:

- Name a given alkene structure
- Draw a structure of an alkene from a given name
- Predict physical properties given an alkene molecule
- Outline Chemical properties of alkenes
- Preparations of alkenes- alcohol dehydration
- Complete chemical equations involving Electrophilic additions
- Determine Oxidation products given reaction conditions
- Identify alkynes
- Describe physical and chemical properties of alkynes

2.1.1 Unit Key Terms

- Unsaturated
- *Cis* and *tran*s isomerism
- Olefin

- Double bond
- Dehydration
- Electrophilic addition
- Markovnikov's rule
- Alkynes

2.2.0 Nomenclature of alkenes.

2.2.1Trivial Names

Simple alkenes have trivial names that are frequently used. Thus ethylene, propylene, butylenes are alkenes with 2, 3 and 4 (straight chain) carbons respectively.

2.2.2 Systematic names

The systematic names of alkenes are derived from those of the corresponding alkanes; the ending **-ane**, in the name of the alkane, being replaced by **-ene**. The rules given in section 1.6.0 also suffice for the alkenes. *Refer to them always*. We will however add a few others, specific to this group.

1. For the alkenes, the longest chain containing the double bond is chosen and numbered according to rule 2 (Section 1.6.0). Here, numbering starts from the end close to the double bond.

2. We have to locate the position of the double bond. Since the double bond is between two carbon atoms, we take the lower number to locate its position.

3. The number locating the double bond is usually inserted in the parent name but before the ending **-ene**. In certain cases though, this number can also be placed before the name of the parent compound.

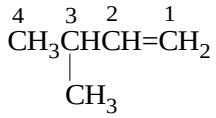
4. We locate any substituent attached to the longest chain by the number of the carbon to which it is attached.

5. When there are more than one double bond, each should be located by its own number and the suffixes diene, triene, etc., are used as appropriate.

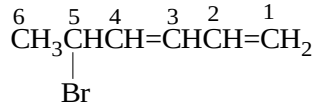
6. Cycloalkenes are numbered starting from the carbon atoms of the double bonds and this goes round the ring so as to include both carbons of the double bond in a numerical sequence. Numbering is also done in such a way that keeps the numbers of the other double bonds in the ring (if there are two or more double bonds), or substituents as small as possible.

7. Since geometrical isomerism (see unit 4) exists in alkenes, the prefix cis or trans is sometimes added to the name as appropriate. We will, however defer this now until we come to unit 4.

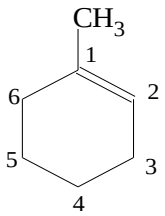
The following examples illustrate this nomenclature system:



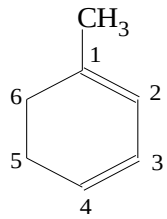
3-methylbut-1-ene
(or 3-methyl-1-butene)



5-bromohexa-1,3-diene
(or 5-bromo-1,3-hexadiene)



1-methylcyclohexene
note: number locating the double bond is not required here.



1-methylcyclohexa-1,3-diene
note: numbers locating the double bonds are required here

2.3.0 Properties of alkenes.

2.3.1 ***Physical:*** The physical properties of alkenes are similar to those of the corresponding alkanes; those with two to four carbon atoms are gases, those with five to seventeen carbons are liquids and those with eighteen or more carbon atoms are solids at room temperature. However, the restricted rotation about the C=C double bond can give rise to two different compounds (isomers), for a given formula. The two compounds are called geometrical or cis-trans isomers (see unit 4), and have different physical and chemical properties.

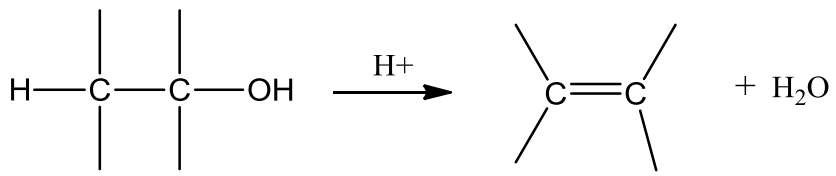
2.3.2 ***Chemical***: *The characteristic feature of the alkenes is the presence of the double bond, which is the functional group of this class of compounds.* Reactions of alkenes are therefore, centered on the double bond. It was also noted earlier that the sideways overlap of the p orbitals is not as effective as say, the end on overlap between two sp^2 hybrid orbital overlap. Consequently, the pi bond is weaker than the sigma bond. In most of the reactions of alkenes, the pi bond is the one that is affected as opposed to the sigma bond. Alkenes undergo several types of reactions, a few of which are described under various subheadings, below.

2.4.0 Preparations of Alkenes

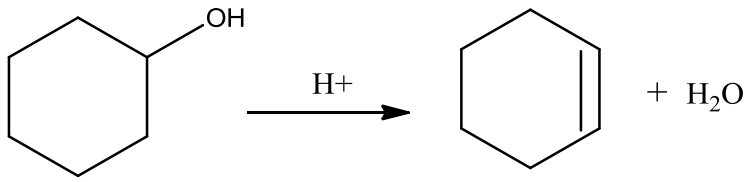
Alkenes can be prepared by the **dehydration** of alcohols and in the process a hydrogen atom, H and hydroxyl group, OH are lost from adjacent carbons. An acid catalyst is required for this reaction.

$$CH_3CH_2OH \xrightarrow{H^+/\ 160^oC} H_2C{=}CH_2 + H_2O$$

In general the dehydration reaction is as follows:

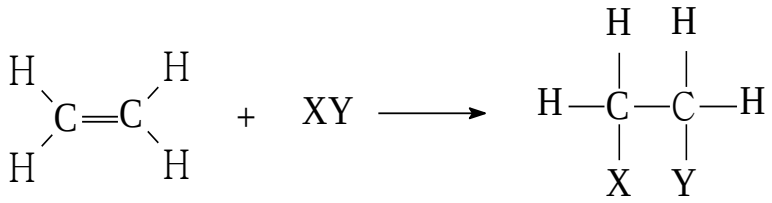


Below is an example of dehydration of a cyclic alcohol which forms a cyclic alkene.



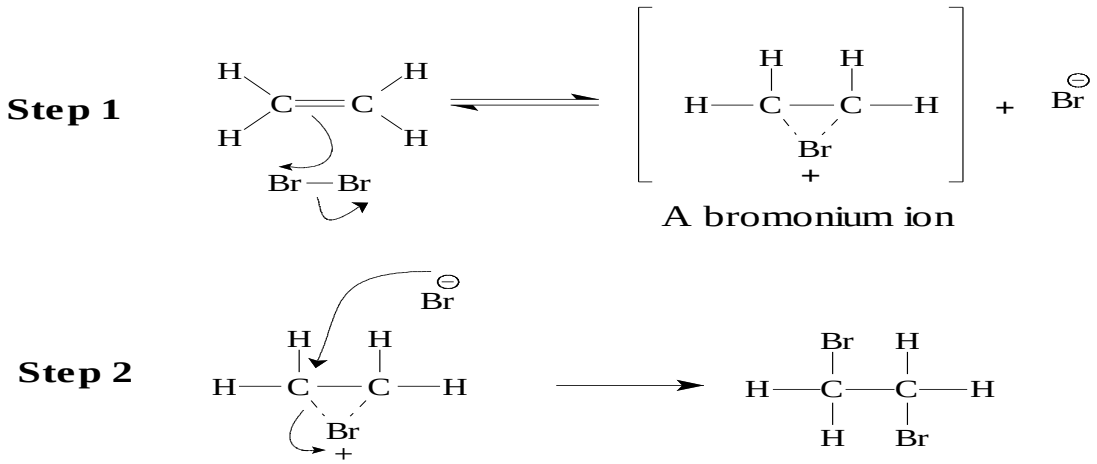
2.5.0. Electrophilic Addition Reactions

Reactions by which an atom (or group of atoms) adds to a molecule are called addition reactions. In these reactions a reagent adds to a molecule without loss of atoms (compare substitution reactions). Alkenes readily undergo addition reactions with various reagents. A generalized addition reaction equation is:



Now recall from section 1.3.0 that because the pi bond is exposed to the outside, there is a region of negative charge that awaits anything that is positively charged. Such a positively charged species we said, is called an **electrophile** (Electro = electron and phile from Greek: *philen* = to love, so ''electron lover''). An electrophile is attracted to the negative region and adds to it. Alkenes therefore undergo electrophilic addition reactions. Some substances that add to the alkenes are:

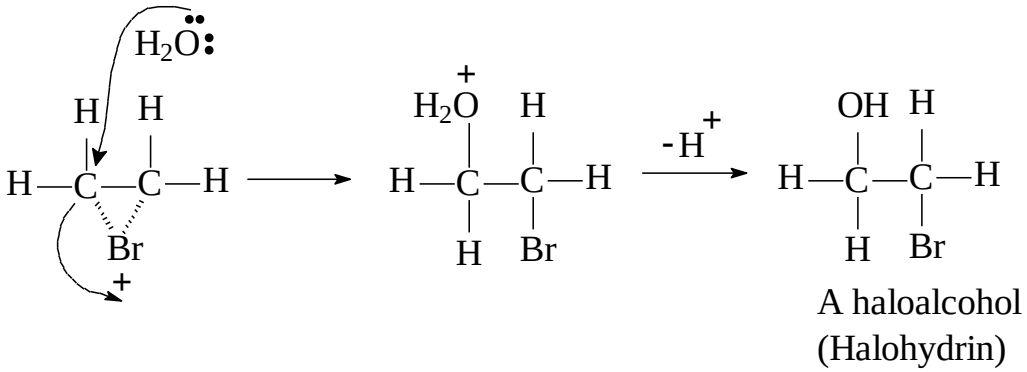
2.5.1 ***Halogens***: *One characteristic test for alkenes is the decolourisation of a bromine solution*. Bromine adds to the double bond of the alkene at room temperature (compare alkanes). The products of the addition depend on whether bromine is dissolved in an inert solvent (e.g., carbon tetrachloride) or a reactive solvent (e.g., water). *The bromine dissolved in carbon tetrachloride is one used to test for the alkenes*. In the first step of the addition process, the pi-bond breaks and one halogen atom becomes attached to both carbons of the double bond resulting in a positively charged cyclic complex (a transition state) termed a **halonium ion**. The halonium ion containing bromine for example is called a bromonium ion. Its formation can be depicted as **in step 1**:



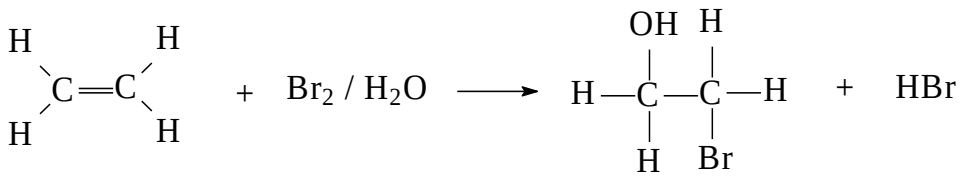
The

curved arrows indicate the movement of electrons. In the second step (step 2 above), the bromide ion attacks one carbon from the other side and this gives the final product, a dibromide. Note that in the final product, the bromine atoms are on opposite sides of the -C-C- single bond. This type of addition is therefore called anti-addition or trans-addition. The overall equation is therefore:

In the equation above, both bromine atoms add to the alkene. When a reactive solvent, such as water is used however, only one bromine atom adds to one of the carbons, and a group from the solvent adds to the other carbon. The first step is similar to that of the addition of bromine dissolved in carbon tetrachloride described earlier (step 1). However, in the second step, water instead of bromide (Br^-), attacks the bromonium ion to give an intermediate which then loses a proton (H^+) yielding the final product called a **halo alcohol** (also called a **halohydrin**):

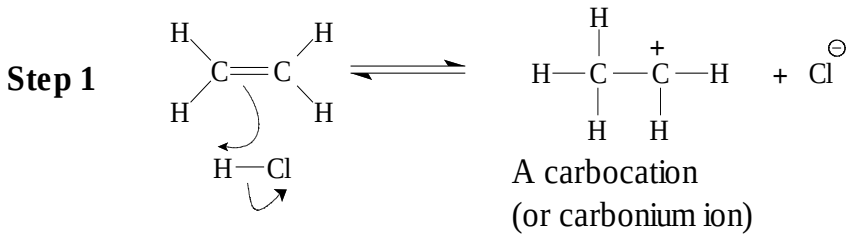


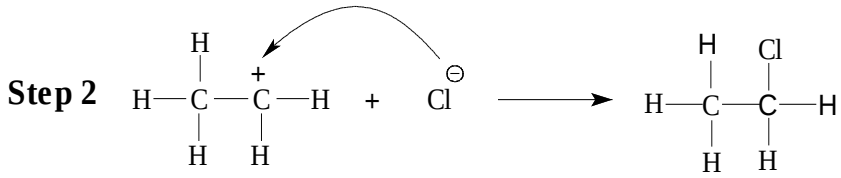
The proton combines with the bromide removed in the first step to form HBr. The overall equation is therefore:



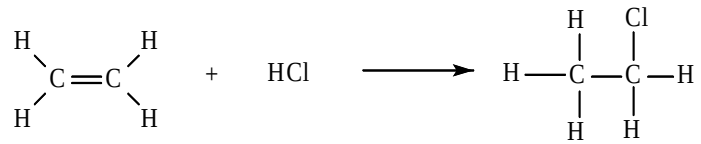
Other halogens, especially chlorine, add in the same way.

2.5.2 ***Halogen acids***: Reagents such as hydrogen halides (HCl, HBr, etc.) also add to the alkenes. Now because the pi-bond is not symmetric (like the sigma bond), there is a region of negative charge around the -C-C- bond. Since the HX molecule (X = halogen) is polarized (see module 1), with a partial negative charge on the halogen and a positive charge centered on the hydrogen atom, the hydrogen atom (of the HX) will be attracted to the region of negative charge of the pi-bond. The addition of HX to an alkene occurs in two steps; in the first step (**step 1**, below), a proton (from HX) is added to one carbon of the double bond yielding a cation termed a **carbocation** (or **carbonium ion**) and a halide. In the second step (**step 2**), the halide combines with the carbocation to give the final product. The addition of HCl to ethene for example, is as follows:





The net equation is therefore:



The order of reactivity of the halogen halides is in the order, HF >HBr >HCl >HI. In the example of ethene shown above, only one product is obtained. When the alkene is unsymmetrical (e.g., propene), two products could be possible, i.e.,

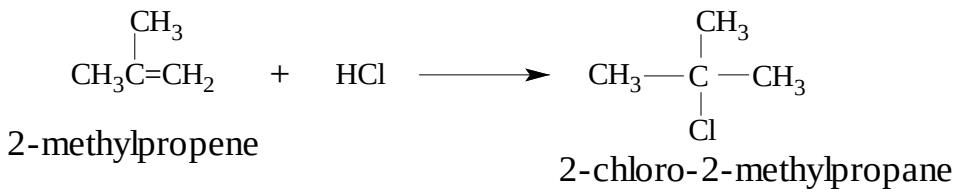
(a) $CH_3CH{=}CH_2 + HCl \longrightarrow CH_3{-}CH(Cl){-}CH_2{-}H$

(b) $CH_3CH{=}CH_2 + HCl \longrightarrow CH_3{-}CH(H){-}CH(Cl){-}H$

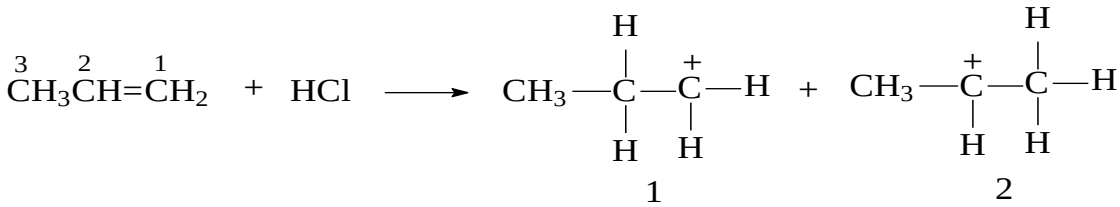
In (**a**), the hydrogen is added to the terminal carbon, while the chlorine is added to the middle carbon. In (**b**), the addition is the opposite of that in (**a**). Vladimir Markovnikov, a Russian chemist, studied many of this type of reactions and he observed a certain pattern in the additions. His observations were formulated into a rule which states as follows: "*when a molecule, HX, adds to an unsymmetrical alkene, the hydrogen atom adds to a carbon atom that already has more hydrogen atoms and the group, in this case, X, adds to the other carbon"*. This rule is generally called **Markovnikov's rule**[16]. Therefore, of the two equations (**a**) and (**b**) above, (**a**) is the correct one. The hydrogen will add to a carbon atom that already has more hydrogen atoms while the other group adds to the carbon atom with few hydrogen atoms. Using this rule, we can

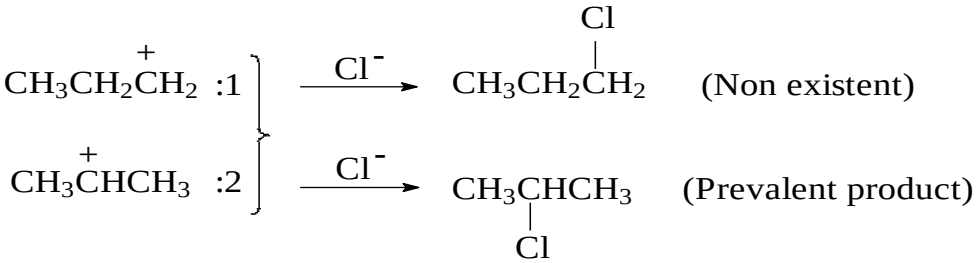
[16] The biblical analogy of this rule is: "*To those who have, more shall be added; to those who do not have, even the little they have, shall be taken away from them*"*(Matthew 13:12).*

straight away predict that in the reaction between 2-methylpropene and hydrochloric acid, the product will be 2-chloro-2-methylpropane, i.e.,



We can understand Markovnikov's rule if we realize that in the addition of say, HCl to an unsymmetrical alkene, such as propene, two carbocations are possible depending on which carbon of the double bond a proton is added. In the example below, notice that in carbocation **1**, the proton is added to carbon 2 while in carbocation **2**, it is added to carbon 1.

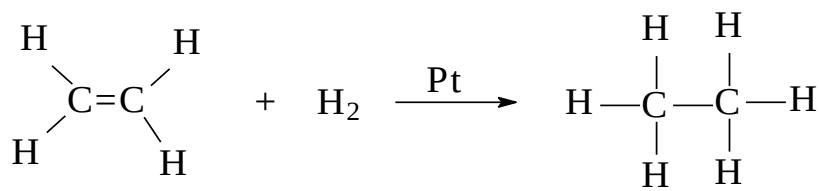




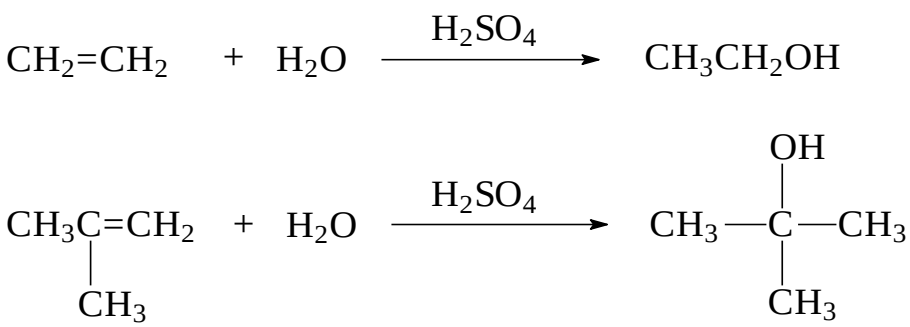
The stability of the carbocation determines which product will prevail. Alkyl groups (e.g., methyl) are known to stabilize carbocations much more than hydrogen atoms do. Therefore, the more alkyl groups attached to a carbon bearing the positive charge, the more stable and more available the cation. Of the two carbocations above, carbocation **2** with two methyl groups

attached to a positive carbon is more stable than carbocation **1**. Consequently, the product from that cation will prevail.

2.5.3 ***Hydrogen (hydrogenation)***: The addition of hydrogen to alkenes is also a very common process. This is an important reaction in that, oils (e.g., mulawe or sunfoil)[17] can easily be converted to a semi solid (fats, e.g., margarine) simply by their reaction with hydrogen. The reaction is effected by metallic catalysts such as nickel (Ni), platinum (Pt) or palladium (Pd). The products are alkanes. For example:



2.5.4 ***Water (hydration)***: The addition of water to a double bond is effected by sulphuric acid which acts as a catalyst. The products obtained are alcohols. For the unsymmetrical alkenes, the addition follows Markovnikov's rule, e.g.,



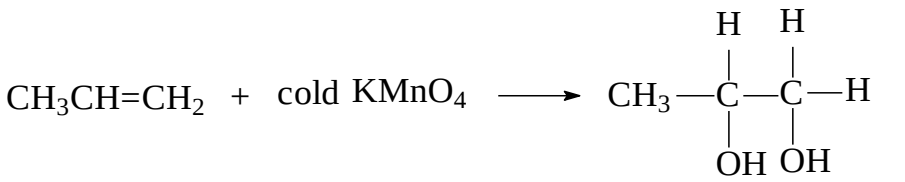
[17]Oils are so called because their molecules contain duoble bonds which greatly lower their melting points. When the double bonds (e.g., in covo)are converted to single bonds (in hydrogenation), the resulting compounds are solids and are called fats (margarine, a fat, is made that way).

2.6.0 Oxidation reactions

Alkenes are oxidized to various products by several oxidizing agents such as potassium permanganate ($KMnO_4$), ozone (O_3) and hydrogen peroxide (H_2O_2).

2.6.1 Oxidation with cold $KMnO_4$

A cold solution of $KMnO_4$ will oxidize alkenes to give 1,2-diols (alcohols with 2 hydroxyl groups on adjacent carbons), e.g.,



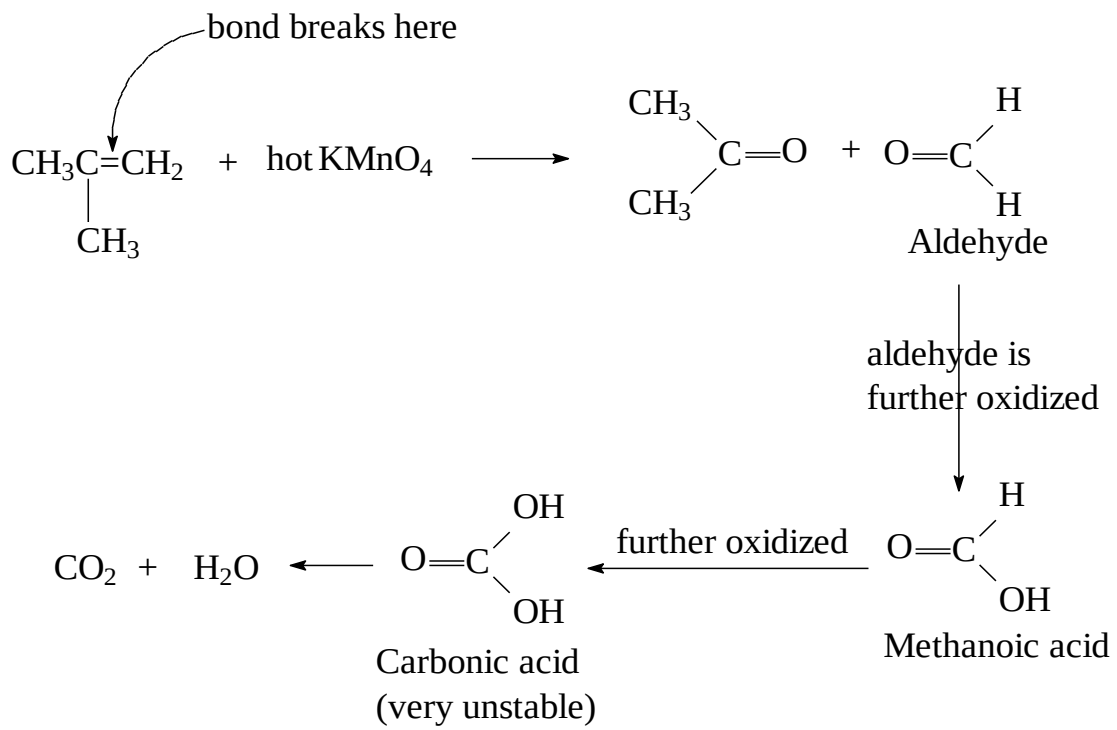
Propan-1,2-diol
(or propylene glycol)

2.6.2 Oxidation with hot $KMnO_4$

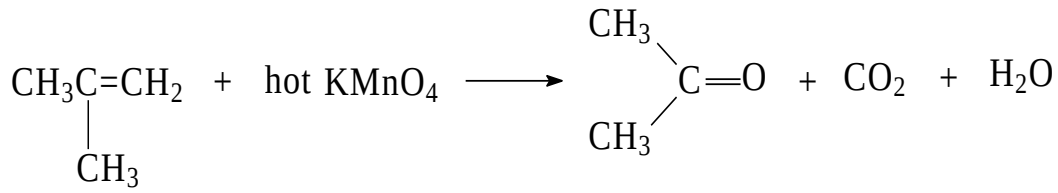
When hot $KMnO_4$ is used as the oxidant, the products obtained depend on whether the double bond is **terminal or internal**.

Terminal double bond

When the double bond is terminal (e.g., $CH_3C(CH_3)=CH_2$), the CH_2 group is oxidized to carbon dioxide and water, through some intermediates as in the example below. Here, imagine the bond breaking where the arrow is pointing (see scheme below) and oxygen, then adds to each carbon to give the final products.



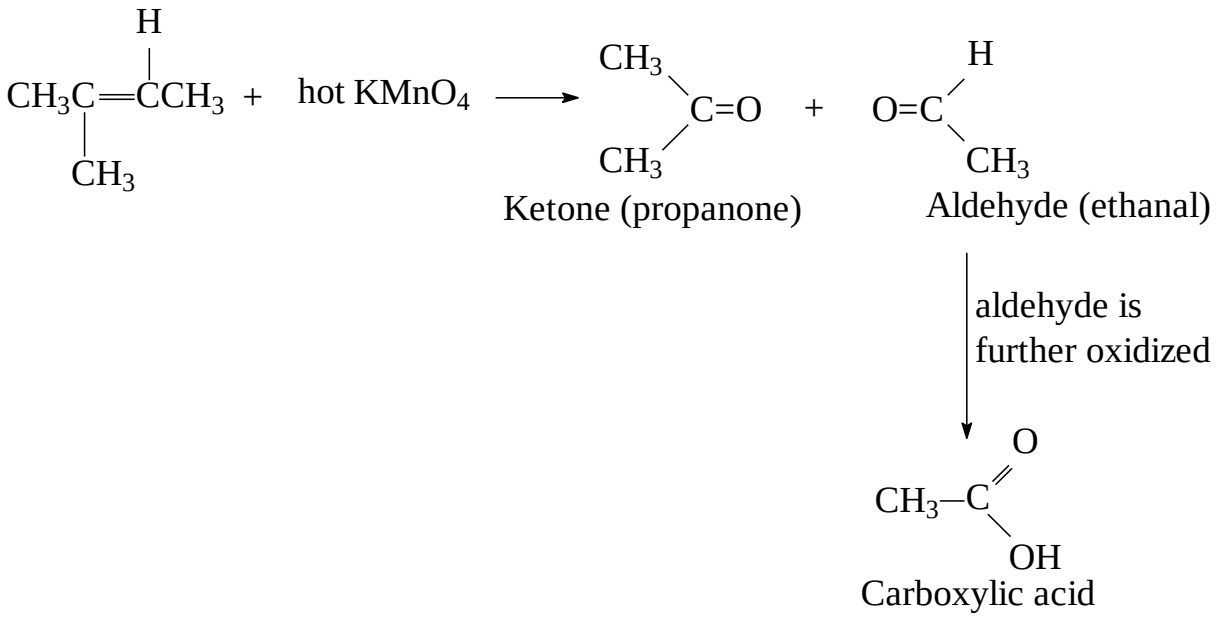
The net equation in the scheme above is therefore:



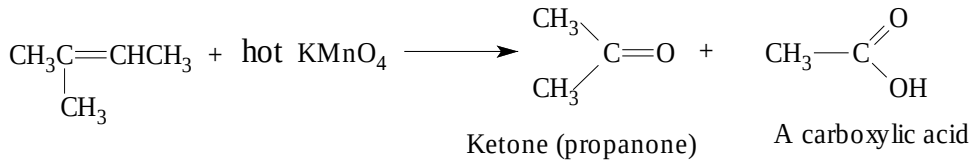
We deliberately put a CH_3 branch on one of the carbons of the double bonds, in the example above, to illustrate a point. *One of the products is a ketone* (i.e., $(CH_3)_2C{=}O$) *and this is not oxidized; the other product, an aldehyde, e.g., methanal, is readily oxidized*. You will learn more on reactions of ketones and aldehydes unit 3.

Internal double bond

Now when the double bond is internal (e.g., $CH_3CR' = CR''CH_3$), oxidation products can be both aldehydes (alkanals), both ketones (alkanones), or an aldehyde and a ketone depending on whether R'R" = H, or R'R" = C or R' = C and R"= H vice versa). If one or both intermediate products are aldehydes, they are further oxidized to carboxylic acids. Any ketone produced is not oxidized any further. The example below illustrates a situation where R' = H and R" = C.



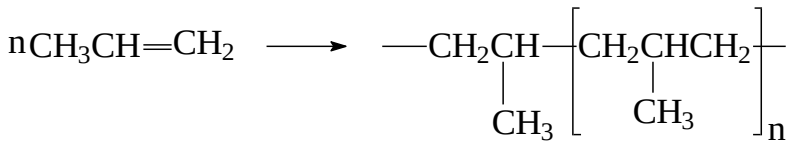
The net equation in the above case is:



2.7.0. Polymerization.

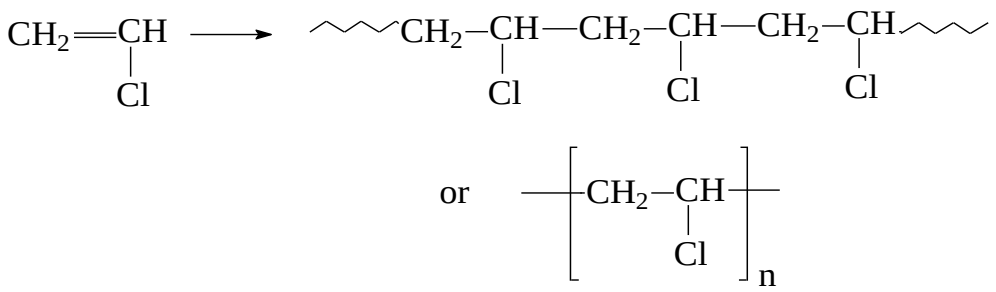
Polymers are long chain molecules with repeating subunits (called monomers). The process by which they are formed is called *polymerization*. If two monomers join, we have a dimer, when three join, then a trimer is formed and so on. Alkenes undergo polymerization reactions, forming a variety of substances, ranging from softer (e.g., those in which clothes or foods are packaged)

to harder materials (e.g., buckets or polyvinylchloride (PVC) pipes). For example propylene polymerizes to give polypropylene, according to the equation:



Polypropylene

The plastics that we are all familiar with such as raincoats, buckets, toys, gramophone records, plastic water pipes (also called PVC pipes) and plastic sandals consists of vinyl chloride units that are joined to form poly(vinyl chloride).

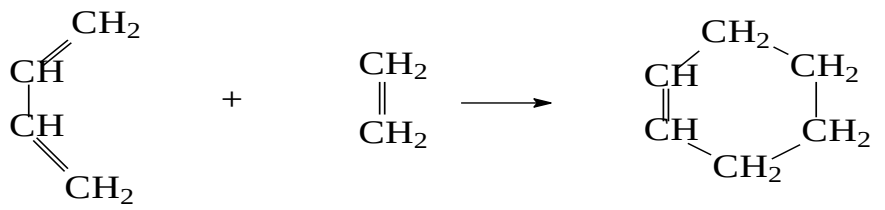


Poly(vinyl chloride)

Both polypropylene and poly (vinyl chloride) are examples of *homopolymers* in which the repeating monomers are of the same type. In other polymers, there are two or more different monomers joined, and these are called *copolymers.* For example, styrene and butadiene, in the ratio of 1:3 respectively, react to give a polymer that is used to make car tyres.

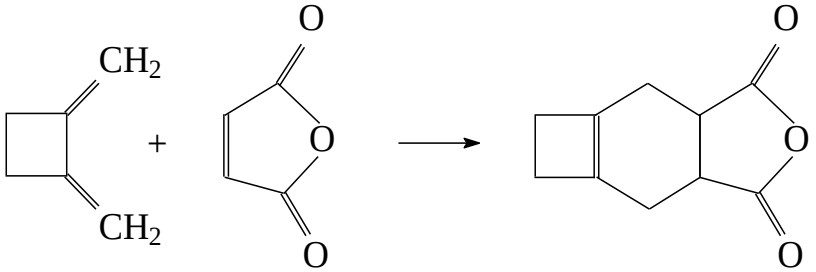
2.8.0 Diels-Alder reaction

The Diels-Alder reaction is a very useful synthetic reaction in that it produces cyclic compounds. In this reaction, a conjugated diene (called a diene) is reacted with a compound containing a double bond (called a dienophile), e.g.,



Buta-1,3-diene (A diene) Ethene (A dienophile) Cyclohexene

The dienophile is more reactive when groups such as $-CO_2H$, -CN or -CHO among others, are attached to the carbon atoms containing the double bond. A good example is the reaction below



A diene A dienophile

2.9.0 ALKYNES

Alkynes are a type of hydrocarbons in which two carbon atoms share three pairs of electrons between them. Therefore the two carbon atoms are bonded by a triple bond (-C≡C-).

The two simplest alkynes are ethyne (acetylene) and propyne.

Acetylene Propyne

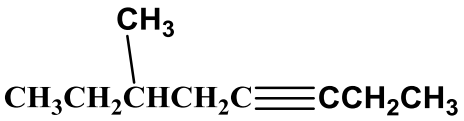
2.10.0 Naming of alkynes

The name of an alkyne is derived from the name of the alkane with the same number of carbons. The ending ‘’-ane’’ in the name of the alkane is replaced by ‘’-yne’’:.

Alkane → Alkyne

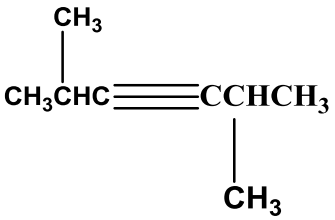
You will notice that the rules are the same as those for naming alkanes or alkenes
For example you must locate position of triple bond by lower number as in alkenes.

Example 1



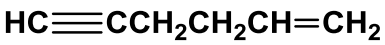
6-Methyloct-3-yne

Example 2



2, 5 – Dimethylhex – 3 – yne

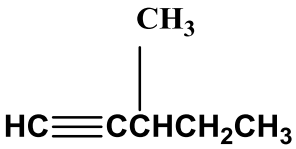
In the case of alkynes with more than one triple bond are called *diynes*, *triynes*, and so on. When you have a compound containing both double and triple bonds it is called an *enyne* (not *ynenes*). Numbering of an enyne chain starts from the end nearer the first multiple bond, whether double or triple. When there is a choice in numbering, double bonds receive lower numbers than triple bonds. For example:

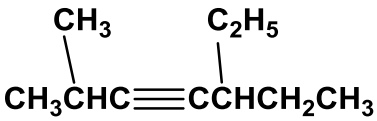


Hex-1-en-5-yne

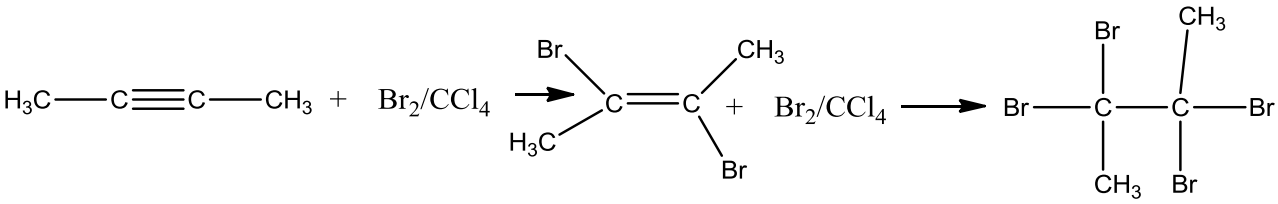
2.10.1 Unit Activity

Name the following compounds:



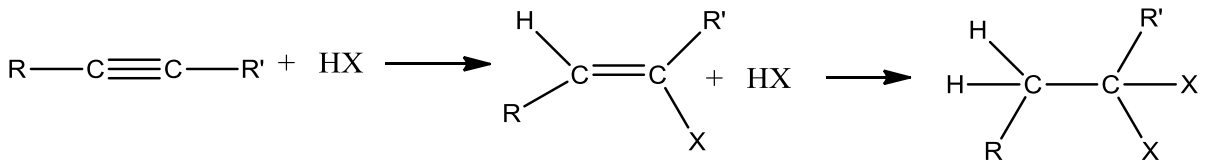


2.10.2 Alkynes undergo electrophilic additions just like alkenes



Addition may occur once or twice, depending on the number of molar equivalents of halogen we employ.

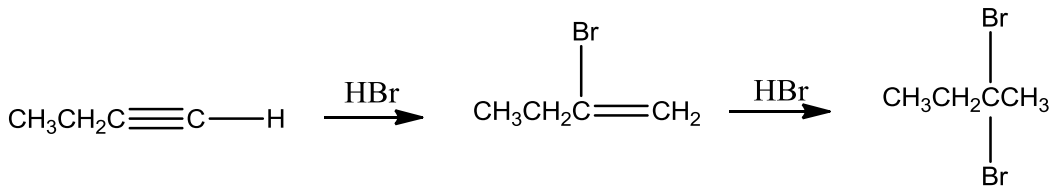
2.10.3 Addition of Hydrogen Halides to Alkynes



-CΞC - + HX → -HC-CX- + HX → H_2C-CX_2

Alkynes react with one molar equivalent of hydrogen chloride or hydrogen bromide to form haloalkenes. When they react with two molar equivalents they form geminal dihalides.

Both additions follow Markovnikov's rule:



But -1- yne 2 - Bromobut - 1 - ene 2, 2 - Dibromobutane

SUGGESTED READING TEXTS

Many texts on organic chemistry have chapters on hydrocarbons. The following list gives just a few. The reader is however encouraged to consult any other text.

1. Graham Solomons T.W. and Fryhle C.B., 2004. *Organic Chemistry, 8th ed.*, John Wiley and Sons. New York.
2. McMurry J. and Castellion ME., 2003. *Fundamentals of general organic and biological chemistry, 4th ed*. Pearson Education Inc. New Jersey.
3. Morrison R.T. and Boyd R.N., 1987. *Organic Chemistry, 5th ed*., Allyn and Bacon Inc.
4. Taylor G.A., 1987. *Organic Chemistry for students of Biology and Medicine, 3rd ed*., Longman,
5. Pine S.H., 1987. *Organic Chemistry, 5th ed*., McGraw-Hill Book Co.
6. Sykes P.,1975. *A Guide Book to Mechanism in Organic Chemistry, 4th ed*., Longman

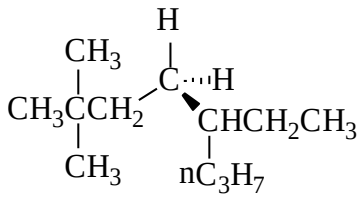
2.11.0 Unit summary

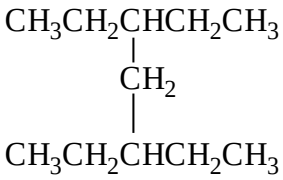
- Alkenes are hydrocarbons that contain a carbon to carbon double bond.
- Alkenes are more reactive than alkanes due to the presence of the double bond.
- The reactions of alkenes are mostly electrophilic additions.
- Alkenes react with hydrogen in a process called hydrogenation to produce margarine.
- Alcohols can also be produced from alkenes through a hydration reaction but this is a waste of an important chemical.
- Alkene compounds containing chlorine(vinylchloride) are used to manufacture plastic materials in a polymerization reaction (PVC}.
- Alkynes with a triple carbon carbon bond are as just reactive as the alkenes

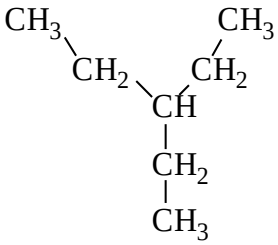
2.12.0 Unit test

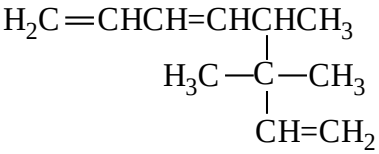
1. Define/explain the following:
   i. Hydration
   ii. Markovnikov's rule
   iii. Dienophile
   iv. Monomer

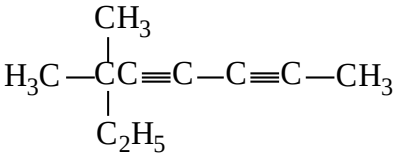
2. Give systematic names of the following compounds

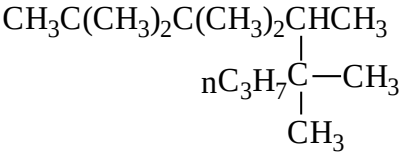
(a) 

(d) 

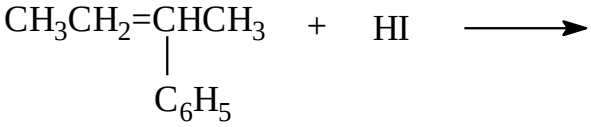
(b) 

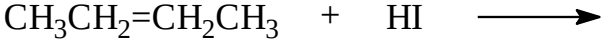
(e) 

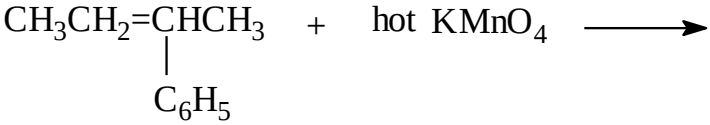
(c) 

(f) 

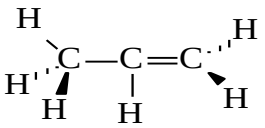
3. Give major products of the following reactions

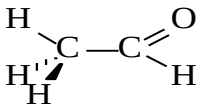


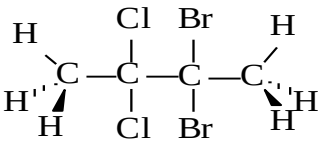


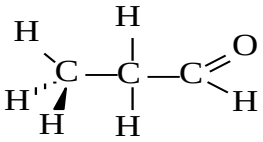


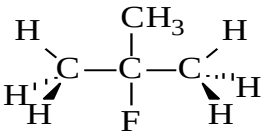
4. Write the Newman projections of the following structures.

(a) 

(c) 

(b) 

(d) 

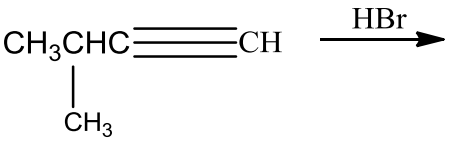
(e) 

5. The commercial insecticide "chlordane" is synthesized by the Diels-Alder addition of cyclopentadiene (acting as a dienophile) to hexachlorocyclopentadiene (acting as a diene), followed by addition of one mole of chlorine to the unsubstituted double bond in the resulting product. From this description,

   (a) Write the srtuctures of the dienophile and the diene.

   (b) Write the equations for the synthesis of chlordane.

6. Complete the following reaction



UNIT 3: OXYGEN CONTAINING ORGANIC COMPOUNDS

3.0.0 Introduction

We now turn our attention to the chemistry of some oxygen-containing organic compounds which will include Alkanals (aldehydes), alkanones (ketones), alkanols (alcohols), alkanoic acids (carboxylic acids) and esters. The aldehydes, ketones and alcohols will be covered in this unit while carboxylic acids and esters will be covered in unit 6.

3.1.0 Unit objectives

By the end of this unit, you should be able to:

- Give systematic names of Aldehydes, Alkanones and alcohols
- State physical properties of Aldehydes, Alkanones and alcohols
- State chemical properties of Aldehydes, Alkanones and alcohols

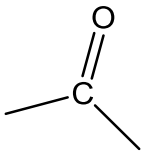
3.1.1Unit Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below.

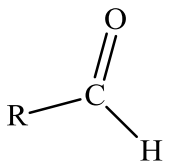
- Addition reaction
- Acetal
- Carbonyl compound
- Electrophilic site
- Grignard compound
- Hemiacetal
- Hemiketal
- Hydration
- Ketal
- Nucleophile
- Nucleophilic addition reaction
- Oxidation reaction

3.2.0 ALKANALS (ALDEHYDES) AND ALKANONES (KETONES)

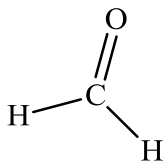
Aldehydes and ketones both contain the carbonyl group (C=O) as the functional group. The structure of the carbonyl group can be presented as below:



When the carbonyl group is attached to a carbon atom and a hydrogen atom (or to two hydrogens), the compound is called an alkanal.

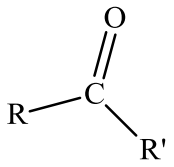


or



ie -CHO

When the carbonyl group is attached to two carbon atoms the compound is an alkanone



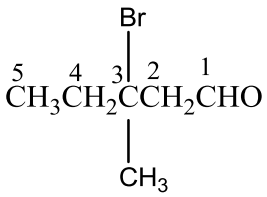
3.2.1 Nomenclature of Alkanals

The IUPAC name of an alkanal has an ending –al. The name is derived from the name of an alkane with the same number of carbon atoms and the ending ‘’e’’ in the name of the alkane is replaced by ‘’al’’ in the name of an alkanal.

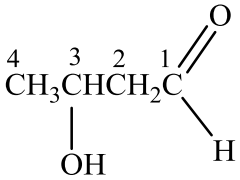
$$Alkane \rightarrow Alkanal$$

Methane becomes methanal, ethane becomes ethanal, etc. Rules for naming the alkanals (aldehydes) are the same as those of alkanes. However, start numbering the longest chain from the carbonyl (-CHO) carbon as number 1. Since the aldehyde group (-CHO) is always at the end, there is no need to indicate its position. The longest chain containing the aldehyde group is selected as the parent compound. Substituents are located by the number of carbons to which they are attached as in alkanes.

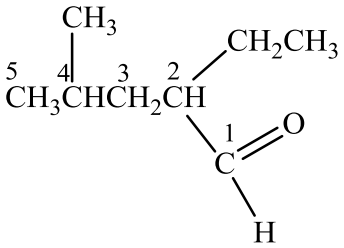
Let us look at the following examples:



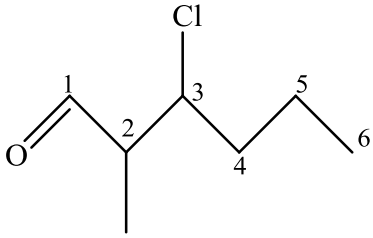
3 - Bromo- -3- methylpentanal



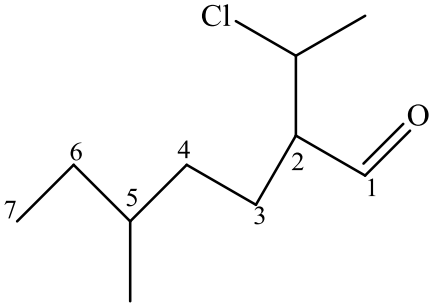
3- Hydroxybutanal



3 - Ethyl - 4 - methylpentanal

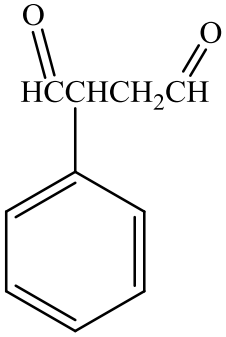


4 - Chloro - 2- methylhexanal



2 - (1 - Chloroethyl) - 5 - methylheptanal

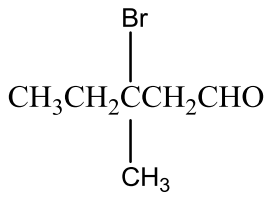
The suffix - *dial* is added to the appropriate alkane name when the compound contains two aldehyde functions. For example:



2 - Phenylbutanedial

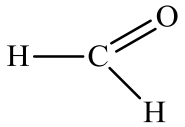
You might have observed that in the name given the final –e of the alkane name has not been dropped. The - *e* ending of an alkane name is dropped before a suffix beginning with a vowel (- *al*) and retained before one beginning with a consonant (- *dial*).

Sometimes Alkanals are named using trivial names. Trivial names are derived from the names of carboxylic acids with the same number of carbon atoms. The ending -ic acid in the name of the acid is replaced by aldehyde. You will learn about carboxylic acids in unit 6. Following is the systematic name as well as trivial name of the given alkanal.

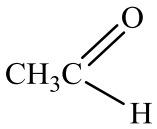


IUPAC: 3 - Bromo- -3- methylpentanal
Trivia name: 3 - Bromo- -3- methylpentanaldehyde

A few simple and well-known aldehydes have common names that are recognized by IUPAC:



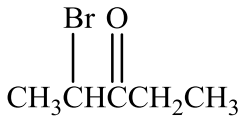
IUPAC: Methanal
Trivial: Methanaldehyde
Common: Formaldehyde



IUPAC: Ethanal
Trivial: Ethanaldehyde
Common: Acetaldehyde

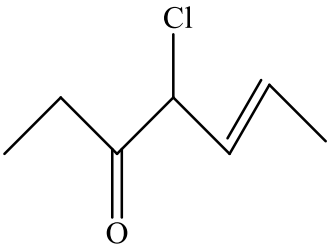
3.2.2 Nomenclature of Alkanones

The systematic names of Alkanones (ketones) have the ending -one. The name is derived from the name of the alkane with the same number of carbon atoms, the ending -e of the alkane being replaced by -one. Rules for naming the Alkanones (ketones) are the same as those of alkanes. However, we number the longest chain starting from the end close to the C=O carbon. Unlike in aldehydes, the position of the carbonyl group is indicated in ketones. Following are examples:



2 - Bromo - 3- pentanone
or 2 - Bromopentan - 3 - one

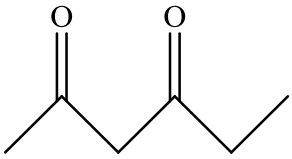
It is recommended to place the locant (number locating the position of C=O group) immediately before the suffix rather than before the parent name. So we recommend to name the structure above as 2 – Bromopentan – 3 – one. Here is another example.



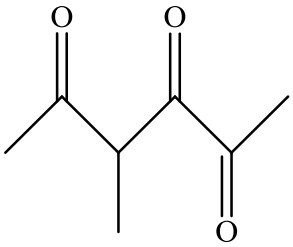
4 - Chlorohept - 5 - en - 3 - one

Notice that in this example both the C=C double and the C=O have been specified.

If the ketone has two or three carbonyl groups, the endings -dione and -trione respectively are used as in structures below.

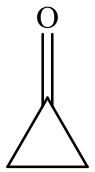


Hexane - 2, 4 - dione

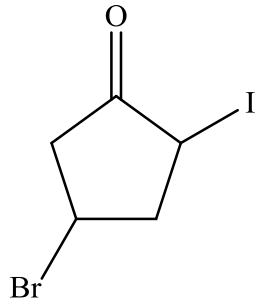


4 - Methylhexane - 2, 3, 5 - trione

Cyclic alkanones are named as cycloalkanones. Examples:

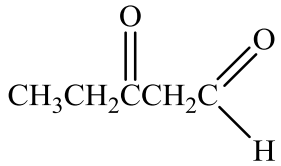


Cyclopropanone



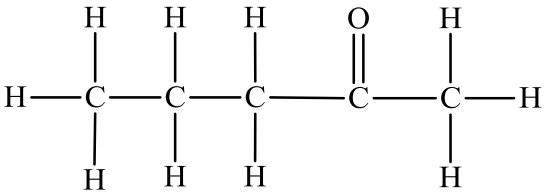
4 - Bromo - 2- iodocyclopentanone

If we want to take the ketone group (C=O) as a substituent on another compound, then we use the prefix “oxo” for this group. For example:

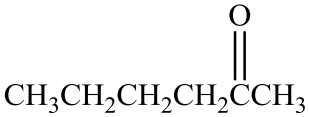


3 - Oxopentanal

Sometimes ketones can be named using trivial names. In trivial names, the groups attached to the carbonyl carbon are specified and the word ketone is added. Examples:

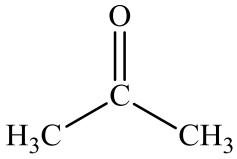


IUPAC: Pentan -2 - one
Trivial: Methyl propyl ketone



IUPAC: Hexan - 2 - one
Trivial: Butyl methyl ketone

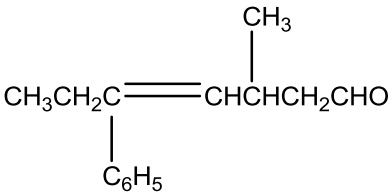
Propanone is well known by its common name acetone such that IUPAC recognises this common.

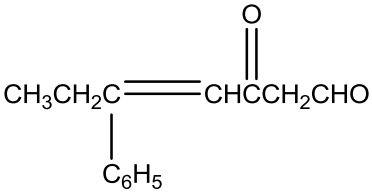


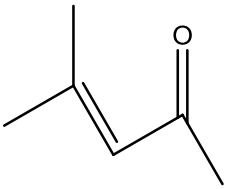
IUPAC: Propanone
Trivial: Dimethyl ketone
Common: Acetone

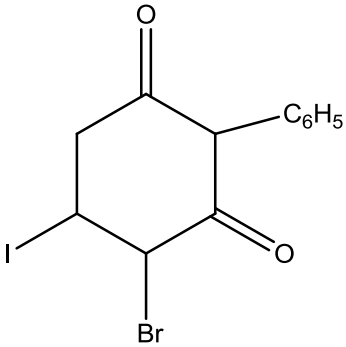
3.2.3 Unit activity

1. Give systematic names of the following compounds:

a. 

b. 

c. 

d. 

2. Write the structures of the following compounds:

   a. 4 – Chloro – 3 – methyl – 2, 4 – pentadienal

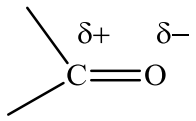
   b. 5 – Bromo – 3, 4 – diethylheptan – 2 – one

   c. 2 – Phenylycyclohexanone

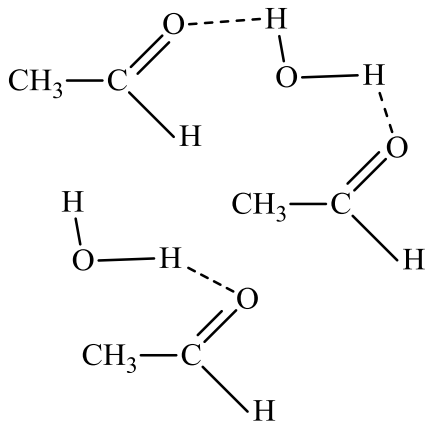
   d. 4 – Methyl – 3 – oxo – 2 – phenylpentanal

3.3.0 Physical properties of Alkanals and Alkanones

Methanol is a gas at room temperature, the other aldehydes and ketones are liquids at room temperature. Aldehydes and ketones have higher boiling points than hydrocarbons with similar number of carbon atoms because of the polar nature of the carbonyl group. We learnt about polar molecules in module 1.



The polar nature of the carbonyl group also means that aldehydes and ketones can form hydrogen bonds with water such that lower molecular weight aldehydes and ketones are soluble in water in all proportions.



Hydrogen bonding between ethanal and water

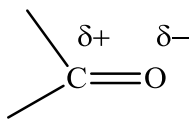
However, solubility decreases with increasing molecular weight. Since aldehydes and ketones do not form hydrogen bonds among themselves, they have lower boiling points than the corresponding alcohols.

3.4.0 Chemical properties of Alkanals and Alkanones

Alkanals and Alkanones undergo various types of reactions. We can place these reactions into three major classes as addition, oxidation and reduction reactions. Let us look at these types of reactions.

3.4.1.0 Addition reactions

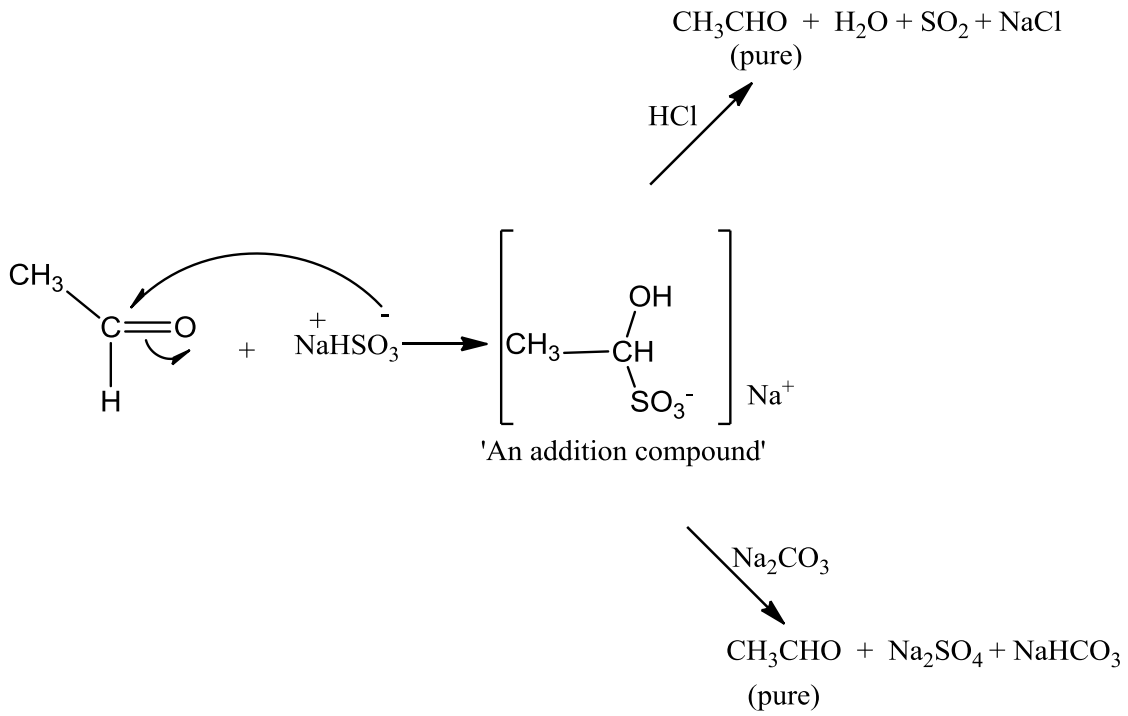
We met addition reactions under alkenes in unit 2. We saw that addition was taking place on the C=C double bond. Alkanals and alkanones contain a double bond just like alkenes do. So they also undergo addition reactions. However, C=C in alkenes is different from the C=O group in alkanals and alkanones and so substances that add to a C=O are not the same as those that add to C=C. In order to appreciate the addition reactions to C=O group, let us consider the polar nature of this group:



The C=O being polar, the carbon atom is partially positively charged while the oxygen has a partial negative charge. This partial positive carbon is the electrophilic site. Anything that is negatively charged will be attracted to this carbon. Such an electron – pair donor that seeks a positive centre in a molecule is called a nucleophile. The alkanals and alkanones undergo nucleophilic addition reactions. Nucleophilic addition reaction is a reaction in which a nucleophile (an electron-pair donor) forms a bond to the carbon of a carbonyl (C=O) group. Several substances add to the carbonyl carbon:

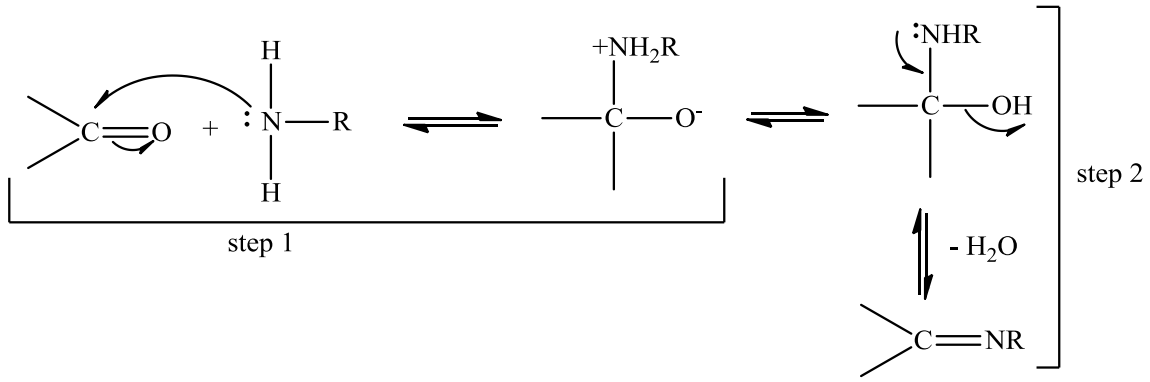
3.4.1.1 Sodium bisulphite ($NaHSO_3$):

Alkanals and alkanones react with a saturated aqueous solution of $NaHSO_3$ to give a ‘’bisulphite addition’’ compound. In this case the sulphite (HSO_3^-) ion is the nucleophile. The product is a salt and is insoluble in organic solution. It precipitates out of solution. Once precipitated, the addition compound can be reconverted to original alkanal or alkanone that is pure by filtering it and reacting it with an acid or base. The reaction equation below shows reaction of ethanal with sodium bisulphite.

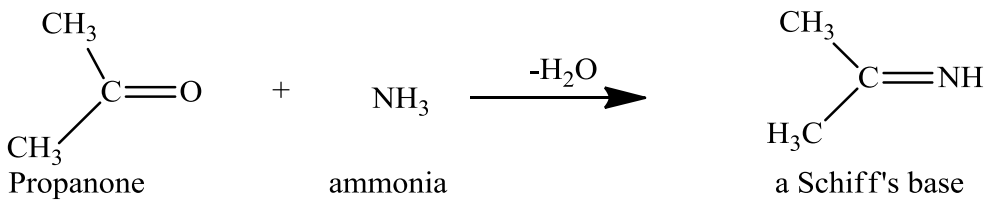


3.4.1.2 2,4 – Dinitrophenyl hydrazine and related compounds:

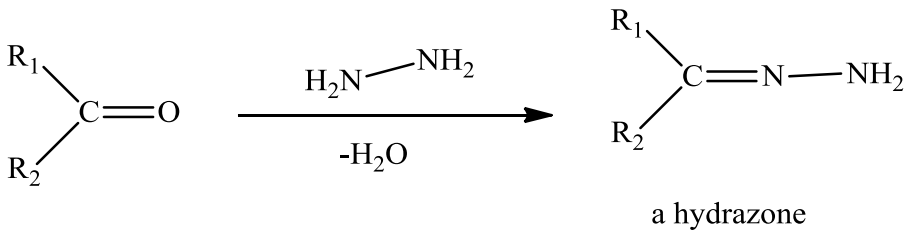
Ammonia (NH_3) and its derivatives add to an alkanal or an alkanone in a similar fashion forming an unstable product which then loses a molecule of water. This is how the reaction takes place: First, the ammonia or related compound (e.g. amine) nitrogen acts as a nucleophile, attacking the carbonyl carbon. Second, water molecule is eliminated and the final product has the C=O double bond replaced by the C=N double bond. The general equation below shows this explanation.



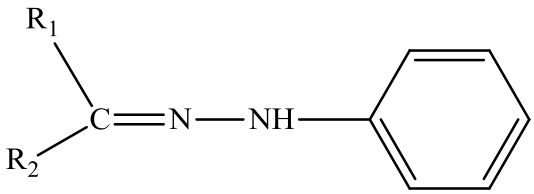
If the reaction is between NH_3 (or amine) and an alkanal or ketone the product is called an imine or shciff's base. You will learn about amines in unit 5. An example of reaction between NH_3 and propanone is shown below:



Replacing one H in NH_3 by NH_2 gives hydrazine, (H_2N-NH_2). Hydrazine reacts with alkanals and alkanones in the same way as NH_3 giving a product called a ''hydrazone''



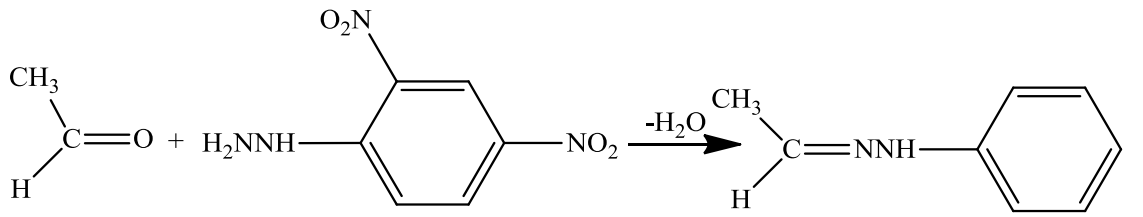
Replacing one H in H_2N-NH_2 by a -C_6H_5 group gives phenyl hydrazine (H_2N-NHC_6H_5) which reacts with alkanals or alkanones in the same way as hydrazine (or NH_3) giving products called ''phenylhydrazones''



a phenylhydrazone

Replacing an H in hydrazine by a 2,4-dinitrophenyl group gives a compound called 2,4-dinitrophynylhydrazine (2,4-DNPH). The 2, 4-DNPH reacts with alkanals and alkanones in the same way giving solid products called 2, 4-dinitrophenylhydrazones which are

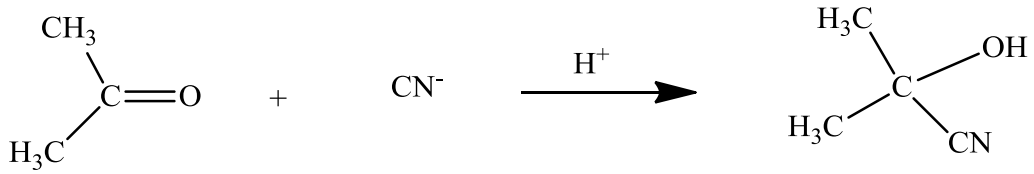
yellow/orange/red in colour depending on the structure of the carbonyl compound. Alkanals and alkanones are differentiated from other compounds by use of 2, 4-DNPH This reaction is used as test for alkanals and Alkanones as you will see in unit 3 of practical module 2. The reaction of 2, 4-DNPH with ethanal is:



Ethanal 2, 4 Dinitrophenylhydrazine Ethanal -2, 4 -dinintrophenylhydrazine

3.4.1.2 Hydrogen cyanide:

Alkanals and Alkanones react with hydrogen cyanide (HCN) to form compounds called cyanogens. The CN^- is the nucleophile which attacks the carbonyl carbon as in the example below:

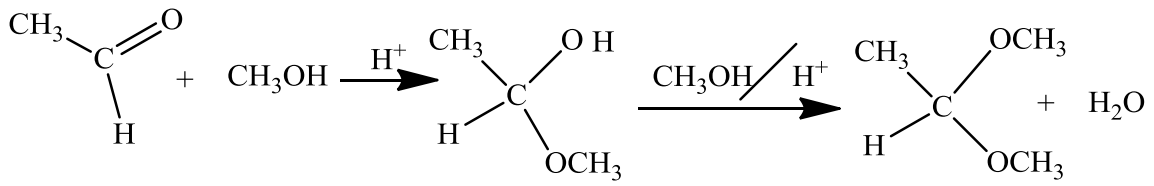


3.4.1.4 Grignard reaction:

The reaction of Alkanals and Alkanones with Grignard reagents will be discussed in section 3.5.5 below.

3.4.1.5 Alkanols (Hemiacetal/ hemiketal formation)

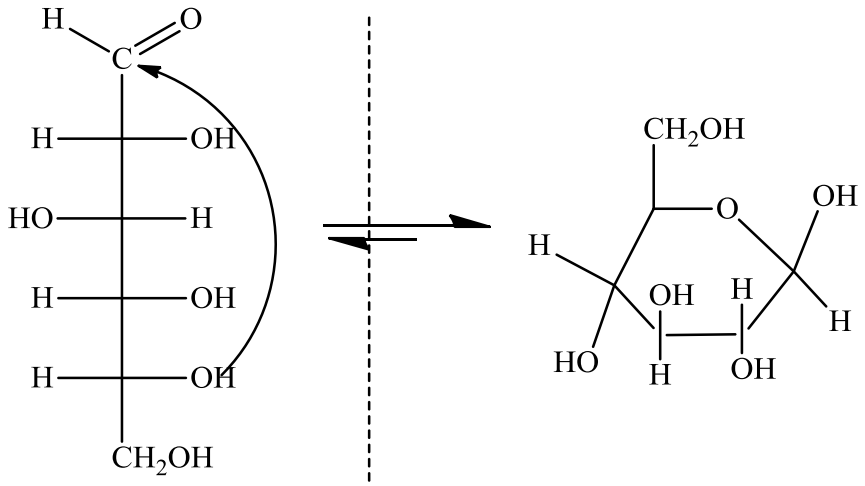
A molecule of alohol reacts with alkanal in the presence of an acid to form a product called hemiacetal. The alcohol oxygen adds to the carbonyl carbon of the aldehyde. The hemiacetal produced reacts with a second molecule of alcohol to give an acetal. Let us look at the reaction of ethanal and methanol (CH_3OH):



Ethanal methanol a hemiacetal an acetal

Notice that in hemiacetal, OH and OR groups are bonded to the same carbon while in acetal there are two OR groups attached to the same carbon atom.

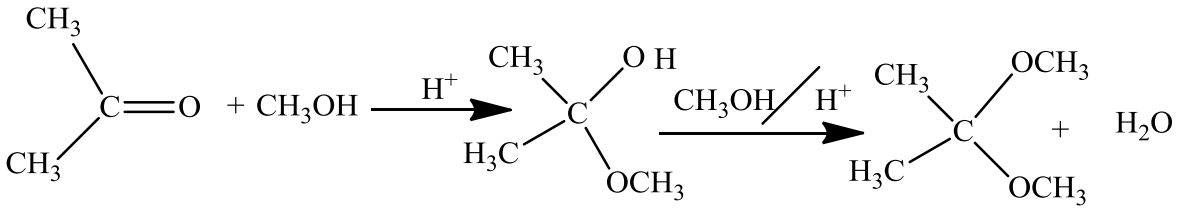
The reaction of aldehydes and alcohols is important in sugars. Sugars react internally giving rings.



Long - chain structure Ring structure

2,3,4,5,6-pentahydroxyhexanal (D-glucose)

Also alkanones react with alcohols in the same way giving hemiketals and then ketals. The reaction of propanone and methanol is shown below.



Propanone methanol a hemiketal a ketal

3.4.2.0 Oxidation reactions

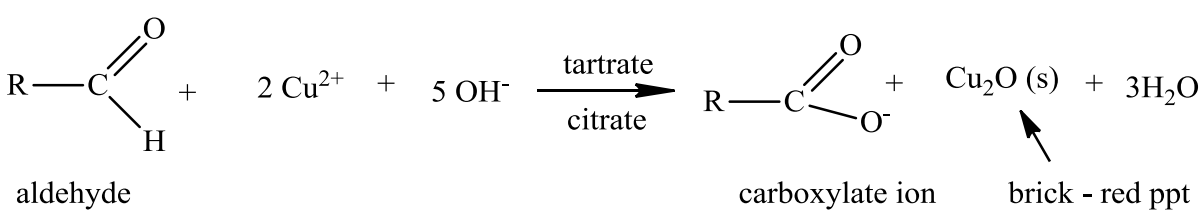
The reactions discussed so far are positive for both alkanals and Alkanones. How then do we distinguish between the two?

Alkanals are readily oxidised to carboxylic acids while alkanones are resistant to oxidation. Reagents like potassium permanganate and chromic acid readily oxidise aldehydes. Remember we met oxidation of aldehydes with hot potassium permanganate under alkenes in unit 2. However, there are two solutions that are commonly used in the oxidation of aldehydes and these are alkaline solution of cupric salts (Cu^{2+}) and alkaline solution of silver nitrate ($AgNO_3$). We use oxidation reactions by these two solutions to distinguish between aldehydes and ketones. The oxidation reaction of Alkanals and Alkanones are also important in connection with sugars.

3.4.2.1 The alkaline solution of cupric salts (Cu^{2+}) - Fehling's solution and Benedict's solution.

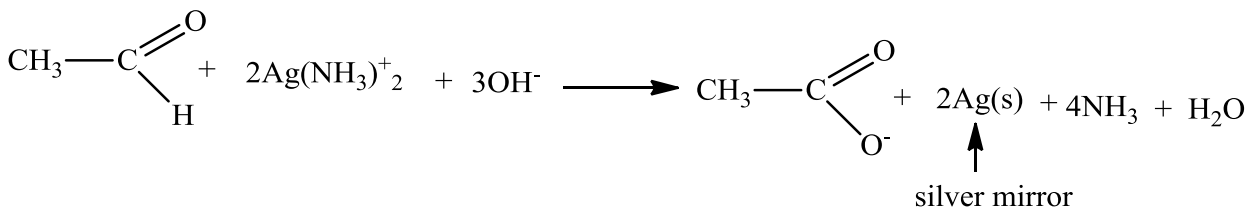
There are two types of solutions of Cupric (Cu^{2+}) ion depending on the counter ion used. When Cu^{2+} is complexed with tartrate the solution is called Fehling's solution and the test is called Fehling's test usually used in Chemistry. If Cu^{2+} is complexed with citrate the solution is called Benedict's solution which is usually used in Biochemistry.

The products of the reaction of aldehydes and Fehling's or Benedict's solution are cuprous oxide (a brick red solid) and a carboxylate ion (a conjugate base of the acid). The general reaction is as follows:



3.4.2.2 Alkaline solution of silver nitrate (Tollens' solution - Tollens' test)

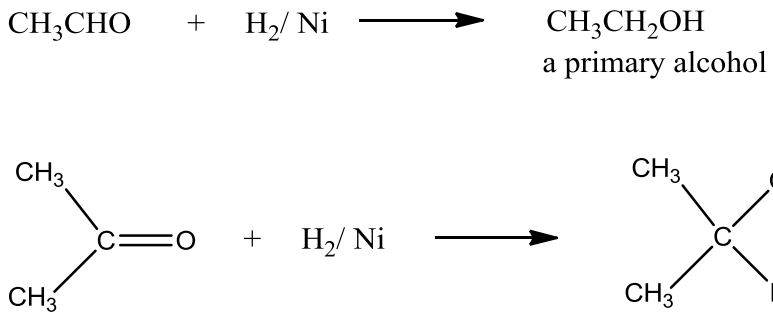
Tollen's reagent is a solution of silver nitrate dissolved in ammonia. The product of the reaction between aldehydes and the Tollen's reagent is shown either by the deposition of metallic silver on the test tube (silver mirror) or the formation of grey/ black precipitate. With Tollen's reagent, the equation for the reaction can be represented as:



Remember we said that ketones are resistant to oxidation. We therefore, expect ketones not to react with Fehling's/ Benedict's and Tollen's solutions (oxidising reagents). However, we should mention here that α-hydroxy ketones do react with these solutions. These α-hydroxy ketones first convert to aldehydes in alkaline solution and it is the aldehydes formed that react with the oxidising reagents. You will perform some experiments using the Fehling's, Benedict's and Tollen's solutions on aldehydes and ketones in Practical module 2 unit 3.

3.4.3.0 Reduction reactions

Alkanals and Alkanones are readily reduced by hydrogen in the presence of catalysts (e.g. nickel, Ni) Alkanals give primary alcohols and Alkanones give secondary alcohols. You will learn about primary and secondary alcohols in section 3.5. 1. Following are reduction equations.

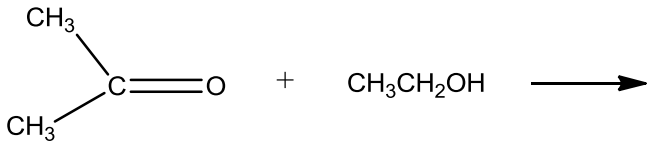


a secondary alcohol

3.4.4 Unit activity

1. Define the following terms:
   a. Electrophile
   b. Alkanone
   c. Acetal
2. State any two physical properties of carbonyl compounds (aldehydes and ketones).
3. Briefly explain why aldehydes cannot form hydrogen bonds with each other.
4. Give structure of the major products (if any) expected from the following reactions:

a.

b. 

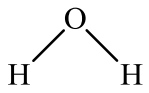
5. What simple test, physical or chemical, would you use to distinguish between the following pairs of compounds? Describe the tests and the results you would observeFor each pair, briefly explain the chosen test, and observations.
   a. Pentanal and pentan-2-one
   b. Cyclohexanone and cychlohexanol
   c. Methanal and methane

3.5.0 ALCOHOLS (ALKANOLS)

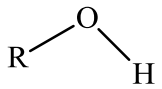
Alcohols are organic compounds that contain a hydroxyl (-OH) functional group. We can represent alcohols as R-OH, where R = alkyl e.g. $CH_3CH_2CH_2$-

If R =aryl, (Aromatic group, e.g. phenyl, C_6H_5-) then it is called a phenol

Alcohols can be viewed structurally as organic derivatives of water in which one of the hydrogen atoms of water is replaced by an alkyl group.



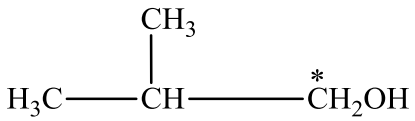
Water



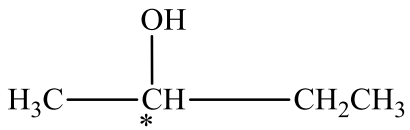
alcohol

3.5.1 Classification of alcohols

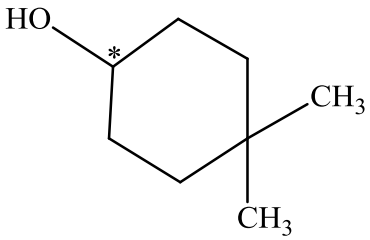
Alcohols are classified as primary (1°), secondary (2°) and tertiary (3°). This classification depends on the number of carbon atoms bonded to the hydroxyl bearing carbon. Primary alcohols have carbon with —OH bonded to one other carbon while secondary alcohols have carbon with —OH bonded to two other carbons. Tertiary alcohols have carbon with –OH bonded to three other carbons. Example of primary, secondary and tertiary alcohols are shown below.



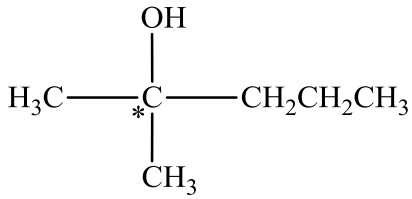
Primary alcohol



Secondary alcohol



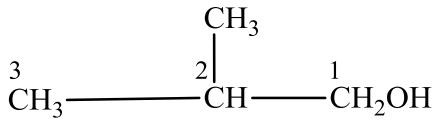
Secondary alcohol



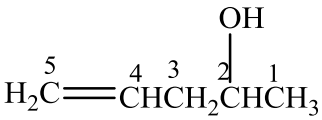
Tertiary alcohol

3.5.2 Nomenclature of Alcohols

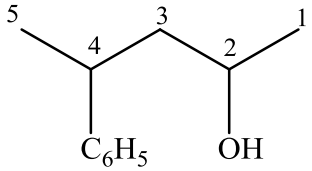
The systematic name of an alcohol is derived from an alkane with similar number of carbon atoms but the final -e in the alkane name is replaced by –ol. For example, methane becomes methanol. Rules for naming the alcohols are the same as those of alkanes. However, select the longest carbon chain containing the –OH group. Number the chain beginning at the end nearer the –OH group and identify the position to which the –OH is bonded. Following alcohols have been named using IUPAC system.



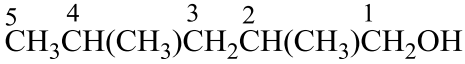
2 - Methyl - 1- propanol or
2 - Methylpropan - 1-ol



Pent - 4 - en - 2 - ol



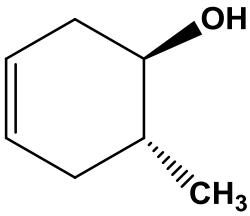
4-phenylpentan -2-ol



2,4-dimethylpentan-1-ol

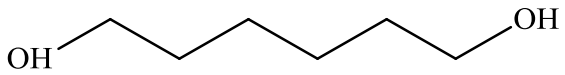
It is recommended to place the locant (number locating the position of –OH group) immediately before the suffix rather than before the parent name.

In cyclic alcohols the first position is always assigned to the carbon carrying the hydroxyl.



Trans-6-methylcyclohex-3-enol

If two or more OH groups are present, use the prefixes di, tri, tetra, etc. The final -e of the alkanes in the last two examples are not deleted because the next letters d and t respectively, are not vowels.

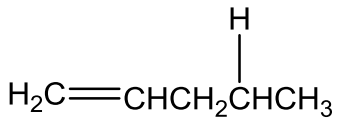


Hexane - 1, 6 - diol

$HOCH_2CH(OH)CH_2OH$

Propane -1, 2, 3 - triol
(Trivial name: glycerol)

If we want to take the OH group as a substituent, we use the name hydroxyl. For example the following structure is named as an alkene.



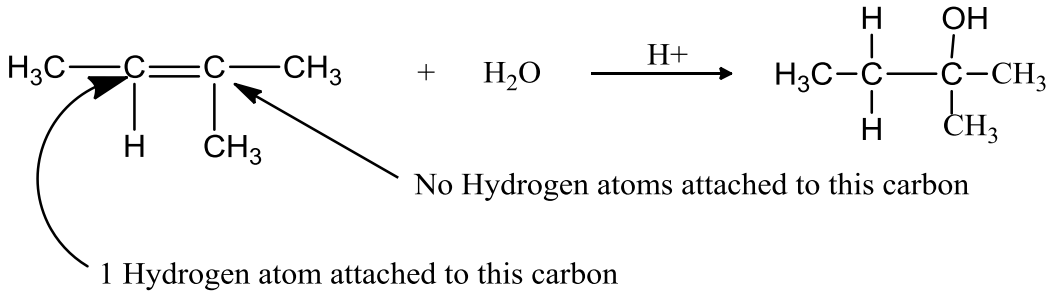
4 - Hydroxypent - 1 - ene

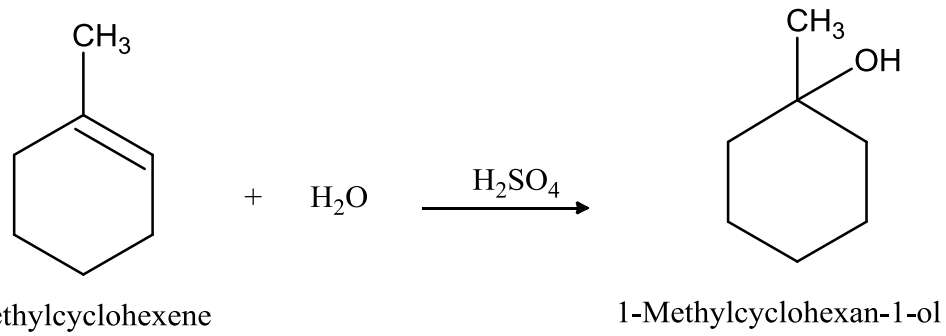
3.5.3.0 Preparation of alcohols

Alcohols are derived from many types of compounds and some typical methods for preparation of alcohols include hydration of alkenes and use of Grignard reagents.

3.5.3.1 Hydration of alkenes

In a hydration reaction, water is added to a double bond and sulphuric acid (H_2SO_4) is used as a catalyst. Can you remember that we discussed hydration of alkenes in unit 2 where we used Markovnikov's rule to predict the major products? The rule predicts that the hydrogen atom from H_2O adds to the carbon atom of the double bond that already has the greater number of hydrogen atoms. The –OH adds to the other carbon of the double bond. Following examples are the hydration reactions.

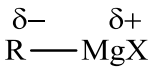




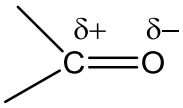
3.5.3.2 Use of Grignard reagents

Grignard reagents are alkyl or aryl magnesium halides represented in the form RMgX, where R = alkyl or aryl; X = F, Cl, Br or I

The organic part of these reagents is negatively charged while the inorganic part is positively charged:

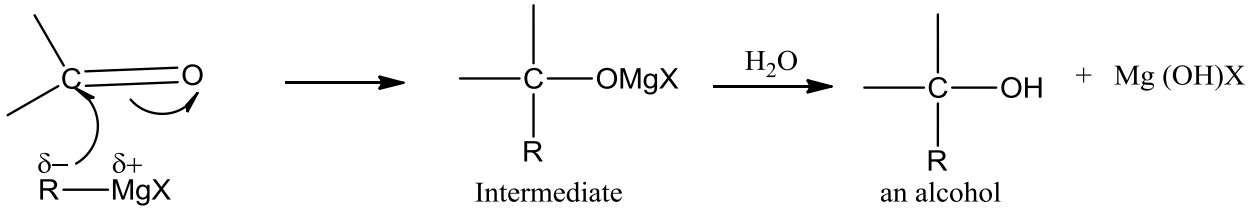


The carbonyl carbon is partially positively charged due to electronegative oxygen attached to it.



The Grignard reagents react with carbonyl compounds (aldehydes & ketones) to produce alcohols as follows:

The negatively charged organic group of the Grignard reagent adds to the positively charged carbon atom of the carbonyl carbon of the aldehyde or ketone to produce an intermediate. When an aqueous acid is added to the intermediate an alcohol is produced.

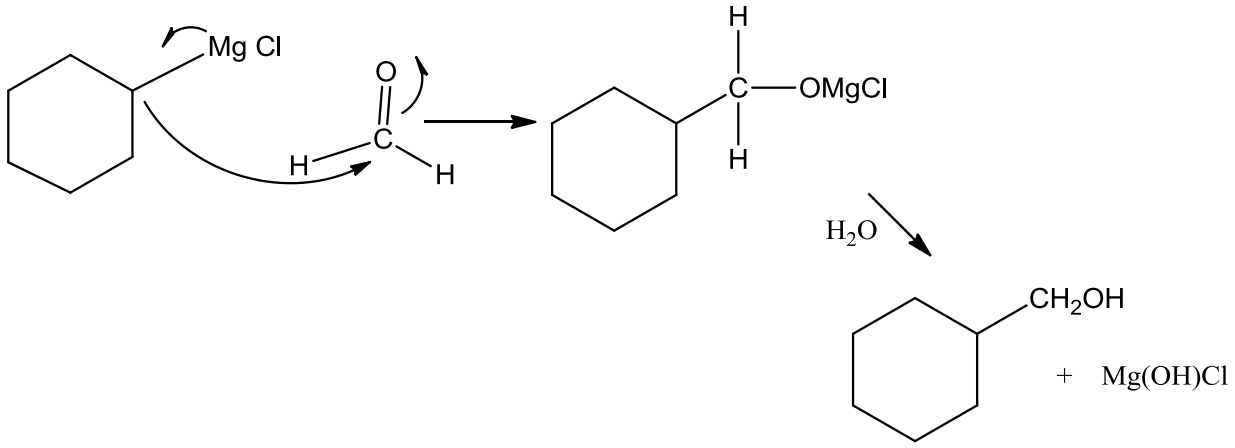


Note that positive seeking groups are called nucleophiles and electron-seeking reagents are called electrophiles. The negatively charged organic group of the Grignard reagent is the nucleophile and the partially positively charged carbonyl carbon is the electrophile.

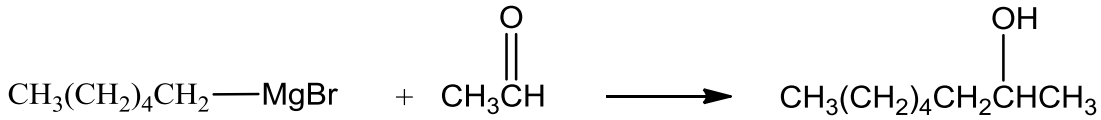
Thus, the addition of a Grignard reagent to an aldehyde or ketone is called nucleophilic addition.

Grignard additions to carbonyl compounds are especially useful because they can be used to prepare primary, secondary, or tertiary alcohols.

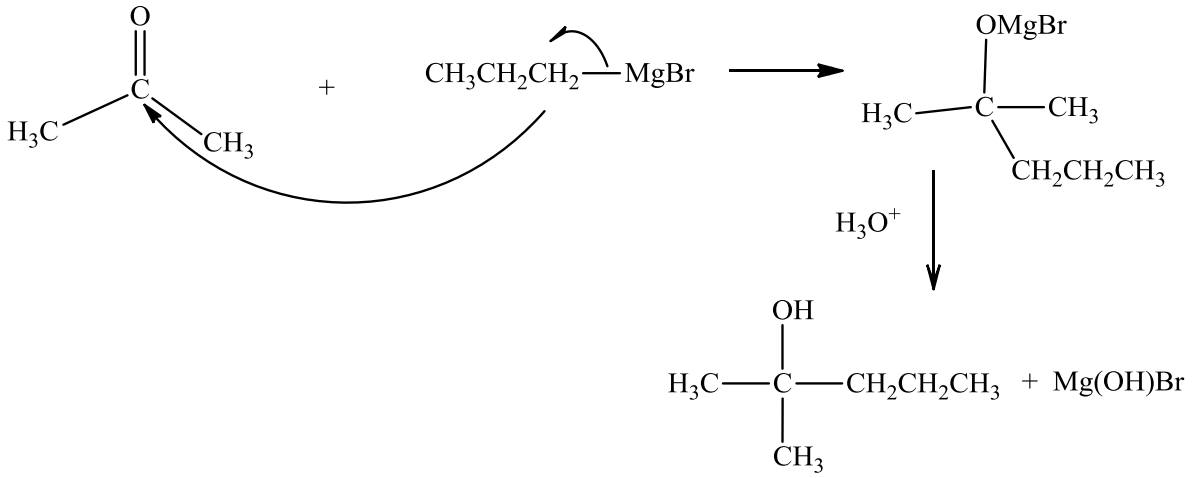
Grignard reagents react with methanal to produce primary alcohols



Reaction of Grignard reagents with other aldehydes produces secondary alcohols



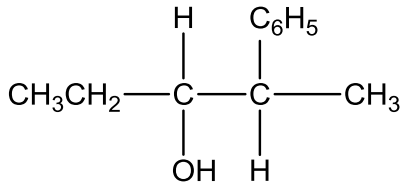
Reaction of Grignard reagents and ketones produce tertiary alcohols



Worked example

Give structures of Grignard reagent and the substrate (aldehyde or ketone) that would yield 2-Phenylpentan-3-ol

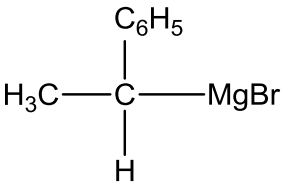
Solution:



This is a 2° alcohol, so needs an aldehyde + Grignard compound

2 - Phenylpentan - 3 - ol

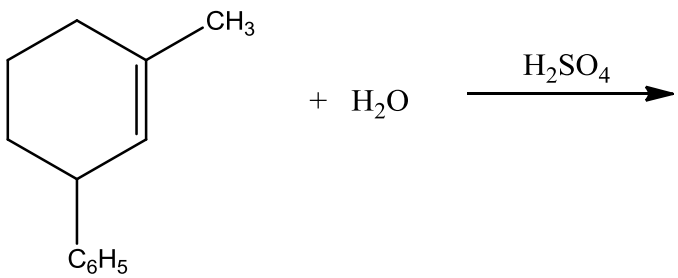
Either CH_3CH_2MgBr + $CH_3CH(C_6H_5)CHO$

Or 

+ CH_3CH_2CHO

3.5.4.0 Unit activity

1. write the structure of the main product for the reaction:

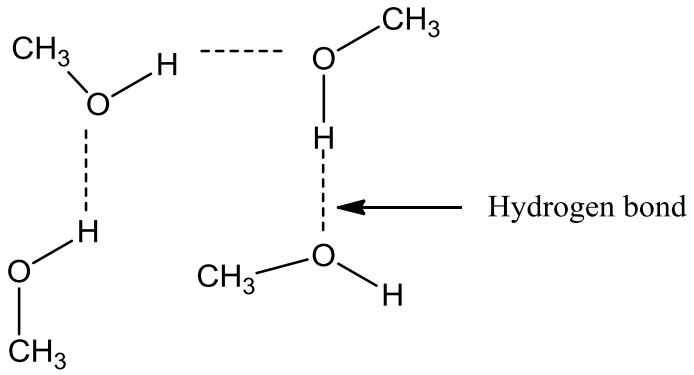


2. Give structures of Grignard reagent and the substrate (alkanal or alkanone) that would yield the following alkanols.
   a. 1 – Methylcyclohexanol
   b. 3 – Methylbutan – 2 – ol
   c. 1 – Phenyl – 1 – pentyn – 3 – ol
   d. 2 – Pentanol
   e. 3 – Methyl – 2 – phenylpentan – 3 - ol

3.5.5.0 Physical properties of Alcohols

3.5.5.1 Boiling points

Alcohols have higher bps than corresponding alkanes with same number of carbon atoms. This is because alcohol molecules can form hydrogen bonds with each other.



Boiling points of alcohols increase with an increase in the number of carbon atoms due to increase in Van der Waals forces. Table 3.1 below shows boiling points of some alcohols.

Table 3.1: Boiling points of some alcohols

Formula	**Name**	**Boiling Point, ^oC**
CH_3OH	Methanol	65
CH_3CH_2OH	Ethanol	78
$CH_3CH_2CH_2OH$	Propan – 1- ol	97
$CH_3CH(OH)CH_3$	Propan – 2 – ol	82
$CH_3CH_2CH_2CH_2OH$	Butan – 1 - ol	118

Can you see that in Table 3.1 propan – 1 – ol has higher boiling points than propan – 2 – ol? If you draw the structure of propan – 1 – ol and propan – 2 – ol you will notice that the two isomeric alcohols are primary (1°) and secondary (2°) respectively. We can therefore, generalise that for isomeric alcohols, 1° alcohols have higher boiling point than 2° alcohols. This is due to uneven packing of the T-shape of the 2° alcohols which decreases the Van der Waals forces.

3.5.5.2 Solubility

Alcohols are soluble in water due to the presence of the –OH group (Like dissolves like). This solubility of the alcohols in water is enhanced by the ability of the alcohol molecules to participate in hydrogen bonding with water. However, solubility decreases as the size of the alkyl group increases - becomes more hydrocarbon-like. The small alcohols (alcohols with fewer numbers of carbon atoms) are miscible in water in all proportions. For isomeric alcohols, tertiary alcohols are more soluble than secondary alcohols and secondary alcohols are more soluble than primary alcohols. Furthermore, if there are many –OH groups in a molecule, solubility increases. For example, hexan -1- ol has water solubility of 0.59 g/100 mL while hexane -1, 6 - diol is miscible in all proportions.

3.5.5.3 Density

All liquid alcohols have densities of approximately 0.8 g/mL and are, therefore, less dense than water.

3.5.6.0 Chemical properties of Alcohols

3.5.6.1 Acid/ base properties

Like water, alcohols act as acids if they are in basic environment and they can act as bases if they are in acid environment

3.5.6.1.1 Alcohols as bases:

Alcohols act as bases in the presence of a strong acid like HCl, H_2SO_4, $HClO_4$. They are very weak bases

$$CH_3OH + HCl \rightleftharpoons CH_3OH_2^+ + Cl^-$$

3.5.6.1.2 Alcohols as acids:

Alcohols are extremely weak acids and will donate protons only to strongest bases to form alkoxide anions ($R\text{-}O^-$)

$$CH_3OH + NaOH \rightleftharpoons CH_3O^-Na^+ + H_2O$$

They also react with metals of group 1a and 2a of the periodic table with evolution of hydrogen gas.

$$R\text{-}OH + Na \rightleftharpoons R\text{-}O^-Na^+ + \tfrac{1}{2} H_2$$

$$2CH_3CH_2OH + 2Na \rightleftharpoons 2CH_3CH_2O^-Na^+ + H_2$$

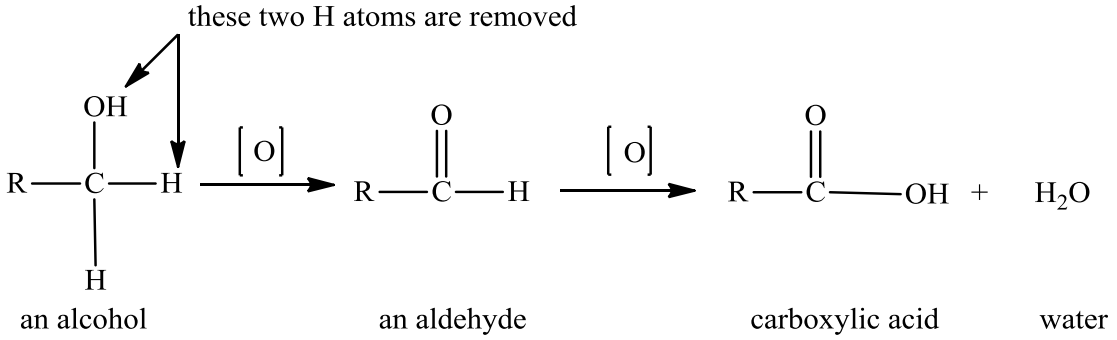
Compounds of the type RO^-Na^+ are called sodium alkoxides and consist of the ions RO^- and Na^+. The RO^- ions are conjugate bases of alcohols and are much stronger bases than hydroxide (OH^-) ion.

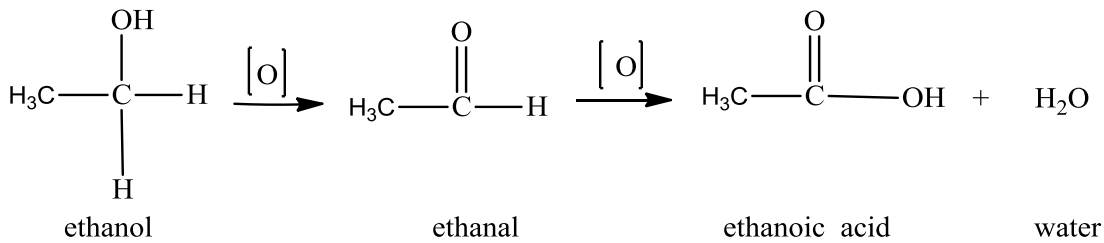
3.5.6.2 Oxidation reactions

Oxidation usually manifests itself in one of two ways (or both): the addition of more oxygen to a molecule, or the loss of hydrogen atoms with their electrons from a molecule. The addition of oxygen usually involves the addition of *one* atom of oxygen at a time while the loss of hydrogen atoms usually involves the loss of *two* hydrogen atoms at one time. Two hydrogen atoms are removed from the molecule. One hydrogen is from the oxygen of the –OH and the other is from the carbon atom bearing the OH group. Any of a large number of reagents that include $KMnO_4$ and $Na_2Cr_2O_7$ is used to oxidise alcohols. Alcohols are oxidised to various products depending on the type of alcohol.

3.5.6.2.1 Oxidation of primary alcohols

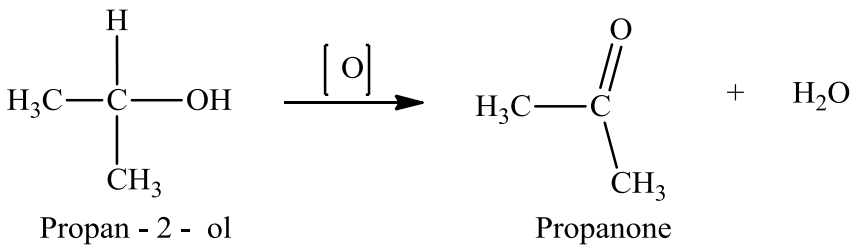
Primary alcohols can be oxidised to either aldehydes or carboxylic acids depending on the reaction conditions. We will use the abbreviation, [O], in the reactions below to denote an unspecified oxidant.





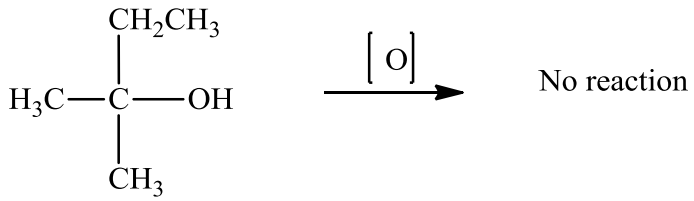
3.5.6.2.2 Oxidation of secondary alcohols

Secondary alcohols are oxidised to ketones as shown in the reaction equation below.



3.5.6.2.3 Oxidation of tertiary alcohols

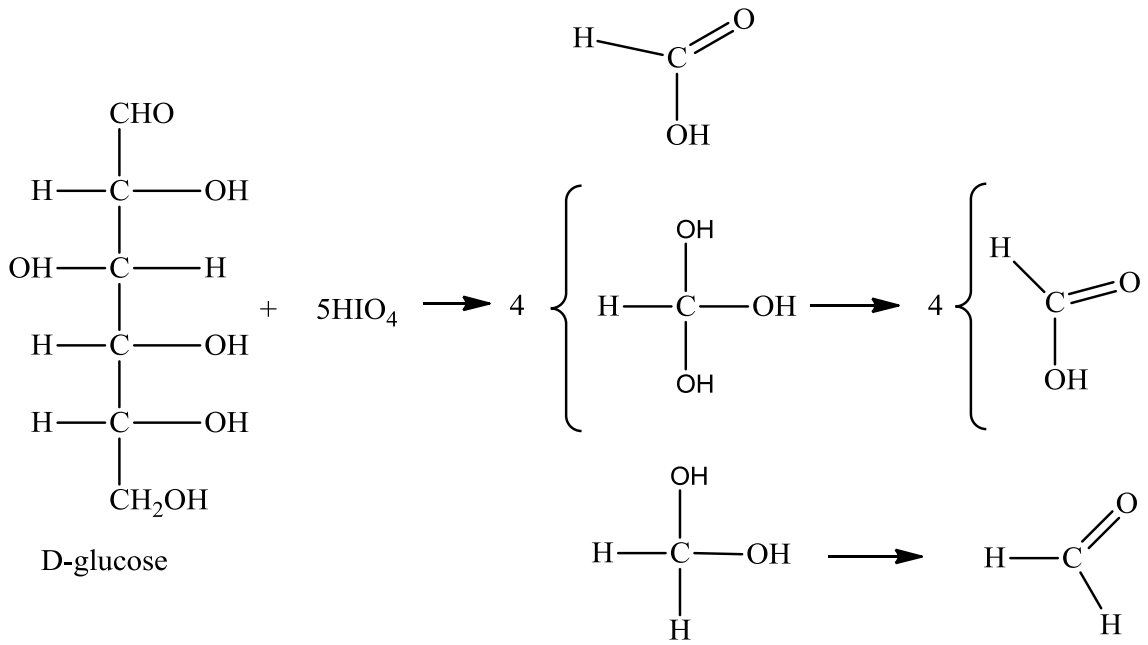
A tertiary alcohol does not have a hydrogen atom bonded to carbon bearing the OH group. Therefore it cannot lose a hydrogen atom and so, it cannot be oxidized in this way.



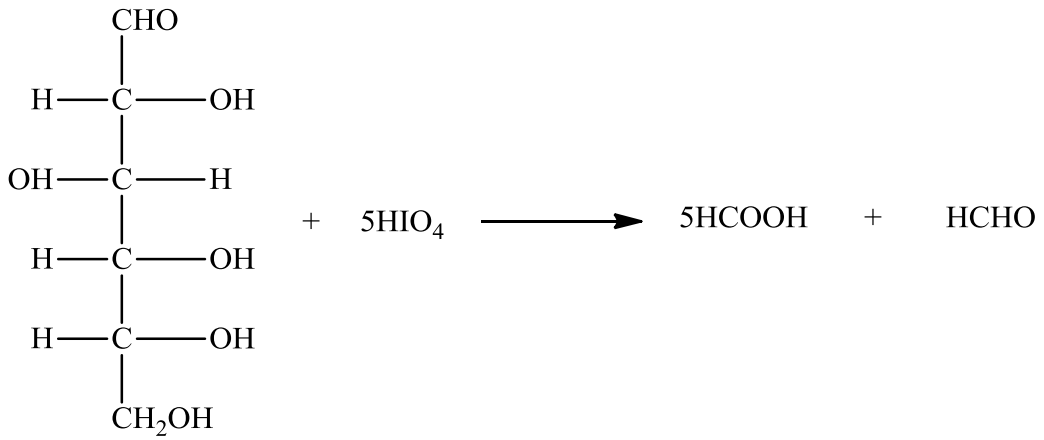
So, oxidation reaction can be used to distinguish primary and secondary alcohols from tertiary alcohols. You will perform some experiments in practical module 2, unit 4 concerning oxidation of alcohols.

3.5.6.2.4 Oxidation of alcohols with OH groups on adjacent carbons by periodic acid (HIO_4)

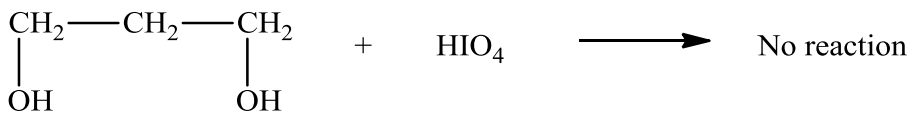
Alcohols with many OH groups on adjacent carbons are readily oxidised by Periodic acid (HIO_4). The periodic acid breaks the C-C bonds if both Cs have OH groups or if one has OH and the adjacent one has aldehyde or ketone group. For each C-C bond broken that C will be oxidised once. Each broken bond will be replaced with an OH group. Any C with two OH groups will lose water and become a carbonyl group.

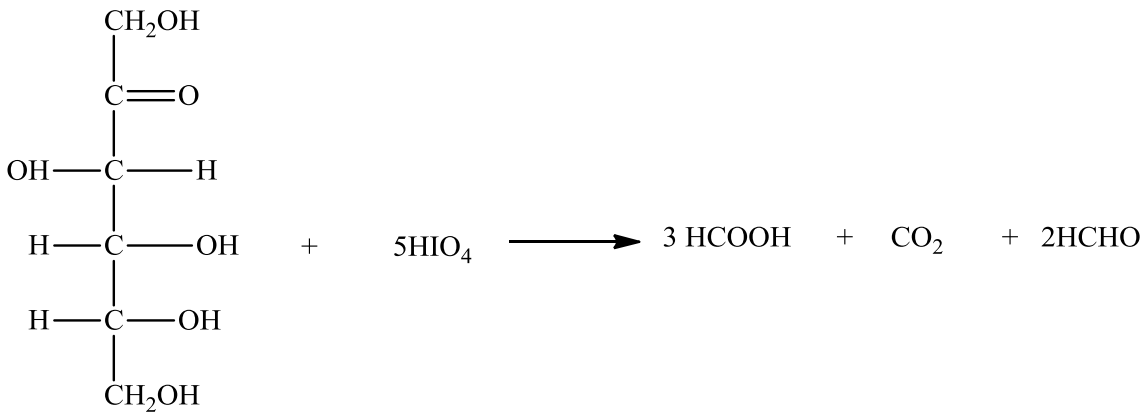


This reaction equation can be summarised as follows:



If the OH groups are not on adjacent carbons, no reaction takes place with the periodic acid

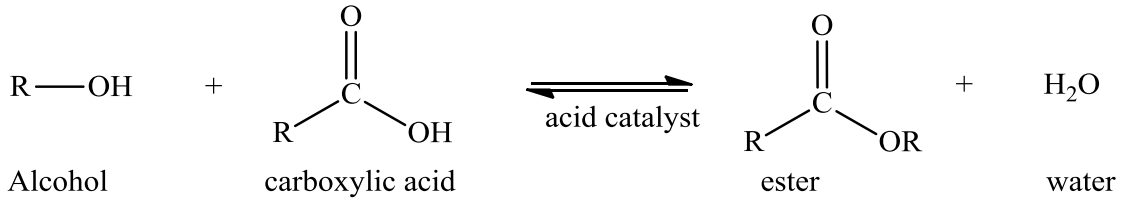




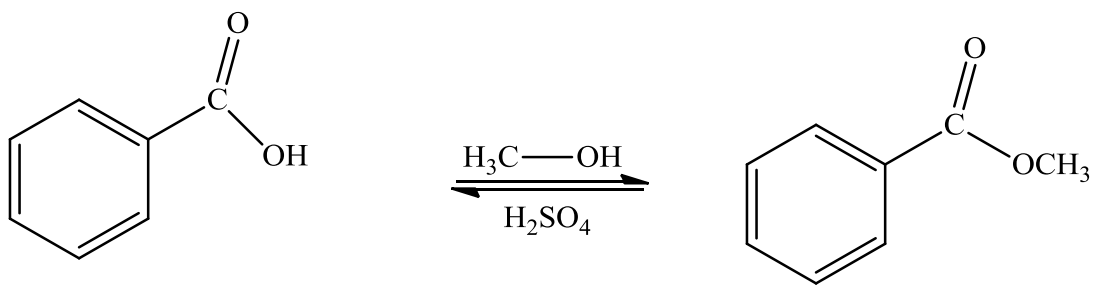
D-Fructose

3.5.7 Formation of esters of carboxylic acids

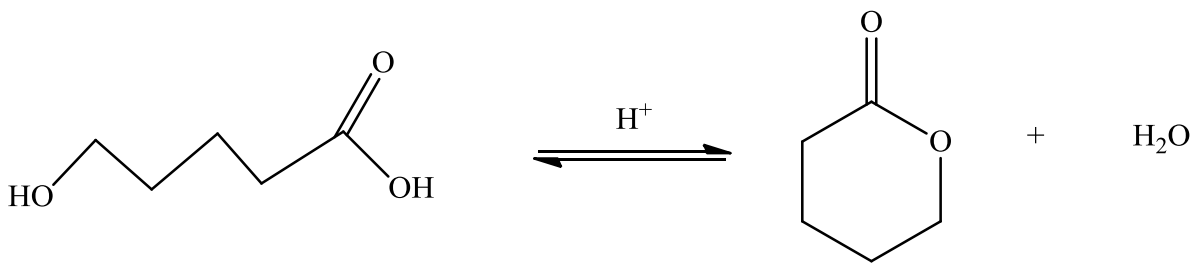
Alcohols react with carboxylic acids in the presence of acid to produce esters (Fischer esterification). In the process water is eliminated between carboxylic acid and the alcohol. Following is a general reaction of an alcohol with a carboxylic acid.



Below is the reaction between benzoic acid and methanol in the presence of sulphuric acid to form an ester.

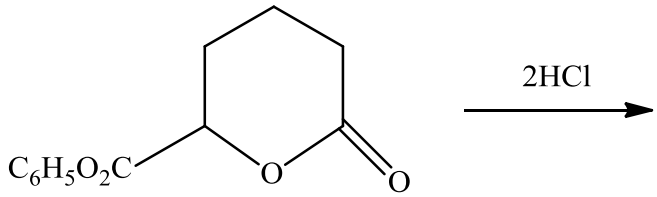


If hydroxyl and carboxyl groups are on the same chain, they can react in the presence of an acid to form a cyclic ester. This is an intramolecular reaction. Cyclic esters are also called lactones.

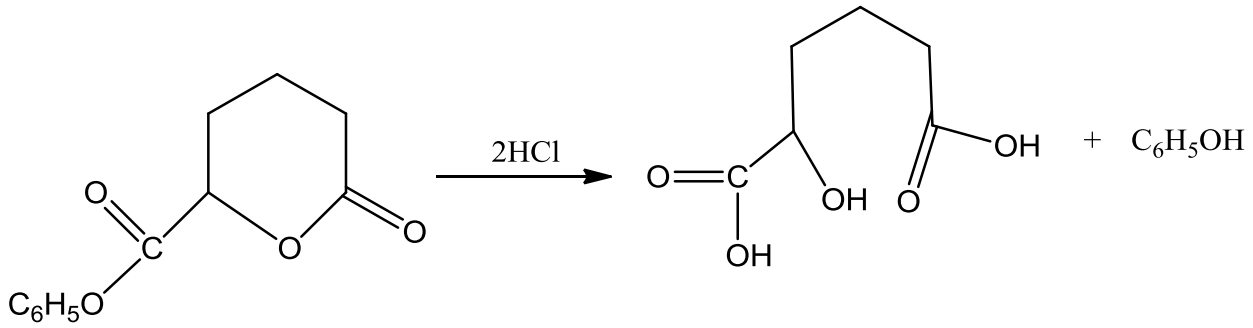


Worked example

Give the products of the following acid hydrolysis:

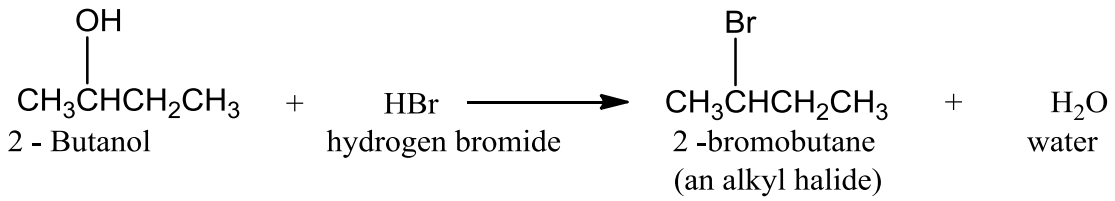


Solution



3.5.8 Preparation of alkyl halides from alcohols

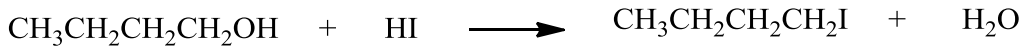
Alcohols react with hydrogen halides to form alkyl halides. Let us look at the reaction between 2-Butanol and hydrogen bromide.



3.5.9 Formation of ethers (Williamson synthesis)

An ether has the general structure: R-O-R, where R=Alkyl or aryl group or both.

The formation of an ether in Williamson synthesis involves two processes. The first process is the formation of an alkyl halide from reaction of alcohol and a hydrogen halide as shown in the reaction below.



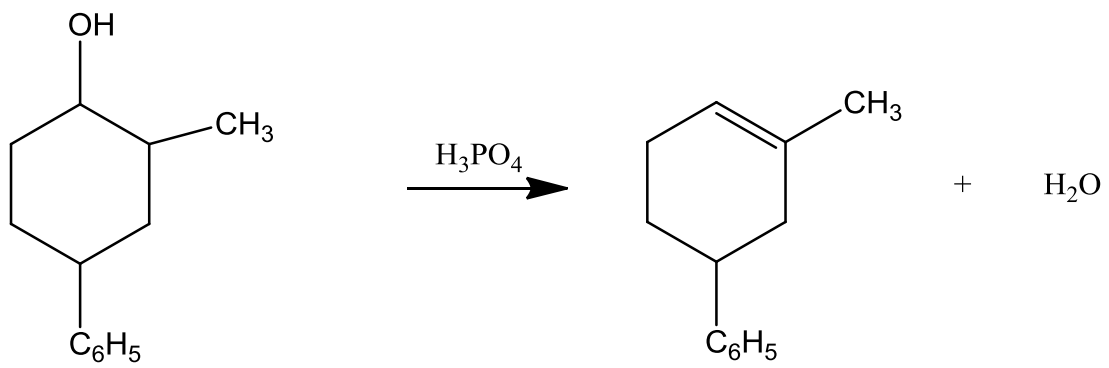
The second process is the removal of a hydrogen atom from an OH group to form an alkoxide ion.

The alkoxide ion formed in the second process and the alkyl halide from the first process then react to form an ether as shown below.

$$CH_3CH_2O^-Na^+ \quad + \quad CH_3CH_2CH_2CH_2I \quad \rightarrow \quad CH_3CH_2OCH_2CH_2CH_2CH_3 \quad + \quad NaI$$

3.5.10 Dehydration

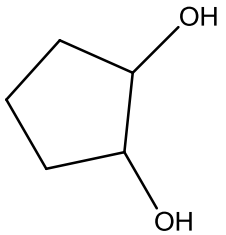
Dehydration is the reverse of hydration. Do you remember hydration of alkenes in the formation of alcohols? If you have forgotten then read section 2.5.4 in unit 2 and section 3.5.4 in this unit. Dehydration reactions are catalysed by sulphuric acid H_2SO_4 or phosphoric acid H_3PO_4. During the reaction water is removed from the alcohol and the products are alkenes. The OH and an H (hydrogen) atom from the carbon adjacent to the carbon with OH are eliminated as water (H_2O). Between the two adjacent carbons it is the carbon with fewer hydrogen atoms that loses the hydrogen atom. Remember in hydration a hydrogen atom is added to the carbon with already more hydrogen atoms. In dehydration the hydrogen atom is removed from the carbon with fewer number of hydrogen atoms. An example of dehydration reaction is shown below.



3.5.11 Unit activity

1. Give structures of major products (if any) expected from the following reactions:

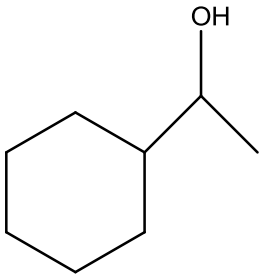
a. 2-Methyl-4-phenylcyclohexanol + H_2SO_4 ⟶

b. 

+ HIO_4 ⟶

c. CH_3CH_2OH $\xrightarrow{[O]}$

.d CH_3CHCH_3 (OH) $\xrightarrow{[O]}$

e 

+ Na ⟶

2. A compound **A** reacts with an alkylmagnesium halide to give the final product **B** to give the final product **C** of formula $\mathbf{C_6H_{12}O}$. Dehydration of **C** gives a compound **D**

which on oxidation gives two products **E** and **F**. The later reacts with 2, 4 –DNPH but **E** does not. However **E** turns blue litmus paper red. Compound **D** decolourises bromine dissolved in carbon tetrachloride.

a) Write structures A and B
b) Using appropriate structures, write the sequence of reactions to the final products E and F
c) Name the compounds A, C, D and F

3.5.0 Unit Summary

- Both aldehydes and ketones have the carbonyl group (C=O) as the functional group.
- When the carbonyl carbon is attached to a carbon atom and a hydrogen atom or two hydrogen atoms then the compound is an aldehyde.
- If the carbonyl carbon is attached to two carbon atoms then the compound is a ketone.
- Systematic names of carbonyl compounds are derived from the alkanes with same number of carbon atoms. The ending “e” in alkane is replaced with “al” in aldehyde and “one” in ketones. For example, an alkane, propane, with three carbons becomes propanal as an aldehyde and propanone as a ketone.
- Both aldehydes and ketones exhibit similar properties. The smaller members are miscible in water in all proportions. Both aldehydes and ketones react with 2, 4-DNPH giving yellow, orange or red precipitates. They also react with sodium bisulphite to yield salts called addition salts.
- Aldehydes and ketones differ in that aldehydes are oxidised to carboxylic acids while ketones are not oxidised.
- On oxidation with Fehling’s or Benedict solution, aldehydes form red precipitates Cu2O while with Tollen’s solution they give a silver mirror or grey precipitate.
- Alcohols can be classified as Primary, Secondary or Tertiary.
- Alcohols form hydrogen bonds with water making alcohols soluble in water.
- Alcohols can be formed from hydration of alkenes and also reaction of carbonyl compounds with Grignard reagents.

- Primary alcohols are oxidised to carboxylic acids, secondary alcohols are oxidised to ketones and tertiary alcohols are not oxidised.
- Alcohols react with carboxylic acid to form esters in a process called esterification.
- Dehydration of alkenes yields alcohols

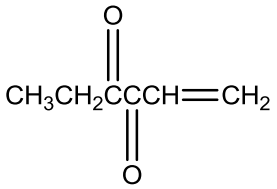
3.6.0 References

1. Graham Solomon T. W., and Fryhle C. B., 2004. Organic *Chemistry, 8th Ed*. John Willey and Sons, New York.

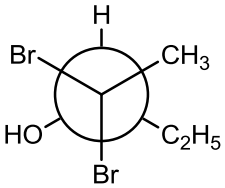
2. McMurry J. and Castellion M. E., 2003. *Fundamentals of general organic and biological chemistry,* 4th Ed., Pearson Education, New Jersey

3.7.0 Unit test

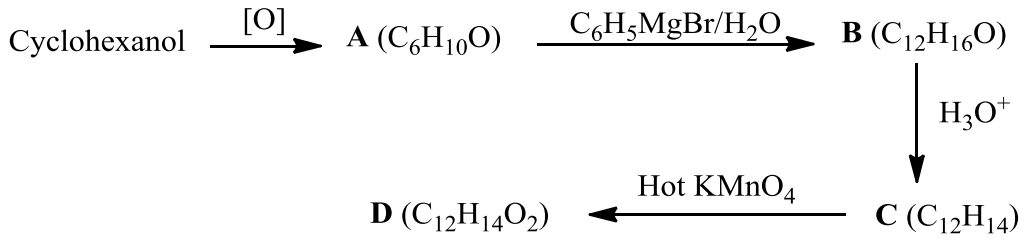
1. Give systematic names of the following compounds:

a. 

b.

c. 

2. Write structures of compounds A – D below.



Unit 4: ISOMERISM

4.0.0 Introduction

In this unit you will learn about the different ways atoms of a particular saturated compound are joined together to form distinct compounds with possible different chemical or physical properties. You will also learn how the atoms are attached differently in space to obtain compounds with unique physical properties. Furthermore, you will learn about how atoms of unsaturated compounds are joined together with respect to the position of the double bond. These types of molecules are commonly referred to as isomers.

4.1.0 Unit Objectives

By the end of this unit you should be able to:

- Write all possible structural isomers given a chemical formula
- Identify the various conditions giving rise to geometric isomers
- Identify optically active compounds
- Show relationships between various stereoisomeric compounds

4.1.2 Unit Key Terms

- Structural isomers
- Geometric isomers
- Optical activity
- Stereoisomers
- Enantiomers

- Diastereoisomers
- Meso compounds

4.2.0 Isomerism

Isomers are different compounds that have the same molecular formula. Isomers contain equal numbers of the same kinds of atoms but the atoms are joined to each other in different ways.

There are two main types of isomerism.

1. Structural (Constitutional) isomerism.
2. Stereoisomerism.

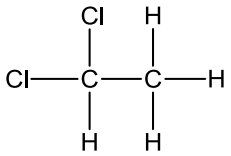
4.2.1 Constitutional isomers.

Constitutional isomers are isomers that have the same molecular formula but different connectivity that is the order in which their atoms are connected is different.

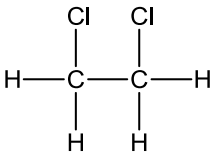
Hence constitutional (structural) isomers:

- May have different chemical properties
- Have different physical properties

Following are examples of constitutional isomers.



1, 1-dichloroethane



1, 2-dichloroethane

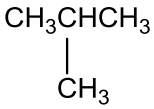
The two structures above have the same molecular formula (**$C_2H_4Cl_2$**) but the order of attachment of the atoms is different (connectivity of atoms is different), hence they have different structural formula. Therefore different structures with the same molecular formula are called **Structural Isomers** (or **constitutional isomers**). This phenomenon is called Structural Isomerism (constitutional isomerism).

Constitutional isomerism can exist in compounds of the same family (alkanes, alkenes, alcohols etc). Such compounds (i.e. isomers) have different physical properties. The chemical properties for isomers of the same family are similar but not identical (e.g., rates of reaction will be different).

Example

$CH_3CH_2CH_2CH_3$ and

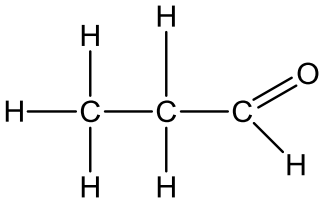
Boiling point = -0.5^{o}C



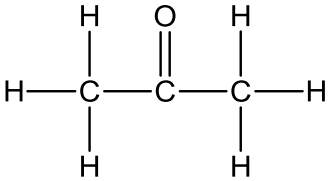
Boiling point = -11.7^{o}C

Constitutional isomerism can also exist in compounds of different families such as between an aldehyde and a ketone.

Example 1



Propanal



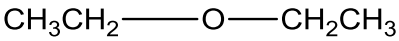
Propanone

For this type of isomers you will see that both physical and chemical properties are different. In the example above propanal has a boiling point of 48.8°C whereas propanone has a boiling point of 56°C. Propanal can be oxidized to propanoic acid (carboxylic acid) whereas propanone cannot be oxidized.

Example 2

$CH_3CH_2CH_2CH_2OH$

Butan-1-ol



Diethyl ether

4.2.2 Unit activity A

Try these:

1. Write all possible structural isomers of butene, C_4H_8
2. Draw and name all structural isomers of C_7H_{16}
3. Draw and name all structural isomers of C_7H_{15} Cl

4.3.0 Stereoisomerism

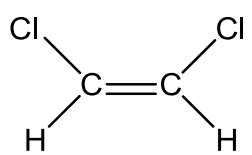
Stereoisomers are isomers that have the same molecular formula, have their atoms connected in the same order but different spatial arrangements of the atoms or groups of atoms. The study of the way a compound's atoms are arranged in space is called stereochemistry. Thus stereochemistry is the chemistry of space.

Two types of stereoisomerism

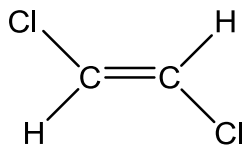
1. Geometric (*cis* and *trans*) isomerism
2. Optical isomerism

4.3.1 Geometric isomerism

Geometric isomers result from the restricted rotation about the carbon to carbon double bond(C=C), other double bonds such as C=N, N=N or the presence of a ring structure. Geometric isomers differ in all their physical properties and in many of their chemical properties Pairs of geometric isomers are referred to as *Cis*- and *Trans* isomers (e.g.,1,2-dichloroethene)



and

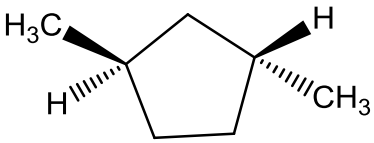


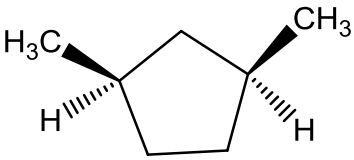
Cis-1,2-Dichloroethene *Trans*-1,2-Dichloroethene

You can notice that a *cis* isomer is one in which the substituents are on the same side (either the two Cl atoms above or two H atoms below) of the carbon to carbon double bond, C=C. A *trans* isomer is one in which the substituents are on the opposite sides (one Cl atom above and another Cl atom below or one H atom above and another H atom below) of the carbon to carbon double bond C=C

If we consider the two isomers above we can conclude that *Cis and Trans* isomers have the same structure since each carbon has exactly the same atoms attached to it. The only difference between the two compounds is the spatial arrangement of the different groups across the double bond. They are classified as Stereoisomers**.**

Let us now consider another type of geometric isomers that result from ring formation which also confers rigidity on the molecular structure (restricted rotation).You may recall that carbon atoms in a chain can close up to form a ring. There is restricted rotation about any C-C ring bond.



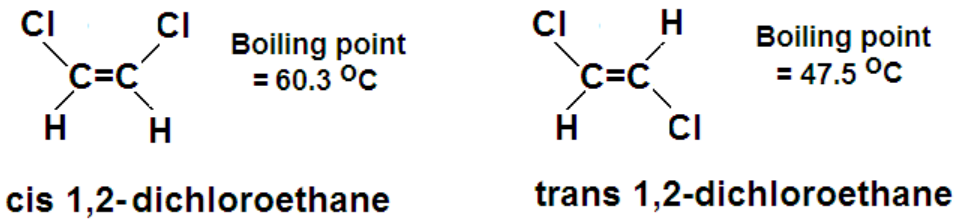


Trans-1, 3-Dimethylcyclopentane *Cis* -1, 3-Dimethylcyclopentane

If we consider the *trans*-1,3-dimethylcyclopentane, we notice that one methyl group is pointing away (dashed lines)from us and another is pointing towards (wedge) us. In the case of *cis*-1,3-dimethylcyclopentane we notice that the two methyl groups are pointing in the same direction (towards us). *Cis* and *trans* isomers cannot be interconverted without breaking one or more bonds.

Geometric (*cis-trans*) isomers have different physical properties, e.g. boiling points, melting points etc**.**

Example

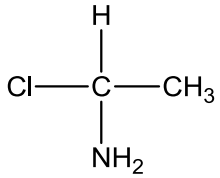


- In contrast geometric (*cis-trans*) isomers have similar chemical properties since they are members of the same family. However, their chemical properties are not identical since their structures are not identical. For example they react at different rates.

4.4.0 Optical isomerism

Another type of stereoisomerism is manifested by the presence or absence of chiral centres in the molecule. A chiral molecule is one that is non-superimposable on its mirror image. One characteristic feature you can easily recognize of a chiral molecule is a carbon atom with four different groups attached to it. Chiral means being 'left handed or right handed'. The opposite of chiral is achiral which means without handedness. Sometimes a chiral molecule is called an asymmetric molecule.

Example



Stereoisomers can be divided into two general groups namely those that are enantiomers(mirror images) of each other and those that are diastereomers (not mirror images) of each other.

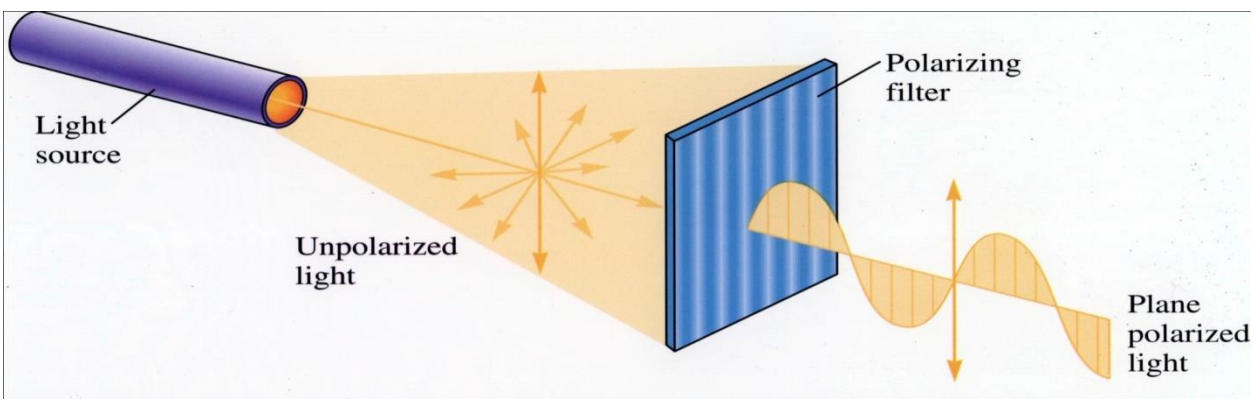
4.4.1 Enantiomers

If we place an imaginary mirror in front of the molecule above we can obtain its mirror image.

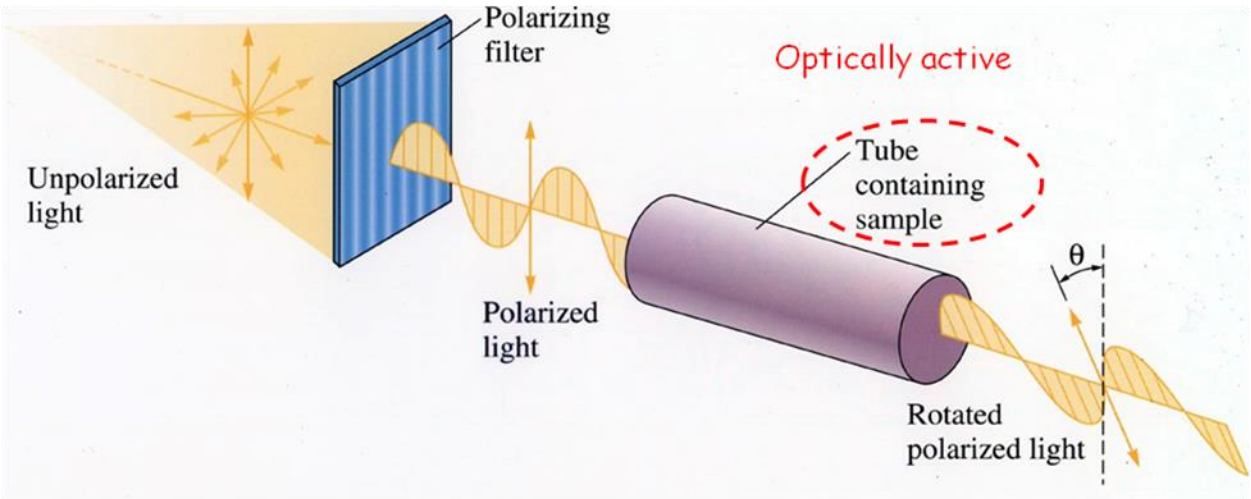
The two compounds above are non-superimposable mirror reflections of each other. It means that one cannot place one compound on the other so that all parts of each coincide (i.e. Cl lie on top of Cl, H on top of H, CH_3 on top of CH_3 and NH_2 on top of NH_2). Compounds related in this way, are called enantiomers. Enantiomers are stereoisomers that are non – superimposable mirror images of each other. So enantiomers have similar melting points, boiling points and reactivity toward symmetric chemical substances. Differences do exist but can be detected only by devices that are themselves asymmetric, the most common being polarized light.

4.4.2 Optical rotation

Optical isomers can be distinguished by the way they interact with plane polarized light. If polarized light passes through an asymmetric compound the plane rotates. You may be asking what is plane polarized light? Polarized light is light that has been passed through a polarizing prism or filter. As a result the light vibrates in a single plane.



If you pass polarized light through an asymmetric (chiral) compound the plane rotates.

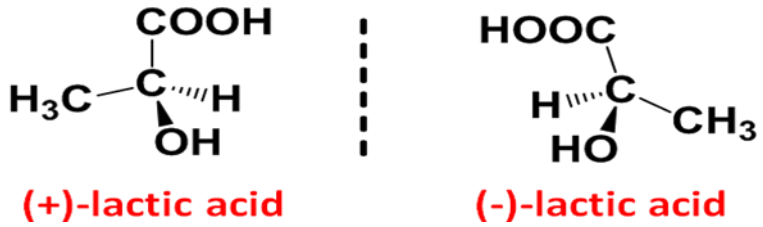


The mirror images will rotate polarized light by the same degrees but in opposite direction. A compound that rotates polarized light is said to be **optically active**.

The angle between the original and final planes of rotation is called the optical rotation. This depends on the following factors:

- Wavelength of light used
- Length of light path
- No of optically active molecules/unit length of light
- Temperature
- Nature of solvent
- Structure of the compound

If the plane of polarization is rotated clockwise the sign of rotation is taken as positive and the compound is said to be dextrorotatory (Latin *Dexter* = right) (e.g., (+)-lactic acid). If the rotation is anticlockwise the sign is negative and the compound is said to be levorotatory (e.g., (-)-lactic acid)

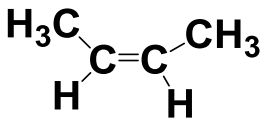


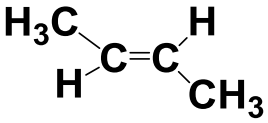
A mixture that contains equal amounts of a pair of enantiomers is called a racemic mixture or racemate. An example would be a (±) mixture, (±)-lactic acid). A racemate is optically inactive

We can compare the optical rotations of two or more compounds, if we use standard conditions. Under standard conditions the optical rotation is called the specific rotation. We can therefore define Specific rotation as the angle in degrees through which polarized light is rotated when it is passed through a 10cm tube of solution of concentration 1g/ml.

4.4.3 Diastereomers

Diastereomers are isomers that are not mirror images of each other. For example *cis*-but-2-ene and *trans*-but-2-ene are stereoisomers that are **diastereomers**.





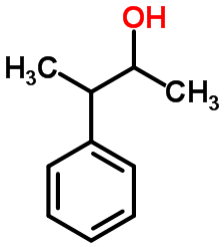
***Cis*-but-2-ene** ***Trans*-but-2-ene**

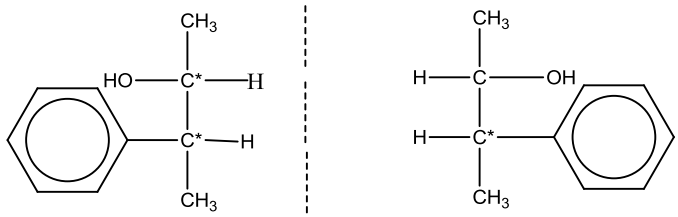
4.4.4 Stereoisomers with multiple asymmetric centres.

So far we have looked at compounds with one chiral centre. You should realize that it is also possible to have compounds that contain two or more asymmetric (chiral) carbons. These exist in more than two stereoisomeric structures. Some are pairs of enantiomers, others may be symmetric and therefore optically inactive.

Many natural biomolecules contain more than one asymmetric centre. It is very common to find two to ten asymmetric centres in biomolecules. Molecules such as carbohydrates and proteins contain hundreds of asymmetric centres. The number of stereoisomers having the same structural formula is called the isomer number for that structure. Generally the isomer number will be equal to 2^n, where n is the number of asymmetric centres.

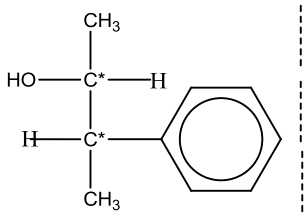
Let us consider 3-phenyl-2-butanol which has two asymmetric centres



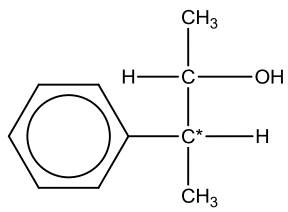


I

II



III

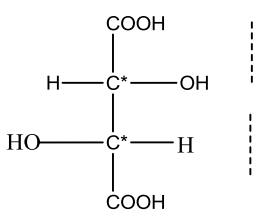


IV

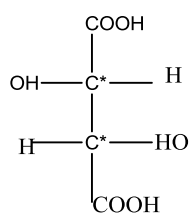
You should notice that compound I is a mirror image of compound II whereas compound III is a mirror image of compound IV. From these four compounds with have two pairs of enantioners. What about if we now compare I and III and II and IV? They are neither superimposable nor mirror images of each other. Therefore I and III and II and IV are called diastereomers that is stereoisomers that are not mirror images.

4.4.5 Molecules with Special Symmetry properties

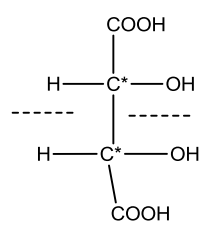
Very often we find substances that have special symmetry properties and when this is so, the 2^n formula does not apply. Symmetric substances that contain asymmetric centres are called meso compounds. Let us look at the structure of tartaric acid, $CO_2HCHOHCHOHCO_2H$:



A



B

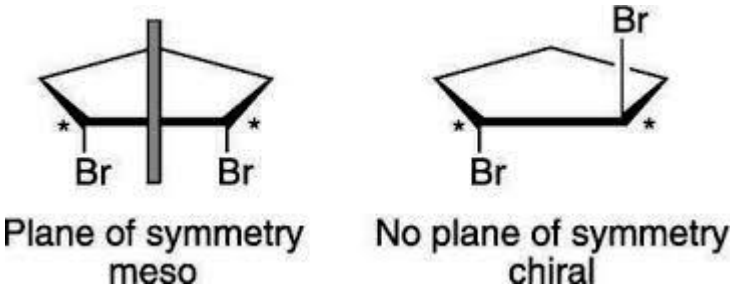


C

Thus, tartaric acid exists in three stereoisomeric forms: two enantiomers (A and B) and one meso form (C). The meso form is the compound which has an internal mirror of symmetry dotted line in compound C). Meso-tartaric acid is a diastereoisomer of either compound A or compound B. Meso compounds are optically inactive

4.4.6 Optical isomerism in cyclic molecules

Consider 1, 2-dibromocyclopentane



Why bother with Isomerism?

Nearly all organic molecules of living organisms have handedness. For example the naturally occurring amino acids are of the L-form.

What difference does this make?

In certain extreme cases this may mean Life or death!!

Each enzyme in the body recognizes one type of substrate shape (enantiomer) through the mechanism of catalysis the so called ‘’Lock and Key.’’

Also many of the drugs we take and biological compounds we have are optically active compounds. The mechanism of drug action is similar to the enzyme catalysis model.

4.5.0 Unit summary

- Structural (Constitutional) isomers are isomers that have the same molecular formula but the order in which their atoms are connected is different.
- Geometric isomers result from the restricted rotation about the carbon to carbon double bond(C=C) and the presence of a ring structure.
- Pairs of geometric isomers are referred to as *Cis*- and *Trans* isomers:
  -*cis* isomer is one in which the substituents are on the same side.
  -*trans* isomer is one in which the substituents are on the opposite sides.
- Optical isomerism is manifested by the presence or absence of chiral centres

in the molecule. A chiral molecule is one that is non-superimposable on its mirror image and is characterized by having four different groups attached to one carbon ato (chiral carbon).

4.6.0 Unit Test

1. Define the following terms:

a. Achiral molecule

b. Specific rotation

c. Meso compound

d. diastereoisomers

2.Write three differences between constitutional and stereoisomerism.

3.(a)Write all the acyclic structural isomers of a compound of formula $C_4H_8Cl_2$.

(b)Which of the structures drawn above in (a) would exhibit stereoisomerism?

(C)For each structure in (b) above write the structures of all the stereoisomers and indicate the ones related as (i) diastereoisomers (ii) meso compound (iii) cis-trans isomers.

4.a) Write all the acyclic isomeric compound with the formula $C_4H_{10}O$.

(b) Write all the structures in (a) that would exhibit stereoisomerism?

(C) For each structure in (b) write the structures that are related as (i) enantiomers (ii) diastereoisomers (iii) meso

5. Indicate whether each of the following statements is true (T) or false (F).

(a) Two structural isomers may be chiral;

(b) Two structural isomers may be diastereoisomers.

(c) A pair of enantiomers always have a mirror image relationship

(d) All chiral molecules have diastereoisomers

(e) If a compound has a diastereoisomer it must be chiral

(f) If a nolecule has has chiral centres it must be optically active

(g) Some diastereisomers can have a mirror image relationship

(h) All compounds with asymmetric carbons are chiral

(i) If a solution is optically inactive it must consist of a symmetric compound

6.(a) Write all the acyclic isomeric structures of chloropentane.

(b)Write the structures in (a) that would show stereoisomerism.

(a) For each structure in (b) write the stereoisomers and label

(i)Pairs of enantiomer

(ii)Pairs related as diastereoisomers

(iii)Meso compounds

UNIT 5: ORGANIC COMPOUNDS CONTAINING NITROGEN

5.0.0 Introduction

In this unit we will discuss the chemical and physical properties of amines and amides. These are compounds characterized by the presence of nitrogen atoms. An understanding of the chemistry of the two groups of compounds is important to the understanding of the chemistry of molecules such as proteins. They are also important in physiology.

5.0.1 Unit Objectives

By the end of this unit, you should be able to

- Write structures and systematic names of amines
- Explain primary, secondary and tertiary amine
- Explain the properties of amines
- Write structures and systematic names of amides
- Explain the properties of Amides

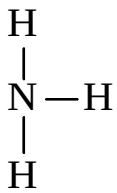
5.0.2 Unit Key Terms

- Amine
- Primary amine
- Secondary amine
- Tertiary amine
- Amide
- Primary amide
- Secondary amide
- Tertiary amide

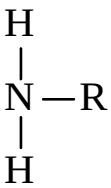
5.1.0AMINES

The oxygen-containing compounds discussed in the last chapter are a few of the many that exist in nature. Nitrogen containing compounds are equally prevalent and among these are the amines which are important as medicines (e.g., quinine, amphetamine, nicotinic acid, etc.), biological compounds (e.g., proteins, nucleic acids, vitamins, etc.) and as poisons (e.g., nicotine). Amines are therefore useful to human beings.

Amines are derived from ammonia by replacing one or more of the hydrogen atoms of ammonia by alkyl or aryl groups:



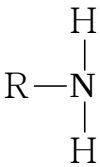
Ammonia



(R= alkyl or aryl)

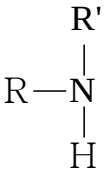
An amine

There are three types of amines; **primary**, **secondary and tertiary** (compare alcohols). Primary amines are formed when only one hydrogen atom in ammonia is replaced by an alkyl or aryl group. When two hydrogen atoms in ammonia are replaced then a secondary amine results, and when all the hydrogen atoms in ammonia are replaced by alkyl or aryl groups, a tertiary amine is formed.



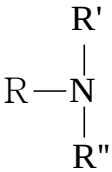
R= alkyl or aryl

Primary



R, R' = alkyl or aryl

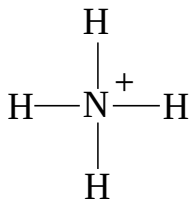
Secondary



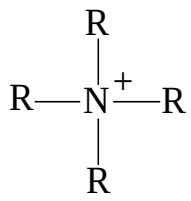
R,R',R" = alkyl or aryl

Tertiary

Attachment of four alkyl or aryl groups to a nitrogen atom produces a quaternary ammonium ion which is an analogue of the ammonium ion.



Ammonium ion



Quarternary ammonium ion

5.2.0 Nomenclature of Amines

5.2.1 **Trivial Names**

Common names of amines are obtained by specifying the alkyl or aryl groups attached to nitrogen and adding the word *''*amine*''* as a suffix. If the groups are the same, the prefixes ''**di**'' (for two) or ''**tr**i'' (for three) are used. *The name is written as one word*. For example:

CH_3NH_2	$CH_3CH_2NH_2$	$CH_3CH_2CH_2NH_2$
Methylamine	Ethylamine	n-Propylamine
$(CH_3)_2NH$	$(CH_3CH_2)_3N$	$C_6H_5NHCH_3$
Dimethylamine	Triethylamine	Methylphenylamine

Secondary or tertiary amines can also be named as N-substituted amines. In this case, the longest chain attached to nitrogen is taken as the parent compound and the other alkyl or aryl groups attached to nitrogen are located by the use of a capital letter N as substituents on the parent compound. *Each substituent must have its own capital letter N*.

$CH_3NHCH_2CH_2CH_3$ $C_6H_5NH(CH_3)CH_2CH_2CH_3$ $(CH_3CH_2)_2NCH_2CH_2CH_2CH_3$

N-Methylpropylamine N-Methyl-N-phenylpropylamine N,N-Diethylbutylamine

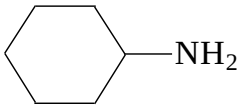
5.2.2 Systematic names

5.2.2.1 Primary amines

The systematic names of amines are derived from the names of alkanes with the same number of carbon atoms. The ending "**e**" in the name of the alkane is replaced by the word "**amine**". For simple primary amines, this is straight forward. For example:

CH_3NH_2
Methanamine

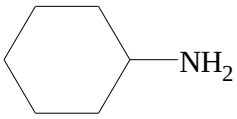
$CH_3CH_2NH_2$
Ethanamine



Cyclohexanamine

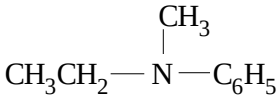
Most often though, amines are named by specifying the groups attached to nitrogen and adding the word amine as suffix. The name is written as one word. For example:

CH_3NH_2
methylamine



cyclohexylamine

$CH_3CH_2NH_2$
ethylamine



ethyl methyl phenylamine

CH_3NHCH_3
dimethylamine

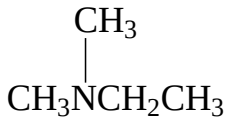
5.2.2.2 Secondary and tertiary amines

For secondary and tertiary amines, first, we locate the longest hydrocarbon chain and this determines the base name of the compound. The other hydrocarbon groups attached to nitrogen are located by a capital letter ‘’N’’ (compare the other hydrocarbons which use numbers). The "N" is placed in front of the name of the substituent followed by the base name of the compound. Each group is located by its own letter "N". If there are two similar groups attached to nitrogen, then two letter "N" are placed together as a prefix, the letters being separated by commas. In such cases the prefix "**di**" is used to show the number of the similar groups. This nomenclature system also applies to aromatic amines which are named as derivatives of aniline. A capital letter N, placed before the name of the substituent, designates its attachment to the nitrogen. For example:

1 $CH_2NHCH_2CH_3$

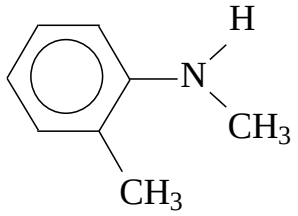
N-Methylethanamine

2



N,N-Dimethylethanamine

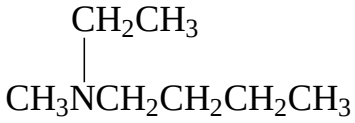
3



N,2-Dimethylaniline

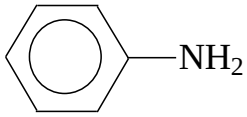
Note: The "2" is locating the methyl group on the benzene ring

4



N-Ethyl-N-methyl-1-butanamine

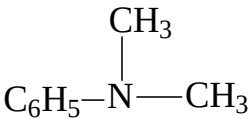
5



or $C_6H_5NH_2$

Aniline

6



N,N-Dimethylaniline

Sometimes, the more complex amines are named as **amino substituted compounds**. In this case, the nitrogen is taken as a substituent on another compound. For example:

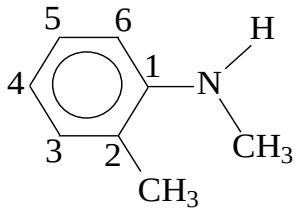
1 $NH_2CH_2CH_2OH$

2-Aminoethanol

2 $CH_3CH_2CH(NHCH_3)CH_2CH{=}CH_2$

4-(N-Methylamino)hex-1-ene
Note: the amino group is on position 4 of the alkene

3

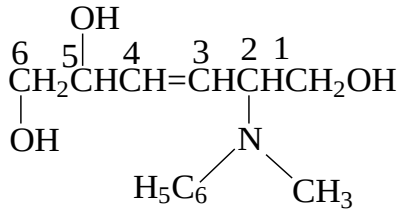


N-Methylamino-2-methylbenzene

4 $CH_3CH(NH_2)CO_2H$

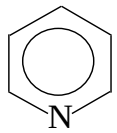
2-Aminopropanoic acid
Trivial name: Alanine
(an amino acid)

5

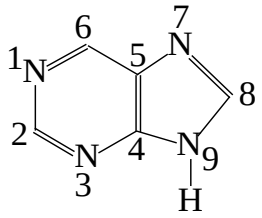


2-(N-Methyl-N-phenylamino)-3-hexen-1,5,6-triol
Note: The "3" is locating C=C

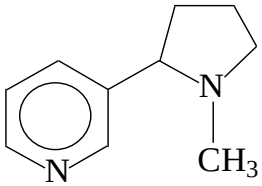
Heterocyclic amines which play important roles in biological processes also have common names. For instance:



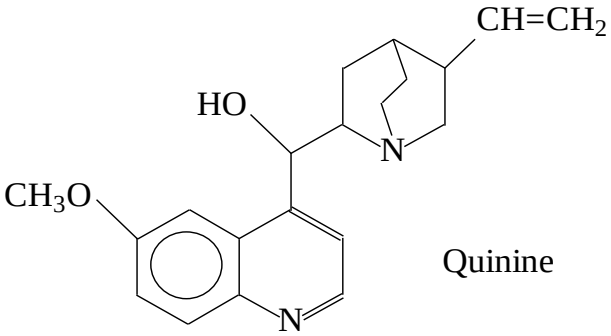
Pyridine



Purine



Nicotine



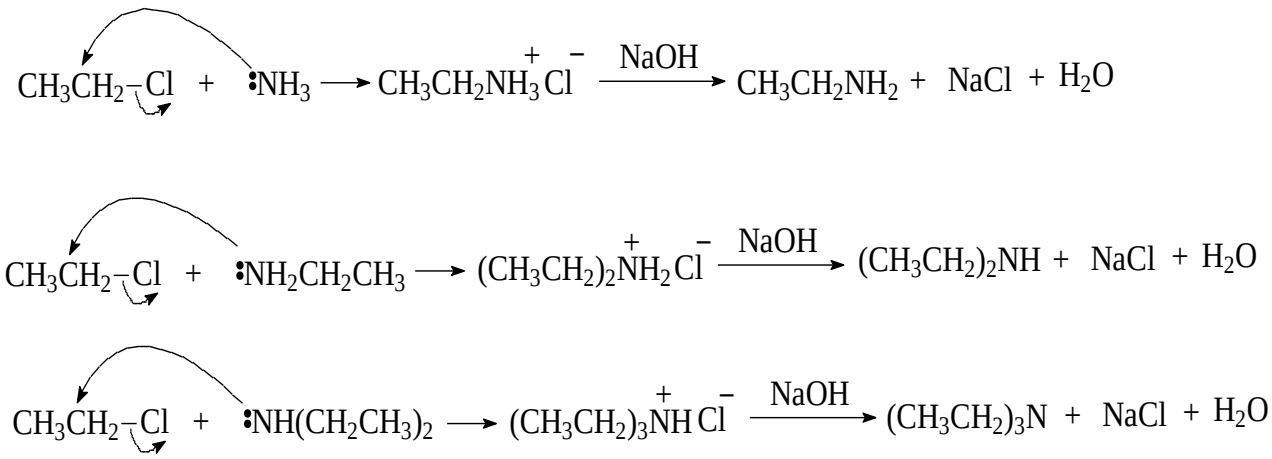
Quinine

5.3.0 Preparation of Amines

Amines can be prepared by a series of methods including the following:

5.3.1 Alkylation

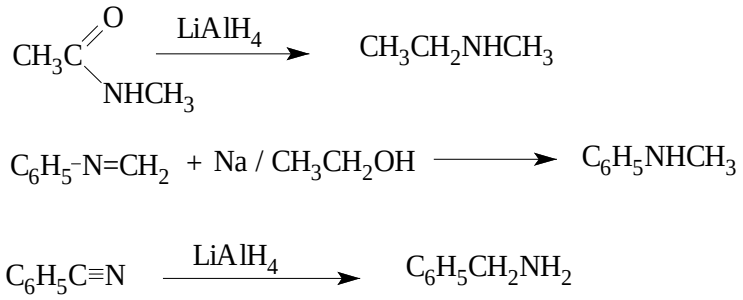
If we put an alkyl group on another substance, we say we are alkylating the substance. The process is called alkylation. An alkyl group can easily be put on nitrogen in ammonia by reacting the latter with an alkylhalide. Since halogens are electronegative atoms, the R-X bond is polar, the organic group (R) being partially positively charged and the halide (X) being partially negatively charged. Thus, the lone pair of electrons on nitrogen in ammonia can be attracted to the organic group of the alkyl halide and displace the halogen from it. This is therefore, a nucleophilic reaction and since the halide is replaced by nitrogen, this type of reaction is termed **nucleophilic substitution**. The ammonium salt that is produced is treated with a base (e.g., NaOH) and the amine is liberated. Unfortunately, further alkylation of the resulting amine occurs leading to a mixture of primary, secondary and tertiary amines. In the examples below, we use curly arrows ( ) to show movement of electrons. Notice in the first equation for example, the arrow is from the electrons on nitrogen to the partially positively charged carbon on the chloroethane. The same applies to all equations where such arrows are used.



In these reactions, a large excess of ammonia favours the formation of a primary amine.

5.3.2 Reduction

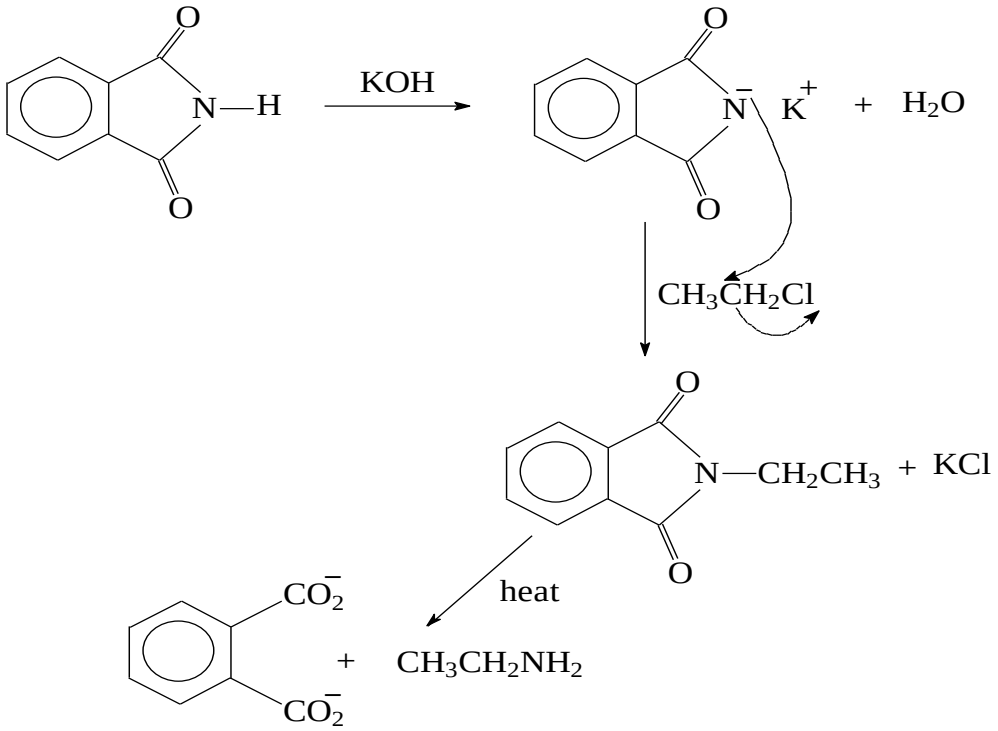
Nitrogen-containing compounds may be reduced by various reducing agents to give primary, secondary or tertiary amines depending on the compound used.



5.3.3 Gabriel's method

The Gabriel synthesis is a useful method for preparing primary amines. The method employs phthalimide as a starting compound. Phthalimide is acidic and so it can easily be converted to its

conjugate base by reaction with a strong base (usually potassium hydroxide is used). The resulting anion can displace a halide from an alkylhalide, and on warming the product, a primary amine is produced:



5.4.0 Properties of Amines

5.4.1 Physical

The lowest molecular weight members among the primary, secondary and tertiary amines are gas at room temperature. Other aliphatic amines are liquids. They have an ammonia-like smell. The presence of N-H bonds in primary and secondary amines means that these amines can hydrogen bond to each other. Tertiary amines do not have N-H bonds and so they cannot form hydrogen bonds to each other; but they can form hydrogen bonds with water molecules. Primary and secondary amines therefore boil at higher temperature than tertiary amines of similar molecular weights.

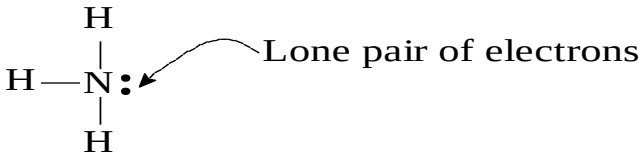
5.4.2 Chemical

5.4.2.1 Basicity

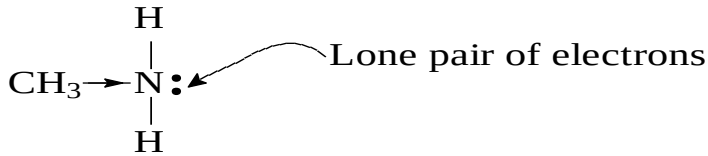
Ammonia is a basic compound. Its derivatives, the amines are similarly basic. Amines are in fact the most common organic bases. They are stronger bases than water but weaker than hydroxide or alkoxide (RO^-) ions. Amines accept protons from acids to form salts just like inorganic bases do. For example:

$$CH_3CH_2NH_2 + HCl \rightleftharpoons CH_3CH_2NH_3^+ Cl^-$$

Primary alkylamines are stronger than ammonia. How do we explain this difference? Let us look at the structures of ammonia and methanamine for example.



Ammonia

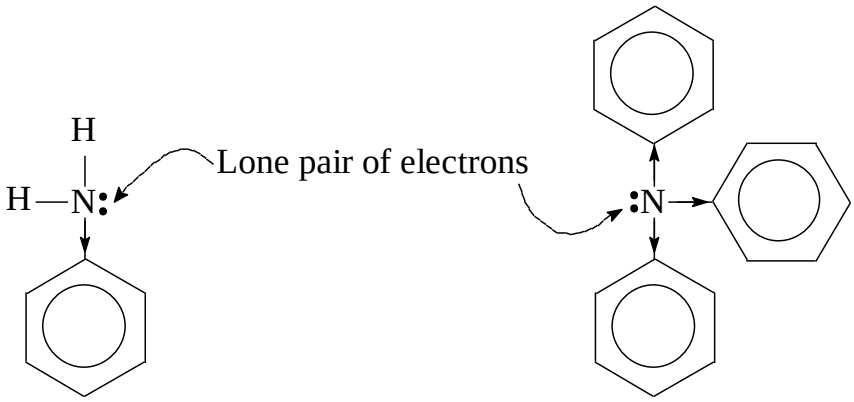


Methanamine

We notice that in both ammonia and methanamine there is a lone pair of electrons on nitrogen. It's this lone pair that is responsible for the basicity of these compounds. The more readily available the lone pair is to an acid, the more basic the compound is. Alky groups are known to donate electrons (i.e, they are electron donors). As such, they tend to donate electrons to the nitrogen (the arrow from methyl (CH_3) to nitrogen (N) shows electron flow in that direction) and this expels the lone pairs of electrons on nitrogen making them more readily available to an acid. Now imagine having two alkyl groups attached to nitrogen, this will expel the lone pair even more. With three alkyl groups attached to nitrogen, the effect will be even more. Therefore, for a

given alkyl group, we would expect the basicity to decrease in the order: tertiary > secondary > primary > ammonia. However, this is only true in gas phase reactions; in solutions, the reverse is true, tertiary amines are much less basic than the other amines (can you think of a reason for this?).

Arylamines on the other hand are considerably weaker bases than the corresponding alkyl amines. We explain this on the basis of the unavailability of the lone pair of electrons on the nitrogen atom. Let us consider aniline **(a)** and N,N-diphenylaniline **(b)** as our examples.



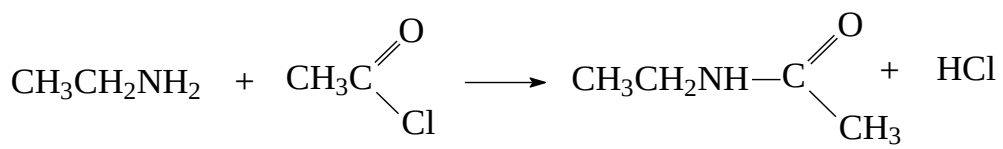
Aniline **(a)** N,N-Diphenylaniline **(b)**

(Neutral, not basic)

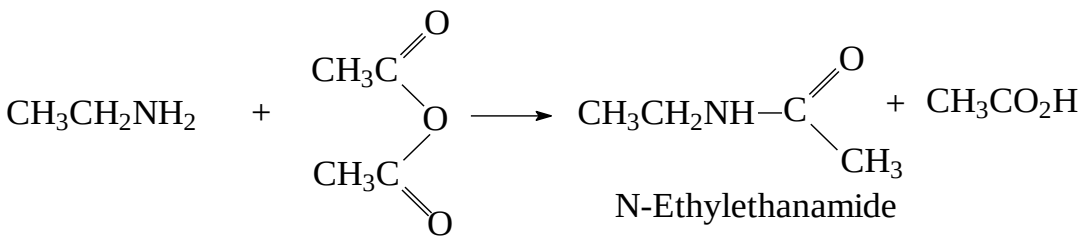
There is still a lone pair of electron on nitrogen. However, unlike alkyl groups, aryl (aromatic) groups tend to pull electrons towards themselves (notice the arrows toward the phenyl groups in both (**a**) and (**b**) in the structures above). They are said to be electron ‘’sinks’’. This tendency makes the lone pair of electrons to be drawn inside the nitrogen. Therefore, this lone pair of electrons is not readily available to an acid. Now if we imagine two phenyl groups attached to nitrogen, the effect will be more and N-phenylaniline will less basic than aniline. With three phenyl groups attached to nitrogen as in structure (**b**) the effect will be even more and to such an extent that N,N-diphenylaniline is a neutral rather than basic compound.

5.4.2.2 Acylation

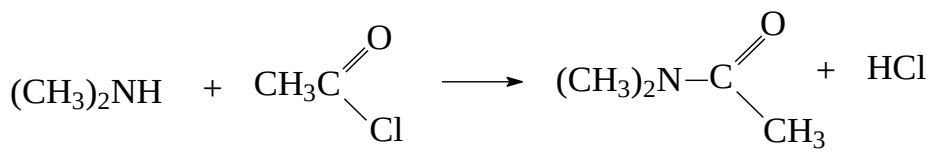
If we put an acyl group on a substance (i.e., R-C=O), we say we are acylating the substance and the process is called acylation. Primary and secondary amines can readily be acylated with various reagents such as acylhalides, acid anhydrides to give products called amides. This is also a nucleophilic substitution reaction. For example:



N-Ethylethanamide



N-Ethylethanamide



Note that in the above reactions, a hydrogen atom of the amine nitrogen is displaced by an acyl group; and since tertiary amines do not have a hydrogen on nitrogen, they are not acylated.

5.4.2.3 Oxidation

Primary and secondary amines are readily oxidized to give hydroxyl amines. For example:

$$CH_3CH_2NH_2 + H_2O_2 \rightarrow CH_3CH_2NH\text{-}OH$$

$$(CH_3CH_2)_2NH + H_2O_2 \rightarrow (CH_3CH_2)_2N\text{-}OH$$

When a tertiary amine is oxidized under similar conditions, an amine oxide is the product:

$$(CH_3CH_2)_3N + H_2O_2 \longrightarrow (CH_3CH_2)_3\overset{+}{N}\text{—}\overset{-}{O}$$

An amine oxide

5.4.2.4 Reaction with metal salts

Amines can form complexes with metal ions, which then precipitate out of solution. This reaction is important in that proteins (which are also amines) can similarly be precipitated. Poisoning by heavy metals in humans can therefore be cured by taking egg white which precipitates the metal out.

$$Hg^{2+} + 4CH_3NH_2 \rightarrow [Hg(NH_2CH_3)_4]^{2+}$$

5.4.2.5 Addition Reaction

This reaction has already been discussed in connection with aldehydes and ketones (see unit 4).

5.4.2.6 Reaction with Nitrous Acid (HNO_2)

Nitrous acid is formed by the reaction of sodium nitrite and hydrochloric acid, i.e.,

$NaNO_2 + HCl \rightarrow HNO_2 + NaCl$, and the two are usually mixed together with the organic reactants. The reaction of an amine with nitrous acid can be used to determine the type of amine, i.e., whether the amine is primary, secondary or tertiary. Primary amines react with nitrous acid to yield a diazonium salt which decomposes spontaneously with evolution of nitrogen gas. Secondary amines produce a yellow oil (a nitrosamine) while tertiary amines do not react at all.

The reaction of a primary amine is depicted as follows:

$$CH_3CH_2CH_2NH_2 + NaNO_2 + HCl \longrightarrow CH_3CH_2CH_2N{=}N{-}OH$$

A diazotic acid

$$\downarrow$$

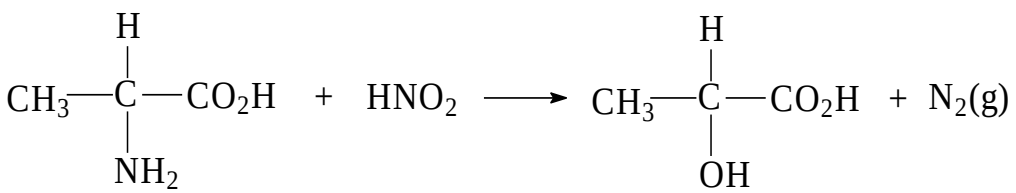
$$CH_3CH_2CH_2\overset{+}{N}{\equiv}N\ Cl^-$$

A diazonium salt

$$\downarrow H_2O$$

$$N_2(g) + CH_3CH_2CH_2OH + CH_3CH{=}CH_2$$

Note in the scheme above that a mixture of sodium nitrite ($NaNO_2$) and hydrochloric acid (HCl), produces nitrous acid (HNO_2), which is then the reactant. The problem with this reaction is that it gives a mixture of products. However, the formation of nitrogen in these reactions is useful in analysis of, for example, biological materials. The chain length of a protein molecule (in terms of the number of amino acids) can be estimated in this way. Hydrolysis of a protein molecule yields the constituent amino acids which can then be reacted with nitrous acid, for example:



Since one mole of amino acid gives one mole of nitrogen, the number of moles of nitrogen gas will be equal to the number of moles of amino acids in the protein molecule. The reaction of a secondary amine to form a nitrosamine is as follows:

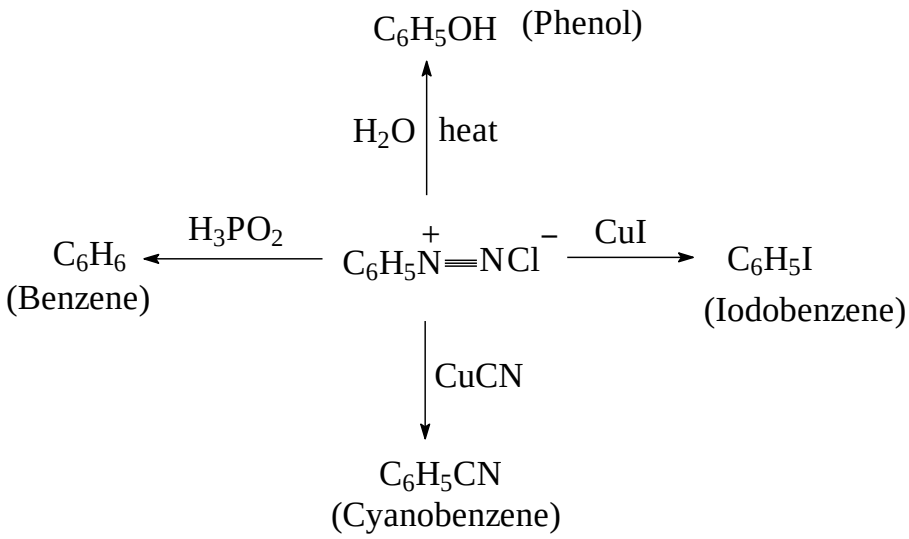
$$(CH_3CH_2)_2NH + HNO_2 \longrightarrow (CH_3CH_2)_2N{-}N{=}O$$

A nitrosamine (oil)

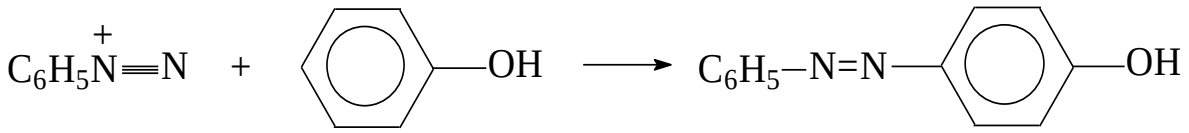
Primary aromatic amines also react with nitrous acid to form diazonium salts (**the process is called diazotisation**).

$$C_6H_5NH_2 + NaNO_2 + HCl \longrightarrow C_6H_5\overset{+}{N}{\equiv}NCl^{-} + H_2O$$

These diazonium salts can be used for the synthesis of various other compounds, e.g.,



Diazonium salts can also act as electrophiles (negative-seeking groups) and can attack other aromatic rings forming azo compounds which are important as dyes and other colouring materials (azo dyes). *The process of forming an azo compound is called diazo coupling*:

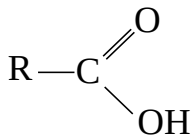


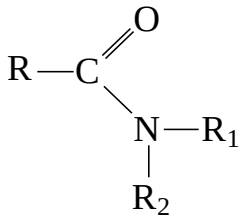
Phenol

An azo compound

5.5.0 AMIDES

The importance of amides is manifested in the chemistry of proteins and in metabolic processes where they appear as intermediates. *Amides can be regarded as derivatives of carboxylic acids in which the hydroxyl group of the acid is replaced by an amino group*.



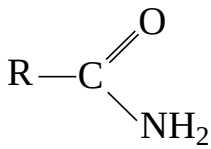


Where R_1, R_2 = H or C or both

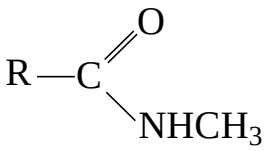
A carboxylic acid

An amide

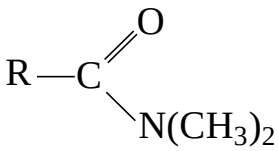
In the same way we write the structure of a carboxylic acid in the form RCO_2H, so too we can write the structure of an amide in the form $R'CONR_2$. Amides are also divided into three categories; *primary, secondary and tertiary*. Primary amides are those in which the nitrogen atom is not substituted by alkyl or aryl groups (in the formula above, both R_1 and R_2 = H). In secondary amides, the nitrogen atom bears only one alkyl or aryl group (in formula above, R_1 = H; R_2 = C or vice versa), and tertiary amides have both hydrogen atoms on nitrogen replaced by alkyl or aryl groups (i.e., R_1R_2 = C). For example:



Primary



Secondary



Tertiary

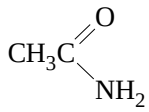
5.5.1 Nomenclature of Amides

5.5.1.2 Trivial names

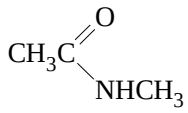
Trivial (or common) names of amides are derived from the common names of related carboxylic acids, the ending **-ic acid** in the name of the acid is replaced by the word **amide**. For substituted amides, a capital letter N is used to locate the substituents on nitrogen (compare trivial names of amines). Some examples are included in parentheses in the systematic names below.

5.5.1.3 Systematic names

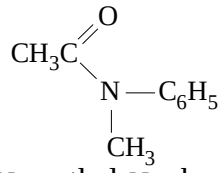
The systematic names of amides are derived from the names of the corresponding carboxylic acids; the ending **-oic acid** or **-ic acid** in the name of the acid, being replaced by the word **amide**. In substituted amides (secondary or tertiary) a capital letter N is placed before the name of the substituent to designate its attachment to the nitrogen atom (compare amines). As in carboxylic acids, numbering starts from the carbonyl carbon as one and proceeds along the parent hydrocarbon chain. The general rules given in unit 1 of this module concerning substituents on the hydrocarbon chain also apply to the amides. ***I urge you to refer to them as you go over the following examples*.**



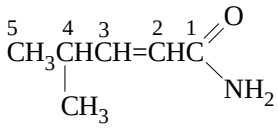
ethanamide
(or acetamide)



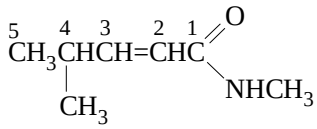
N-methyl ethanamide
(or N-methyl acetamide)



N-methyl-N-phenyl ethanamide



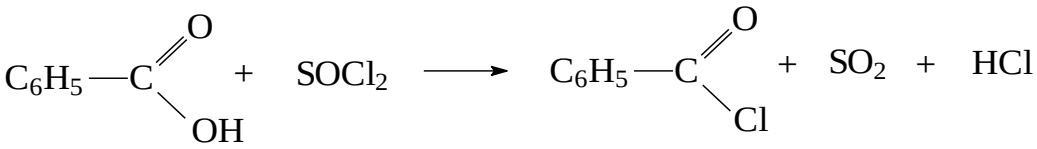
4-methyl-2-pentenamide
(**note:** 2 is locating the C=C)



N,4-dimethyl-2-pentenamide

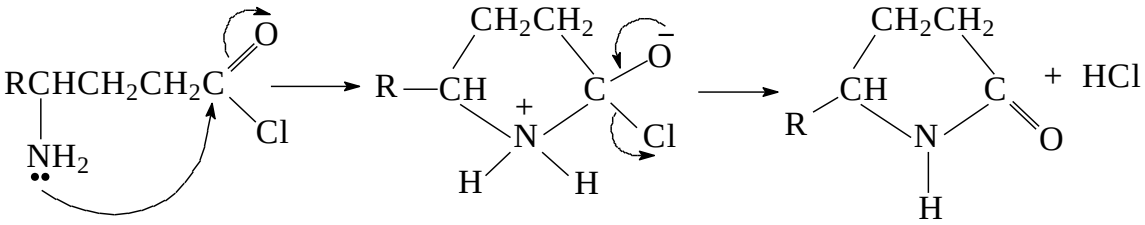
5.5.1.4 Preparation of Amides

There are several methods of preparing amides. We discussed the acylation of ammonia and amines in Section 5.4.2.2. The acid chloride used in acylation can be formed from the reaction of a carboxylic acid and thionyl chloride as in the example below.



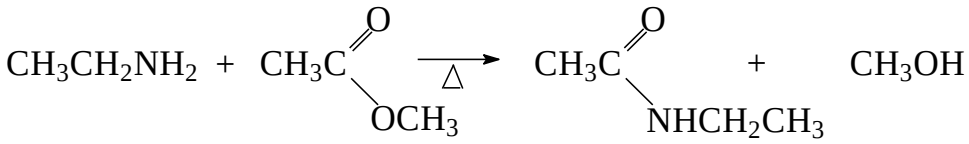
Benzoic acid Benzyl chloride

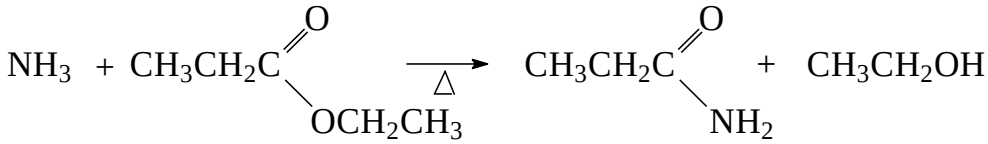
Now imagine the amino and acyl halide groups to be on the same chain. They can also react (internally) and the product will be a cyclic amide. *A cyclic amide is called* as *a lactam*. As in lactones, the size of the ring is also designated by Greek letters, β(beta, 4-membered ring), γ(gamma, 5-membered ring), δ(delta, 6-membered ring), etc. Thus a γ-lactam has five atoms in a ring. For example:



A γ-lactam

The most common lactams are γ- and δ-lactams although β-lactams are also encountered in some antibiotics such as penicillin. Esters can also be used as acylating agents on the ammonia or amines. The solutions here, must be heated.





5.5.1.5 **Properties of Amides**

5.5.1.5.1 **Physical**

Except for methanamide which is a liquid at room temperature, the other members are solids. Primary and secondary amides can readily form hydrogen bonds to each other; consequently, these have high melting and boiling points. Tertiary amides, on the other hand, have no hydrogen attached to nitrogen and as a result, they cannot hydrogen-bond to each other and their melting and boiling points are consequently lower than those of the corresponding primary amides with the same number of carbons. The lower members are also soluble in water.

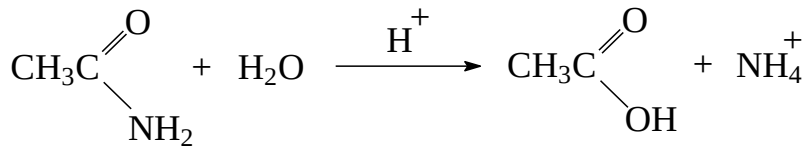
5.5.1.5.2 Chemical

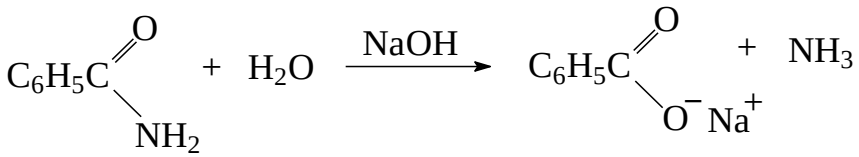
5.5.1.5.2.1 Basicity

Although ammonia and amines are basic, amides are not; in fact their pK_b values are so low that for practical purposes, they can be regarded as neutral compounds.

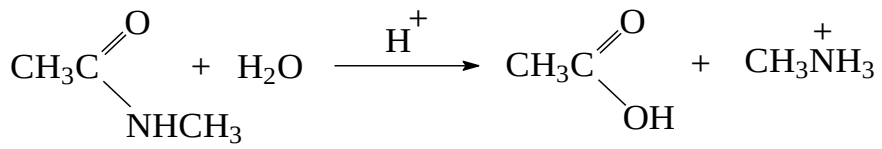
5.5.1.5.2.2 Hydrolysis

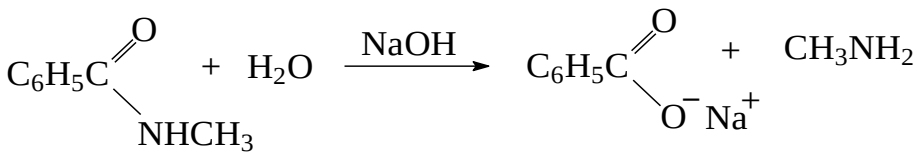
Amides can be hydrolyzed by using either acid or base, as catalysts. For primary amides, acid hydrolysis gives a carboxylic acid and an ammonium ion, while base hydrolysis produces a salt of the carboxylic acid and ammonia. The examples below show acid hydrolysis of ethanamide and base hydrolysis of benzanamide.





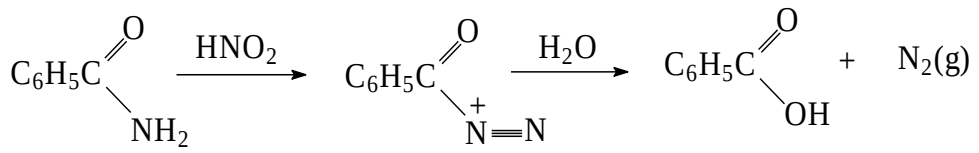
Secondary and tertiary amides, on the other hand, produce a carboxylic acid and an alkyl ammonium ion when hydrolyzed by acid while base hydrolysis gives an amine and a salt of the carboxylic acid.





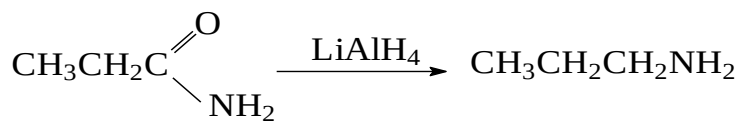
5.5.1.5.2.3 Reaction with nitrous acid

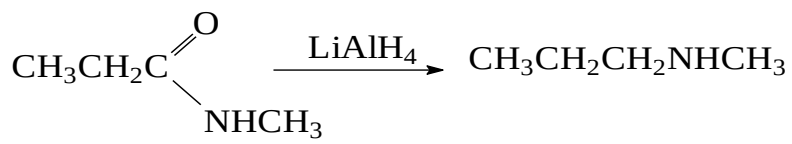
In section 5.4.2.6 we described the reaction of an amine with nitrous acid. Primary amides undergo a similar reaction with nitrous acid which produces the corresponding carboxylic acid and nitrogen gas.



5.5.1.5.2.4 Reduction

Amides are readily reduced to the corresponding amine by strong reductants such as lithium aluminum hydride, e.g.,





Note in the equations above that the C=O is converted to CH_2. Since oxygen is removed from the carbonyl group, this is a reduction process (see module on oxidation and reduction).

SUGGESTED READING TEXTS

1. Graham Solomons T.W. and Fryhle C.B., 2004. *Organic Chemistry, 8th ed.*, John Wiley & Sons. New York.
2. McMurry J. and Castellion M.E., 2003. *Fundamentals of general organic and biological Chemistry, 4th ed.*, Pearson Education Inc.New Jersey.
3. Taylor G.A., 1987. *Organic Chemistry for students of Biology and Medicine 3rd ed.*, Longman.
4. Morrison R.T. and Boyd R.N., 1987. *Organic Chemistry, 5th ed.*, Allyn and Bacon, Inc., .
5. Pine S.H., 1987. *Organic Chemistry, 5th ed.*, McGraw-Hill Book Co.

5.6.0 Unit summary

- In this Unit we have discussed the types of amines and their properties.
- We have seen that there are three types of amines, primary, secondary and tertiary.
- We have also discussed the naming of these compounds. Finally we have learned the physical and chemical properties of amines.
- We then looked at the types of amides. We also found that there are three types of amides. You have learnt the naming of these compounds.
- We finished the module by looking at the physical and chemical properties of amides.

5.7.0 Unit test

1. Define the following terms/phrases

(a) Amine

(b) Secondary amine

(c) Tertiary amide

(d) δ - Lactam

(e) Nucleophilic substitution reaction

2. Give systematic names of the following compounds

(a) $CH_3CH_2CH_2CH_2NH_3$

(b) $CH_3CH_2CH_2CH_2NHC_6H_5$

(c) $CH_3CH_2CH_2CH_2N(C_6H_5)C_6H_5$

(d) $CH_3CH_2CH_2CONH_2$

(e) $CH_3CH_2CH_2CONHC_6H_5$

3. Write structures of the major products of the following reactions

(a) $CH_3CH_2CH_2CH_2NH_3 + C_6H_5C(=O)Br \longrightarrow$

(b) $CH_3CH_2CH_2CH_2NH_3 + C_6H_5C(=O)H \longrightarrow$

(c) $C_6H_5C(=O)NH_2 + H_2O \xrightarrow{H^+}$

(d) $C_6H_5C(=O)NHC_6H_5 + H_2O \xrightarrow{NaOH}$

UNIT 6: CARBOXYLIC ACIDS AND ESTERS

6.0.0 Introduction

In unit 3 we studied organic compounds with oxygen -containing functional groups that included aldehydes, ketones and alcohols. In this unit we will look at carboxylic acids and esters which also contain oxygen in their structures.

6.1.0 Unit Learning Outcomes

By the end of this unit, you should be able to:

- Give systematic names of Carboxylic acids and esters
- State physical properties of Carboxylic acids and esters
- State chemical properties of Carboxylic acids and esters

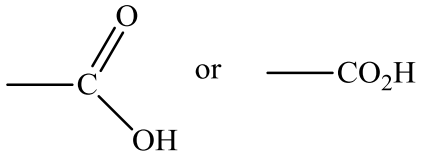
6.1.1 Unit Key Terms

Ensure that you understand the key terms or phrases used in this unit as listed below.

- Acid hydrolysis
- Alkanoic acid
- Base hydrolysis
- Carboxylic acid
- Carboxylic ion
- Ester
- Saponification

6.2.0.0 Carboxylic acids

The carboxylic acids also called alkanoic acids, contain the carboxyl group (-COOH) as the functional group. The –COOH functional group can also be represented as:

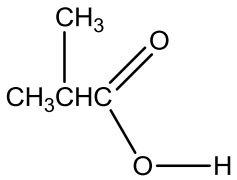


6.2.1 Nomenclature of carboxylic acids

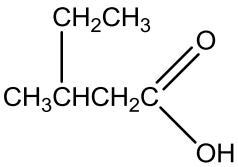
The systematic name of a carboxylic acid ends in oic acid. The name is derived from the name of an alkane with the same number of carbon atoms and the ending ‘’e” in the name of the alkane is replaced by ‘’oic acid” in the name of a carboxylic acid.

Alkane → Alkanoic acid

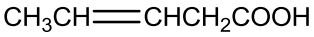
Methane becomes methanoic acid, ethane becomes ethanoic acid, etc. Rules for naming the carboxylic acids are the same as those of alkanes. However, start numbering the longest chain from the carboxyl (-COOH) carbon as number 1. Since the carboxyl carbon is always at position 1, there is no need to put this number in the name.



Methylpropanoic acid



3 - Methylpentanoic acid

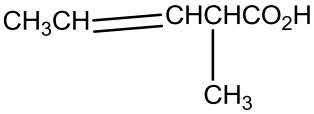


Pent - 3 - enoic acid

or 3 - Pentenoic acid

$HO_2CCH_2CH_2CO_2H$

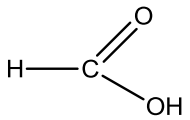
Butane - 1, 4 - dioic acid



2 - Methylpent - 3 - enoic acid

or 2 - Methyl - 3 - pentenoic acid

Some carboxylic acids are known by their common names. Common names of some carboxylic acids are put in brackets in the structures below.

	CH_3CO_2H	CH_3CH_2COOH	$CH_3CH_2CH_2COOH$
Methanoic acid	ethanoic acid	propanoic acid	butanoic acid
(Formic acid)	(acetic acid)	(propionic acid)	(butyric acid)

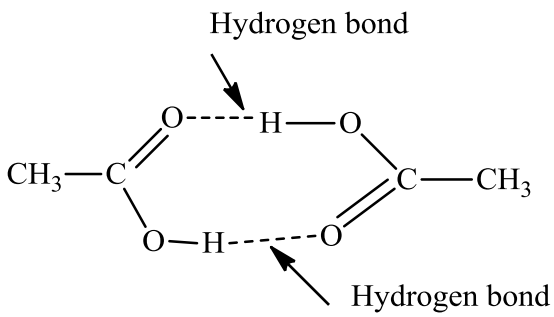
6.2.2 Preparation of carboxylic acids

Carboxylic acids can be prepared by a variety of methods such as oxidation of alkenes, alcohols and Alkanals. In unit 2 we saw that some of the products formed when alkenes were oxidised by hot potassium permanganate ($KMnO_4$) were carboxylic acids. In unit 3 we learnt that oxidation of primary alcohols produced alkanals which on further oxidation yielded carboxylic acids. We also learnt in the same unit that alcohols with OH groups on adjacent carbons were oxidised by periodic acid to yield carboxylic acids as some of the products.

6.2.3 Physical properties of carboxylic acids

6.2.3.1 Boiling and melting points

Carboxylic acids have higher boiling points than alkanes and alcohols with same number of carbon atoms. Carboxylic acids have hydroxyl and carbonyl groups making these compounds very polar in nature. This makes carboxylic acids possible to form two hydrogen bonds unlike one hydrogen bond in alcohols. Below are structures of two ethanoic acid molecules forming hydrogen bonds with each other.



The boiling points of carboxylic acids increase as the molecules get bigger.

- Methanoic acid $101^{o}C$
- Propanoic acid $141^{o}C$
- Pentanoic acid $186^{o}C$

6.2.3.2 Solubility

Smaller carboxylic acids (C_1 to C_5) are soluble in water, whereas larger carboxylic acids (C_6 and above) are less soluble due to the increasing hydrophobic nature of the hydrocarbon chains.

6.2.4 Chemical properties of carboxylic acids

6.2.4.1 Acidity and basicity of carboxylic acids

Carboxylic acids are weak acids. They partially dissociates into H^+ cations and $RCOO^-$ anions in neutral aqueous solvents such as water.

$$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$$

Carboxylic acids react with bases in the same way as mineral acids do. This is neutralisation reaction and an example is shown below.

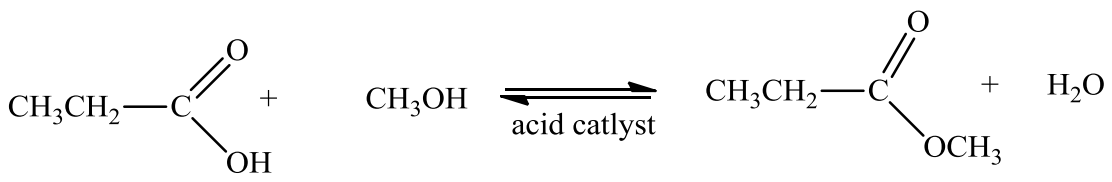
$$CH_3COOH + NaOH \rightarrow CH_3COO^- Na^+ + H_2O$$

Carboxylic acids are also weak bases as they do accept protons only from the strongest acids as shown below.

$$CH_3CO_2H + HClO_4 \rightarrow CH_3CO_2H_2 + ClO_4^-$$

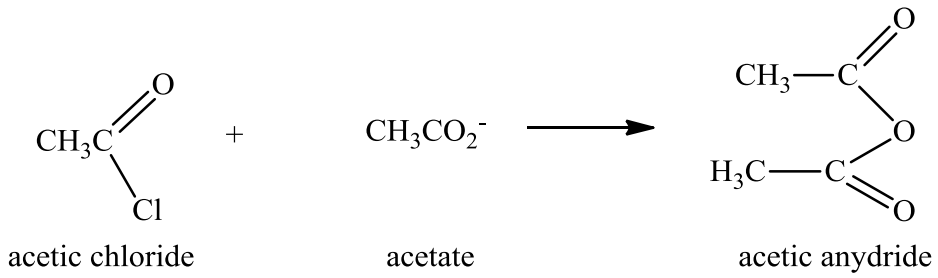
6.2.4.2 Formation of esters

Can you recall that in unit 3.7.2 we learnt about formation of esters of carboxylic acids? We learnt that esters are formed when carboxylic acids react with alcohols in the presence of acid. We called this reaction Fischer esterification. Following is a reaction of propanoic acid with methanol in the presence of an acid catalyst.



6.2.4.3 Formation of acid anhydrides

The reaction of carboxylate ions with acid chlorides produces acid anhydrides as shown in the reaction below.



6.2.5 Unit activity

1. Give systematic names of the following compounds:

a. $CH_3CH_2\overset{\overset{Br}{|}}{C}HCH_2COOH$

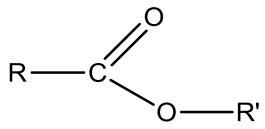
b. $CH_3COCH_2CO_2H$

c. $CH_3CH(CH_2CH_3)CH_2COOH$

2. Explain why carboxylic acids have higher boiling points than alcohols of the same number of carbon atoms.

6.3.0 ESTERS

Esters are derivative from carboxylic acids, prepared by the reaction of a Carboxylic acid and an alcohol. In ester, the H in the hydroxyl group is replaced with an alkyl group as shown in the general formula of the ester:



where R = H or C

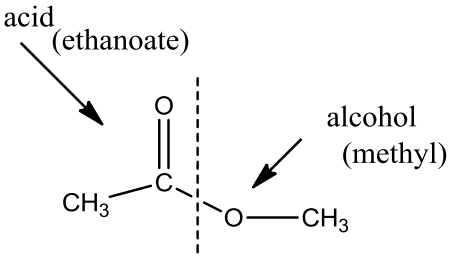
R' = alkyl or aryl

Esters of carboxylic are particularly important because the fats such as margarine and cooking oil such as Kazinga are esters of long chain carboxylic acids and an alcohol called glycerol (1, 2, 3 – propanetriol). In plants, esters give flowers and fruits their pleasant fragrance and flavours.

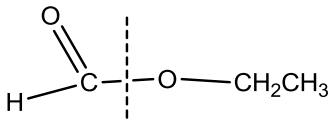
6.3.1 Nomenclature

Esters of carboxylic acids are named as alkyl or aryl carboxylate. The systematic name is derived from the name of carboxylic acid with the same number of carbon atoms. The ending “oic” acid in the name of the acid is replaced by “oate”. Numbering starts from the carbonyl carbon as

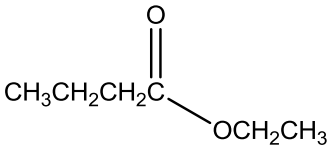
position 1 and proceeds along the chain. The alcohol gives the first part of the name, and the carboxylic acid gives the second part.



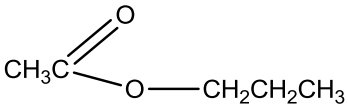
Methyl ethanoate



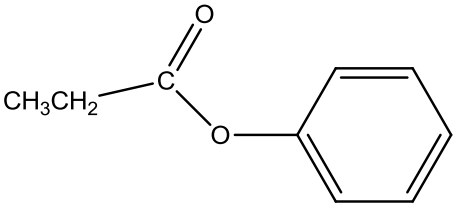
Ethyl methanoate



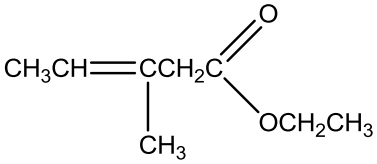
ethyl butanoate



Propyl ethanoate



Phenyl propanoate

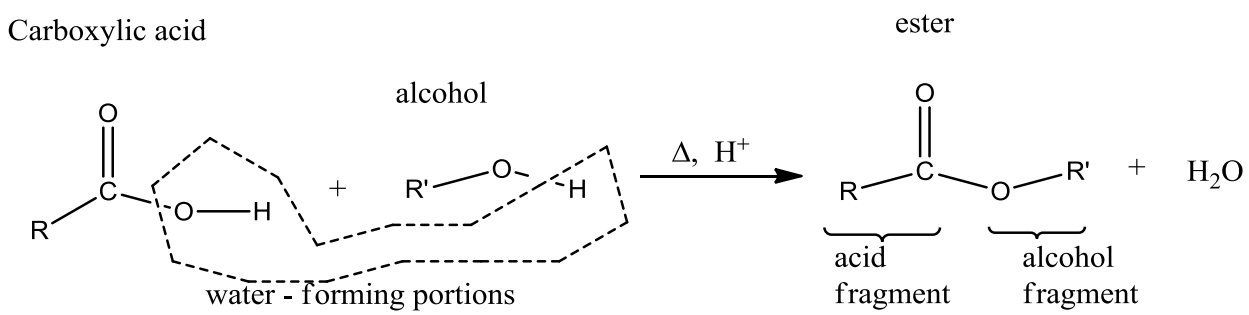


Ethyl - 3 - methyl - 3 - pentenoate

6.3.2.0 Preparation of esters

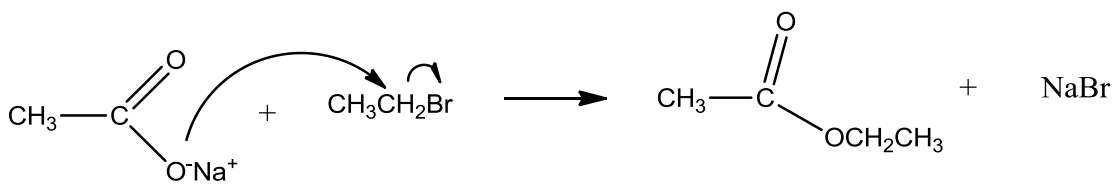
6.3.2.1 Esterification

Esterification is the reaction of a carboxylic acid and an alcohol in the presence of an acid catalyst to produce an ester. We discussed esterification in unit 3 and also section 6.2.4.2. Therefore, here we only give a general formation of an ester from the reaction of a carboxylic acid and an alcohol.



6.3.2.2 Displacement

Esters can also be prepared by the displacement of a halogen from an alkylhalide by a carboxylate ion. The following is an example.



6.3.3.0 Physical properties of esters

6.3.3.1 Boiling point and melting point

Lower molecular weight esters are liquids at room temperature with pleasant smell. The higher ones are solids. Esters have no H atom attached to the O atoms and so there are no hydrogen bonding between the ester molecules. As such esters have lower boiling and melting points than corresponding carboxylic acids.

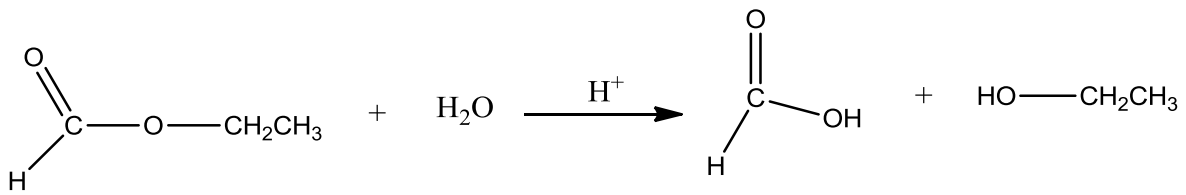
6.3.3.2 Solubility

We said that ester molecules do not form hydrogen bonds amongst themselves. Can you see that these esters can form hydrogen bonds with water? The O in ester will form hydrogen bond with H from water molecule. As a result, esters are relatively miscible with water but solubility decreases with increasing mass of the molecule.

6.3.4.0 Chemical properties esters

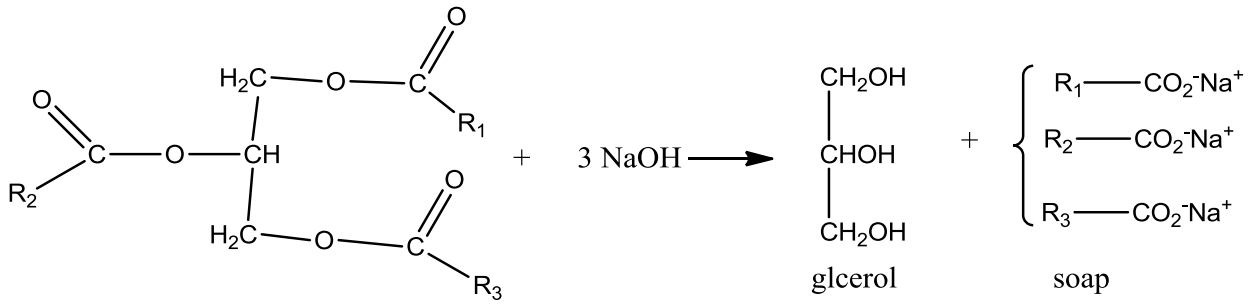
6.3.4.1 Acid hydrolysis

Esters react with water using an acid catalyst and in the process the ester is split into carboxylic acid and alcohol. Can you see that in this case an ester is converted back to carboxylic acid and alcohol by hydrolysing it with an acid? We call this acid hydrolysis. Acid hydrolysis is reversible.



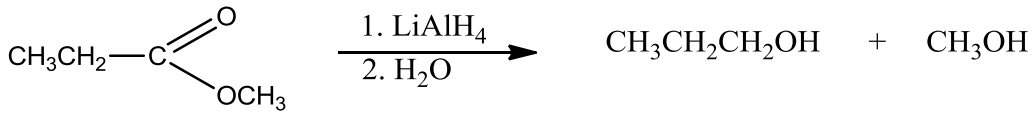
6.3.4.2 Base hydrolysis (Saponification)

An ester can react with a strong base to produce a salt of carboxylic acid and an alcohol. This base hydrolysis of an ester is also called saponification and is irreversible. Soaps (for bathing, washing etc) are formed industrially by the base hydrolysis. When an ester of a long chain carboxylic acid and an alcohol, glycerol, is hydrolysed by sodium hydroxide, the sodium carboxylates (salts) formed are the soaps. Below is an example.



6.3.4.3 Reduction

Esters can be reduced to the corresponding alcohols by using strong reducing reagents like lithium aluminium hydride (LiAlH4), followed by hydrolysis.



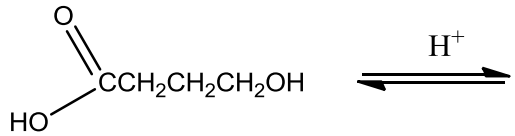
6.3.4.4 Formation of amides

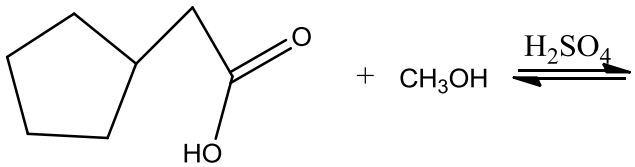
Amines and ammonia react with esters in a condensation type of reaction in which alcohol is eliminated and amides are formed. We discussed amines, ammonia and amides in unit 5.

6.3.5 Unit activity

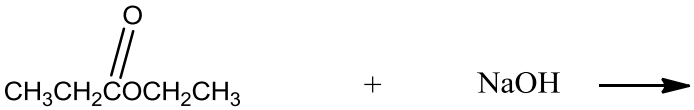
1. Define the following terms
   a) Saponification
   b) Esterification

2.Write the structures of the products in the following reactions

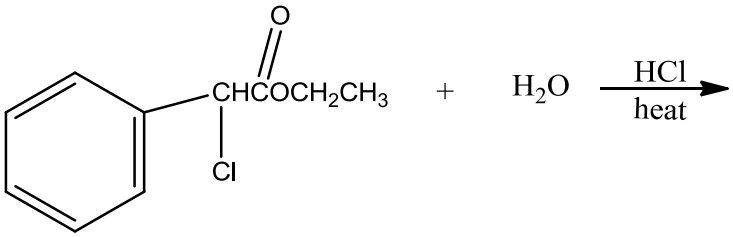
a. 

b. 

3. Give the product of the following base hydrolysis:



4. Give the product of the following acid hydrolysis:



6.4.0 Unit Summary

Carboxylic acids are organic compounds which contain a carboxyl (-COOH) functional group.

The systematic names of carboxylic acids are derived from the name of alkanes with the same number of carbon atoms and the ending “e” in the alkane name is replaced by “oic acid” in the carboxylic acid.

Carboxylic acids can be prepared by oxidation of alkanes, alcohols and aldehydes.

Carboxylic acids have higher boiling points than corresponding alkanes and alcohols.

Carboxylic acids form hydrogen bonds with water and smaller carboxylic acids are soluble in water.

Reaction of carboxylic acids and alcohols in the presence of an acid catalyst produces esters.

The functional group of esters is R-COO-R.

When naming an ester, the alcohol gives the first part of the name (alkyl or aryl) and the carboxylic acid gives the second part. The ending “oic acid” of the carboxylic acid is replaced by “oate”. For example: methyl ethanoate.

Esters can be hydrolysed by an acid to produce carboxylic acids and alcohols.

Esters can also be hydrolysed by strong base and the process is called saponification.

6.5.0 References

1. Graham Solomon T. W., and Fryhle C. B., 2004. Organic *Chemistry, 8th Ed*. John Willey and Sons, New York.

2. McMurry J. and Castellion M. E., 2003. *Fundamentals of general organic and biological chemistry,* 4th Ed., Pearson Education, New Jersey